

AP Calculus AB/BC

Lecture Notes

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Contents

0	Preliminary	5
0.1	Trigonometry	5
0.1.1	The Unit Circle and Special Angles	5
0.1.2	Periodic Properties of Sine and Cosine	6
0.1.3	Core Identities	7
0.2	Advanced Knowledge of Functions	8
0.2.1	Translations and Dilations	8
0.2.2	Exponential Functions	9
0.2.3	Logarithmic Functions	9
0.2.4	Exponentials and Logarithms as Inverses	10
1	Limits and Continuity	11
1.1	Introducing Calculus: Can Change Occur at an Instant?	11
1.2	Defining Limits and Using Limit Notation	12
1.3	Estimating Limit Values from Graphs	13
1.4	Estimating Limit Values from Tables	13
1.5	Determining Limits Using Algebraic Properties of Limits	14
1.6	Determining Limits Using Algebraic Manipulation	15
1.7	Selecting Procedures for Determining Limits	15
1.8	Determining Limits Using the Squeeze Theorem	16
1.9	Connecting Multiple Representations of Limits	17
1.10	Exploring Types of Discontinuities	17
1.11	Defining Continuity at a Point	18
1.12	Confirming Continuity over an Interval	18
1.13	Removing Discontinuities	19
1.14	Connecting Infinite Limits and Vertical Asymptotes	19
1.15	Connecting Limits at Infinity and Horizontal Asymptotes	20
1.16	Working with the Intermediate Value Theorem (IVT)	21
2	Differentiation: Definition and Fundamental Properties	23
2.1	Defining Average and Instantaneous Rates of Change at a Point	23
2.2	Defining the Derivative of a Function and Using Derivative Notation	24
2.3	Estimating Derivatives of a Function at a Point	25
2.4	Connecting Differentiability and Continuity: Determining When Derivatives Do and Do Not Exist	25
2.5	Applying the Power Rule	26
2.6	Derivative Rules: Constant, Sum, Difference, and Constant Multiple	27
2.7	Derivatives of $\cos x$, $\sin x$, e^x , and $\ln x$	27
2.8	The Product Rule	28
2.9	The Quotient Rule	28
2.10	Finding the Derivatives of Tangent, Cotangent, Secant, and/or Cosecant Functions	29

3	Differentiation: Composite, Implicit, and Inverse Functions	30
3.1	The Chain Rule	30
3.2	Implicit Differentiation	31
3.3	Differentiating Inverse Functions	32
3.4	Differentiating Inverse Trigonometric Functions	33
3.5	Selecting Procedures for Calculating Derivatives	34
3.6	Calculating Higher-Order Derivatives	35
4	Contextual Applications of Differentiation	36
4.1	Interpreting the Meaning of the Derivative in Context	36
4.2	Straight-Line Motion: Connecting Position, Velocity, and Acceleration . .	37
4.3	Rates of Change in Applied Contexts Other Than Motion	38
4.4	Introduction to Related Rates	38
4.5	Solving Related Rates Problems	39
4.6	Approximating Values of a Function Using Local Linearity and Linearization	41
4.7	Using L'Hospital's Rule for Determining Limits of Indeterminate Forms .	42
4.7.1	A Rigorous Proof of L'Hospital's Rule	43
5	Analytical Applications of Differentiation	48
5.1	Using the Mean Value Theorem	48
5.2	Extreme Value Theorem, Global Versus Local Extrema, and Critical Points	49
5.3	Determining Intervals on Which a Function is Increasing or Decreasing .	51
5.4	Using the First Derivative Test to Determine Relative (Local) Extrema .	52
5.5	Using the Candidates Test to Determine Absolute (Global) Extrema . . .	52
5.6	Determining Concavity of Functions over Their Domains	53
5.7	Using the Second Derivative Test to Determine Extrema	54
5.8	Sketching Graphs of Functions and Their Derivatives	56
5.9	Connecting a Function, Its First Derivative, and Its Second Derivative . .	57
5.10	Introduction to Optimization Problems	57
5.11	Solving Optimization Problems	58
5.12	Exploring Behaviors of Implicit Relations	59
6	Integration and Accumulation of Change	61
6.1	Exploring Accumulations of Change	61
6.2	Approximating Areas with Riemann Sums	62
6.3	Riemann Sums, Summation Notation, and Definite Integral Notation . .	63
6.4	The Fundamental Theorem of Calculus and Accumulation Functions . . .	64
6.5	Interpreting the Behavior of Accumulation Functions Involving Area . . .	65
6.6	Applying Properties of Definite Integrals	65
6.7	The Fundamental Theorem of Calculus and Definite Integrals	66
6.8	Finding Antiderivatives and Indefinite Integrals: Basic Rules and Notation	67
6.9	Integrating Using Substitution	68
6.9.1	Products of Trigonometric Powers	69
6.9.2	Trigonometric Substitution	70
6.9.3	Antiderivatives Involving Inverse Trigonometric Functions	71
6.10	Integrating Functions Using Long Division and Completing the Square .	72
6.11	Integrating Using Integration by Parts	73
6.12	Integrating Using Linear Partial Fractions	74
6.13	Evaluating Improper Integrals	74

6.13.1	Improper Integrals with a Discontinuous Integrand	75
6.13.2	The p -Integral: A Convergence Test We Can Prove Completely	76
6.14	Selecting Techniques for Antidifferentiation	78
6.14.1	Summary: A Complete Table of Common Antiderivatives	79
7	Differential Equations	82
7.1	Modeling Situations with Differential Equations	82
7.2	Verifying Solutions for Differential Equations	84
7.3	Sketching Slope Fields	85
7.4	Reasoning Using Slope Fields	87
7.5	Approximating Solutions Using Euler's Method	88
7.6	Finding General Solutions Using Separation of Variables	90
7.7	Finding Particular Solutions Using Initial Conditions and Separation of Variables	92
7.8	Exponential Models with Differential Equations	93
7.9	Logistic Models with Differential Equations	95
7.9.1	Beyond the AP Exam: Deriving the Explicit Logistic Solution	97
8	Applications of Integration	100
8.1	Finding the Average Value of a Function on an Interval	100
8.2	Connecting Position, Velocity, and Acceleration Using Integrals	102
8.3	Using Accumulation Functions and Definite Integrals in Applied Contexts	104
8.4	Finding the Area Between Curves Expressed as Functions of x	105
8.5	Finding the Area Between Curves Expressed as Functions of y	106
8.6	Finding the Area Between Curves That Intersect at More Than Two Points	107
8.7	Volumes with Cross Sections: Squares and Rectangles	108
8.8	Volumes with Cross Sections: Triangles and Semicircles	110
8.9	Volume with Disc Method: Revolving Around the x - or y -Axis	111
8.10	Volume with Disc Method: Revolving Around Other Axes	112
8.11	Volume with Washer Method: Revolving Around the x - or y -Axis	114
8.12	Volume with Washer Method: Revolving Around Other Axes	115
8.13	The Arc Length of a Smooth, Planar Curve and Distance Traveled	116
9	Parametric Equations, Polar Coordinates, and Vector-Valued Functions	119
9.1	Defining and Differentiating Parametric Equations	119
9.2	Second Derivatives of Parametric Equations	122
9.3	Finding Arc Lengths of Curves Given by Parametric Equations	123
9.4	Defining and Differentiating Vector-Valued Functions	125
9.5	Integrating Vector-Valued Functions	126
9.6	Solving Motion Problems Using Parametric and Vector-Valued Functions	127
9.7	Defining Polar Coordinates and Differentiating in Polar Form	128
9.8	Find the Area of a Polar Region or the Area Bounded by a Single Polar Curve	131
9.9	Finding the Area of the Region Bounded by Two Polar Curves	133

10 Infinite Sequences and Series	136
10.1 Defining Convergent and Divergent Infinite Series	136
10.2 Working with Geometric Series	138
10.3 The n th Term Test for Divergence	140
10.4 Integral Test for Convergence	141
10.5 Harmonic Series and p -Series	144
10.6 Comparison Tests for Convergence	145
10.7 Alternating Series Test for Convergence	147
10.8 Ratio Test for Convergence	149
10.9 Determining Absolute or Conditional Convergence	150
10.10 Alternating Series Error Bound	152
10.11 Finding Taylor Polynomial Approximations of Functions	153
10.12 Lagrange Error Bound	154
10.13 Radius and Interval of Convergence of Power Series	156
10.14 Finding Taylor or Maclaurin Series for a Function	157
10.15 Representing Functions as Power Series	158
10.15.1 Summary: Common Power Series and Their Radii of Convergence	160

Preliminary

Remark

None of the material in this chapter is directly assessed on the AP Calculus Exam, but every topic in it is used constantly, without re-explanation, starting in Unit 1. Trigonometric identities appear throughout differentiation and integration; fluency with function transformations, exponentials, and logarithms is assumed from the first day of the course.

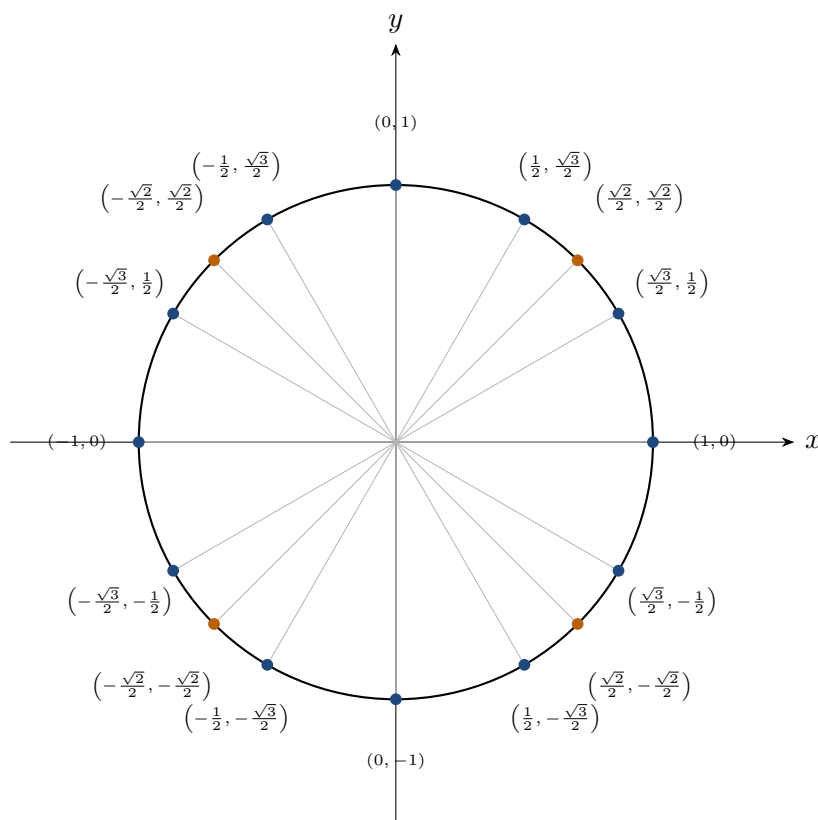
0.1 Trigonometry

0.1.1 The Unit Circle and Special Angles

Definition 0.1.1: Unit Circle Definition of Sine and Cosine

For an angle θ measured counterclockwise from the positive x -axis, let (x, y) be the point where the terminal ray meets the unit circle $x^2 + y^2 = 1$. Then

$$\cos \theta = x, \quad \sin \theta = y, \quad \tan \theta = \frac{y}{x} = \frac{\sin \theta}{\cos \theta} \quad (x \neq 0).$$

**Remark**

By symmetry, every angle on the circle has the same $|\cos \theta|$, $|\sin \theta|$ magnitudes as its first-quadrant reference angle (30° , 45° , or 60°); only the sign changes according to the quadrant (ASTC: All / Sine / Tangent / Cosine positive, quadrants I–IV respectively).

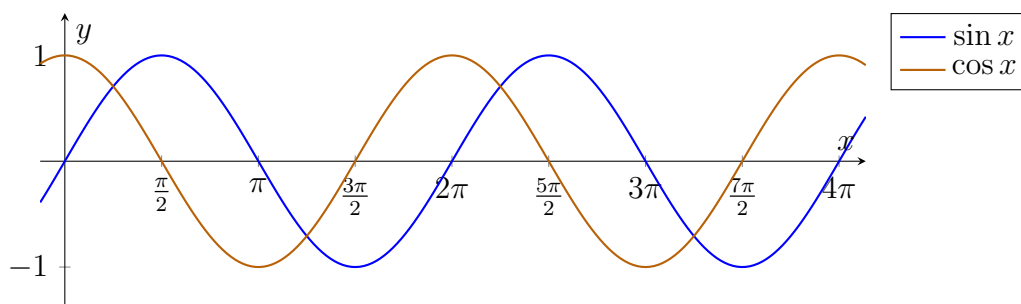
0.1.2 Periodic Properties of Sine and Cosine**Theorem 0.1.1: Periodicity, Symmetry, and Range**

For all x and all integers k :

$$\sin(x + 2\pi k) = \sin x, \quad \cos(x + 2\pi k) = \cos x \quad (\text{period } 2\pi),$$

$$\sin(-x) = -\sin x \text{ (odd)}, \quad \cos(-x) = \cos x \text{ (even)},$$

$$-1 \leq \sin x \leq 1, \quad -1 \leq \cos x \leq 1 \quad \text{for all } x \in \mathbb{R}.$$



0.1.3 Core Identities

Theorem 0.1.2: Reciprocal and Quotient Identities

$$\csc x = \frac{1}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}, \quad \tan x = \frac{\sin x}{\cos x}.$$

Theorem 0.1.3: Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x.$$

Theorem 0.1.4: Co-function Identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x, \quad \cos\left(\frac{\pi}{2} - x\right) = \sin x, \quad \tan\left(\frac{\pi}{2} - x\right) = \cot x.$$

Theorem 0.1.5: Sum and Difference Formulas

$$\begin{aligned}\sin(a \pm b) &= \sin a \cos b \pm \cos a \sin b, \\ \cos(a \pm b) &= \cos a \cos b \mp \sin a \sin b, \\ \tan(a \pm b) &= \frac{\tan a \pm \tan b}{1 \mp \tan a \tan b}.\end{aligned}$$

Theorem 0.1.6: Double-Angle and Half-Angle Formulas

$$\begin{aligned}\sin 2x &= 2 \sin x \cos x, \\ \cos 2x &= \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x, \\ \tan 2x &= \frac{2 \tan x}{1 - \tan^2 x}, \\ \sin \frac{x}{2} &= \pm \sqrt{\frac{1 - \cos x}{2}}, \quad \cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}}.\end{aligned}$$

Theorem 0.1.7: Sum-to-Product Formulas

$$\begin{aligned}\sin A + \sin B &= 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right), \\ \sin A - \sin B &= 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right), \\ \cos A + \cos B &= 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right), \\ \cos A - \cos B &= -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right).\end{aligned}$$

Theorem 0.1.8: Product-to-Sum Formulas

$$\begin{aligned}\sin A \sin B &= \frac{1}{2} [\cos(A - B) - \cos(A + B)], \\ \cos A \cos B &= \frac{1}{2} [\cos(A - B) + \cos(A + B)], \\ \sin A \cos B &= \frac{1}{2} [\sin(A + B) + \sin(A - B)].\end{aligned}$$

Example 0.1.1: Evaluating with the Sum Formula

Find the exact value of $\sin 75^\circ$.

Solution. Write $75^\circ = 45^\circ + 30^\circ$:

$$\sin 75^\circ = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Example 0.1.2: Sum-to-Product Simplification

Simplify $\frac{\sin 5x + \sin x}{\cos 5x + \cos x}$.

Solution. Apply sum-to-product to numerator and denominator:

$$\sin 5x + \sin x = 2 \sin 3x \cos 2x, \quad \cos 5x + \cos x = 2 \cos 3x \cos 2x.$$

The $2 \cos 2x$ factors cancel, leaving $\frac{\sin 5x + \sin x}{\cos 5x + \cos x} = \frac{\sin 3x}{\cos 3x} = \tan 3x$.

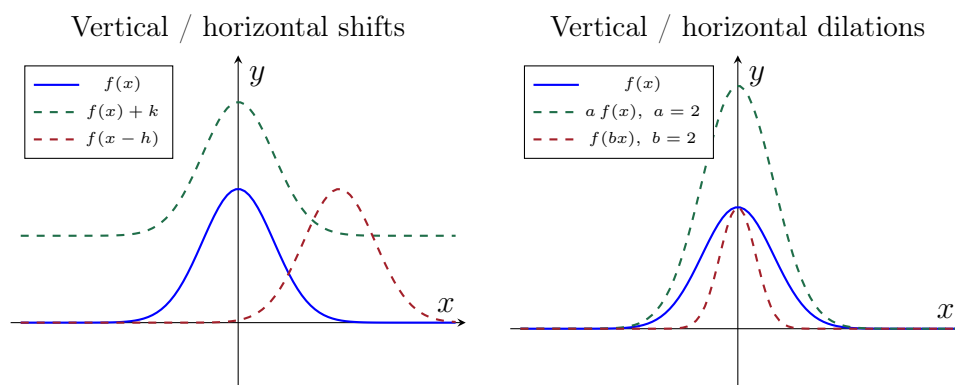
0.2 Advanced Knowledge of Functions

0.2.1 Translations and Dilations

Theorem 0.2.1: Transformations of $f(x)$

Let f be a function and h, k, a, b be constants with $a, b \neq 0$.

Transformation	Effect on the graph of $y = f(x)$
$y = f(x) + k$	vertical shift, up k units ($k > 0$)
$y = f(x - h)$	horizontal shift, right h units ($h > 0$)
$y = a f(x)$	vertical dilation by factor $ a $; reflects over x -axis if $a < 0$
$y = f(bx)$	horizontal dilation by factor $1/ b $; reflects over y -axis if $b < 0$



Example 0.2.1: Combining Transformations

Describe how the graph of $y = -3(x + 1)^2 + 4$ is obtained from $y = x^2$.

Solution. Starting from $f(x) = x^2$: shift left 1 (replace x with $x + 1$), vertically dilate by a factor of 3 and reflect over the x -axis (multiply by -3), then shift up 4. In transformation order applied to the graph: reflect/stretch first, then translate.

0.2.2 Exponential Functions

Definition 0.2.1: Exponential Function

For a base $b > 0$, $b \neq 1$, the **exponential function** is $f(x) = b^x$, with domain $\mathbb{R}$ and range $(0, \infty)$. It is increasing if $b > 1$ and decreasing if $0 < b < 1$; every exponential curve passes through $(0, 1)$ and has horizontal asymptote $y = 0$.

Theorem 0.2.2: Laws of Exponents

For $b > 0$ and real x, y :

$$b^x b^y = b^{x+y}, \quad \frac{b^x}{b^y} = b^{x-y}, \quad (b^x)^y = b^{xy}, \quad b^{-x} = \frac{1}{b^x}.$$

0.2.3 Logarithmic Functions

Definition 0.2.2: Logarithm as the Inverse of an Exponential

For $b > 0$, $b \neq 1$, and $x > 0$,

$$y = \log_b x \iff b^y = x.$$

$\log_b x$ has domain $(0, \infty)$, range $\mathbb{R}$, and vertical asymptote $x = 0$. The special case $b = e$ is written $\ln x$ (the natural logarithm).

Theorem 0.2.3: Laws of Logarithms

For $x, y > 0$ and any real r :

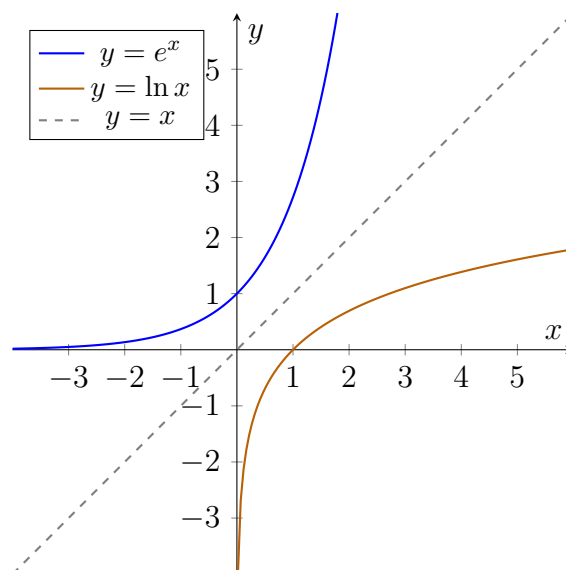
$$\log_b(xy) = \log_b x + \log_b y, \quad \log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y, \quad \log_b(x^r) = r \log_b x.$$

0.2.4 Exponentials and Logarithms as Inverses**Theorem 0.2.4: Inverse Cancellation**

For $b > 0$, $b \neq 1$:

$$b^{\log_b x} = x \quad (x > 0), \quad \log_b(b^x) = x \quad (\text{all } x).$$

In particular, $e^{\ln x} = x$ for $x > 0$, and $\ln(e^x) = x$ for all x .

**Remark**

Because $f(x) = b^x$ and $g(x) = \log_b x$ are inverses, their graphs are reflections of one another across the line $y = x$ — visible above for e^x and $\ln x$. This graphical relationship, together with the cancellation identities, is used repeatedly in Units 3, 6, and 7 to differentiate, integrate, and solve equations involving exponential and logarithmic functions.

Example 0.2.2: Solving an Exponential Equation with Logarithms

Solve $3e^{2x-1} = 21$ for x .

Solution. Isolate the exponential: $e^{2x-1} = 7$. Apply $\ln$ to both sides: $2x - 1 = \ln 7$.

Solve: $x = \frac{1 + \ln 7}{2}$.

UNIT 1

Limits and Continuity

AP Exam Weighting	10–15% (AB)	5–10% (BC)	
Big Ideas	Change (CHA)	Limits (LIM)	Analysis of Functions (FUN)

Limits formalize the distinction between evaluating a function *at* a point and describing the value a function *approaches* near that point. This distinction is what allows us to make sense of instantaneous change, and it underlies every major construction that follows: the derivative (Unit 2), the definite integral (Unit 6), and infinite series (Unit 10, BC). Everything in this unit is built to answer one guiding question: *how does knowing the value of a limit — or knowing that a limit does not exist — help us understand the behavior of a function?*

1.1 Introducing Calculus: Can Change Occur at an Instant?

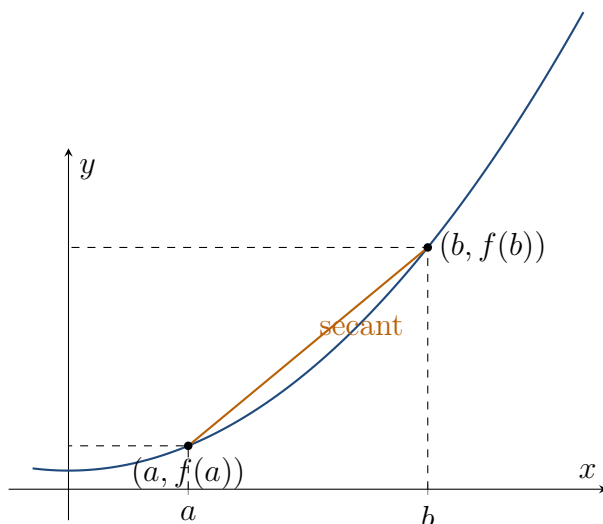
Average rate of change over an interval is familiar from precalculus: it is a slope. But questions in calculus ask about change *at an instant* — at a single value of x , where there is no interval left to compute a slope from directly. Calculus resolves this using limits.

Definition 1.1.1: Average Rate of Change

The **average rate of change** of f over $[a, b]$ is

$$\frac{f(b) - f(a)}{b - a},$$

the slope of the secant line through $(a, f(a))$ and $(b, f(b))$. It is undefined when $b = a$.



As b is chosen closer and closer to a , the secant line rotates toward a limiting position: the **tangent line** at a . The slope of that tangent line is the *instantaneous* rate of change at a — a quantity that cannot be computed by plugging $b = a$ into the average-rate-of-change formula (this gives $\frac{0}{0}$), but *can* be defined as a limit:

$$\text{instantaneous rate of change at } a = \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a}.$$

Remark

The average rate of change is undefined exactly where the independent variable's change is zero — i.e., at $b = a$. The entire purpose of the limit is to assign meaning to this expression anyway, by examining its behavior as $b \rightarrow a$ rather than evaluating at $b = a$. This is the single idea that launches all of differential calculus.

1.2 Defining Limits and Using Limit Notation

Definition 1.2.1: Limit of a Function (Informal)

Let f be a function defined near c (though not necessarily at c itself). We say

$$\lim_{x \rightarrow c} f(x) = L$$

if $f(x)$ can be made arbitrarily close to the real number L by taking x sufficiently close to c , but not equal to c .

Remark

The ε - δ definition of a limit is *not* assessed on the AP Calculus AB or BC Exam, though it may be introduced at a teacher's discretion. The informal, working definition above is what students are expected to reason with.

Limit notation separates *two-sided* limits from *one-sided* limits:

$$\lim_{x \rightarrow c^-} f(x) \quad (\text{left-hand limit, } x \rightarrow c \text{ with } x < c), \quad \lim_{x \rightarrow c^+} f(x) \quad (\text{right-hand limit, } x \rightarrow c \text{ with } x > c).$$

Theorem 1.2.1: Existence of a Two-Sided Limit

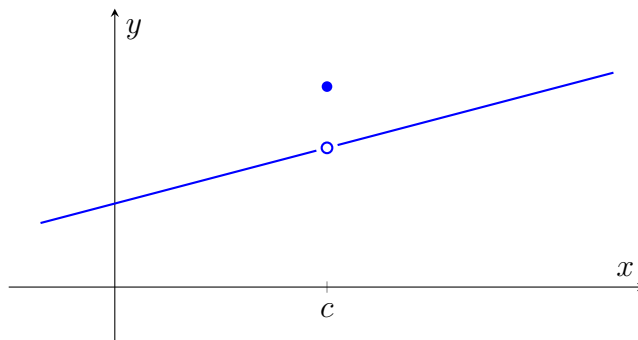
$\lim_{x \rightarrow c} f(x) = L$ if and only if

$$\lim_{x \rightarrow c^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c^+} f(x) = L.$$

If the one-sided limits disagree, or if either fails to exist, then $\lim_{x \rightarrow c} f(x)$ does not exist.

1.3 Estimating Limit Values from Graphs

Graphs reveal limit behavior directly, but two cautions matter: the value of the limit (if it exists) need not equal $f(c)$, and viewing windows can hide important behavior (oscillation, thin spikes) purely due to scale.



Here $\lim_{x \rightarrow 2} f(x) = 2.5$ (both one-sided limits agree, shown by the open circle at the limiting height), even though $f(2) = 3.6$ (the filled dot). This is the central lesson of Topic 1.3: *the limit describes the hole in the graph the curve is heading toward, not necessarily the plotted value of the function there.*

Example 1.3.1: Reading a Limit from a Piecewise Graph

Using the graph above, evaluate $\lim_{x \rightarrow 2^-} f(x)$, $\lim_{x \rightarrow 2^+} f(x)$, $\lim_{x \rightarrow 2} f(x)$, and $f(2)$.

Solution. Tracing the left branch as $x \rightarrow 2^-$, the curve approaches height 2.5: so $\lim_{x \rightarrow 2^-} f(x) = 2.5$. Tracing the right branch as $x \rightarrow 2^+$, the curve also approaches height 2.5: so $\lim_{x \rightarrow 2^+} f(x) = 2.5$. The one-sided limits agree, so $\lim_{x \rightarrow 2} f(x) = 2.5$. The filled dot shows the actual function value, $f(2) = 3.6$, which is unrelated to the limit.

Common ways a limit *fails* to exist at $x = c$: the function is unbounded near c (Topic 1.14), the function oscillates infinitely often near c , or the one-sided limits disagree (a jump).

1.4 Estimating Limit Values from Tables

Numerical (tabular) data can approximate a limit by evaluating f at x -values that approach c from both sides.

Example 1.4.1: Estimating a Limit from a Table

Estimate $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$ using the table below.

x	0.9	0.99	0.999	1.001	1.01
$f(x)$	1.0536	1.0050	1.0005	0.9995	0.9950

Solution. As $x \rightarrow 1^-$ and $x \rightarrow 1^+$, $f(x)$ approaches 1 from both sides. So $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1} \approx 1$. (Note: a table only lets us *estimate* a limit — it cannot prove the limit equals a value analytically.)

Remark

A table can strongly suggest a limit but never proves it, and can also mislead if the chosen x -values happen to miss erratic behavior between them. Tables and graphs are exploratory tools; Topics 1.5–1.8 give the algebraic tools that make evaluation rigorous.

1.5 Determining Limits Using Algebraic Properties of Limits

Theorem 1.5.1: Limit Laws

Suppose $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, both finite. Then

$$\begin{aligned} \lim_{x \rightarrow c} [f(x) \pm g(x)] &= L \pm M, & \lim_{x \rightarrow c} [f(x) \cdot g(x)] &= L \cdot M, \\ \lim_{x \rightarrow c} [k \cdot f(x)] &= kL \quad (k \in \mathbb{R}), & \lim_{x \rightarrow c} \frac{f(x)}{g(x)} &= \frac{L}{M} \quad (M \neq 0). \end{aligned}$$

If g is continuous at L , then $\lim_{x \rightarrow c} g(f(x)) = g(L)$ (limits of composites).

Example 1.5.1: Applying Limit Laws Directly

Given $\lim_{x \rightarrow 2} f(x) = 5$ and $\lim_{x \rightarrow 2} g(x) = -3$, find $\lim_{x \rightarrow 2} [2f(x) - g(x)^2]$.

Solution. $\lim_{x \rightarrow 2} [2f(x) - g(x)^2] = 2(5) - (-3)^2 = 10 - 9 = 1$.

If direct substitution gives a real number, that number *is* the limit — this is the content of Topic 1.5. The harder cases, where substitution yields an indeterminate form like $\frac{0}{0}$, are the subject of Topic 1.6.

1.6 Determining Limits Using Algebraic Manipulation

When direct substitution produces $\frac{0}{0}$, the expression can often be rewritten into an equivalent form (valid for $x \neq c$) for which substitution succeeds.

Example 1.6.1: Factoring

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

Solution. Direct substitution gives $\frac{0}{0}$. Factor:

$$\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3 \quad (x \neq 3).$$

Since the two expressions agree everywhere except at $x = 3$, their limits as $x \rightarrow 3$

agree: $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$.

Example 1.6.2: Rationalizing (Multiplying by the Conjugate)

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x + 4} - 2}{x}$.

Solution. Multiply by the conjugate $\frac{\sqrt{x + 4} + 2}{\sqrt{x + 4} + 2}$:

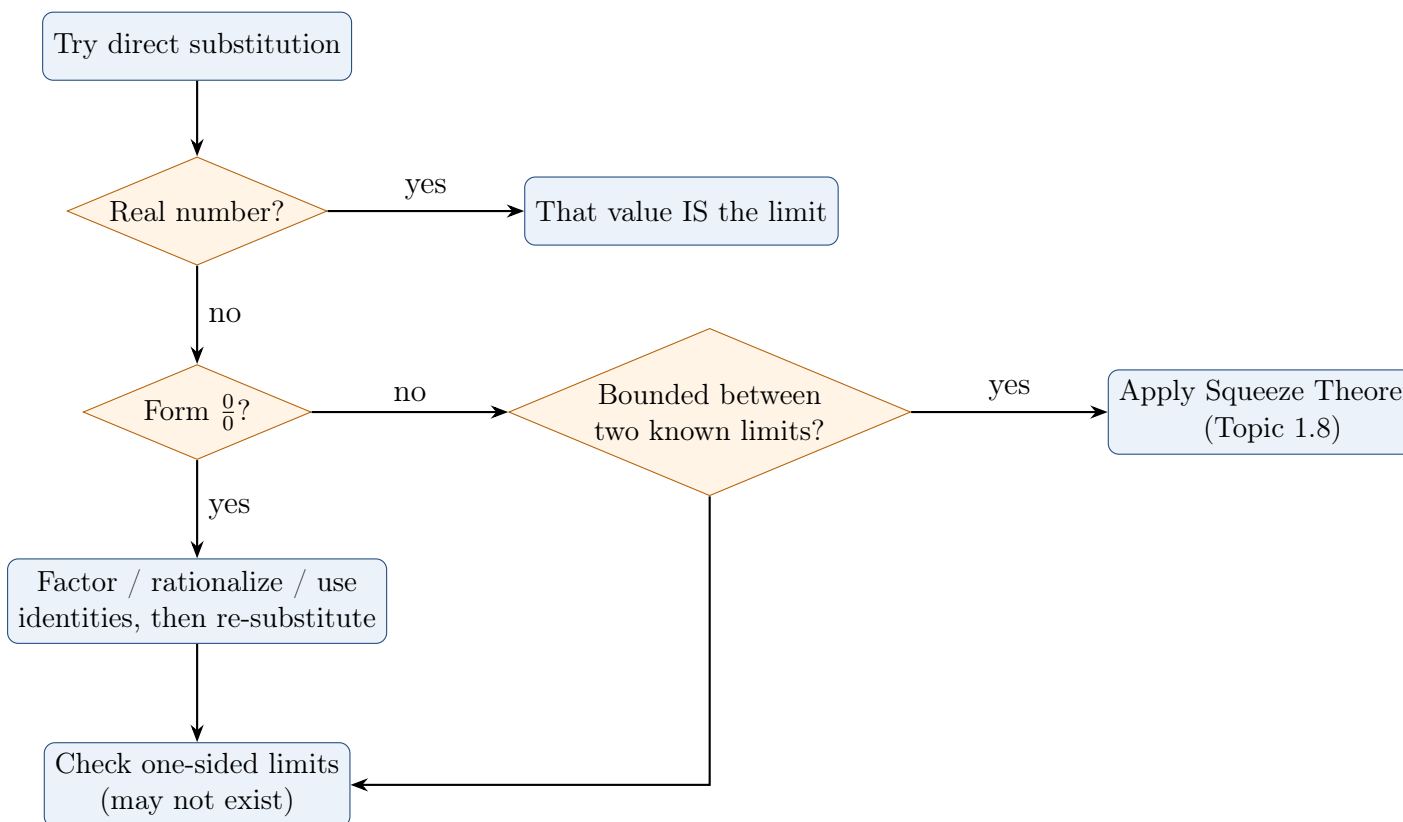
$$\frac{\sqrt{x + 4} - 2}{x} \cdot \frac{\sqrt{x + 4} + 2}{\sqrt{x + 4} + 2} = \frac{(x + 4) - 4}{x(\sqrt{x + 4} + 2)} = \frac{x}{x(\sqrt{x + 4} + 2)} = \frac{1}{\sqrt{x + 4} + 2}.$$

Now substitution is valid: $\lim_{x \rightarrow 0} \frac{1}{\sqrt{x + 4} + 2} = \frac{1}{4}$.

Other standard manipulations: combining fractions over a common denominator, and rewriting trigonometric expressions using identities before substituting.

1.7 Selecting Procedures for Determining Limits

Topic 1.7 asks students to *choose* the correct strategy, rather than execute one that's already specified. A practical decision process:

**Remark**

There is rarely only one valid method. The purpose of Topic 1.7 is fluency: given an unfamiliar limit expression, quickly classify its form and select an efficient path, rather than blindly trying every tool in sequence.

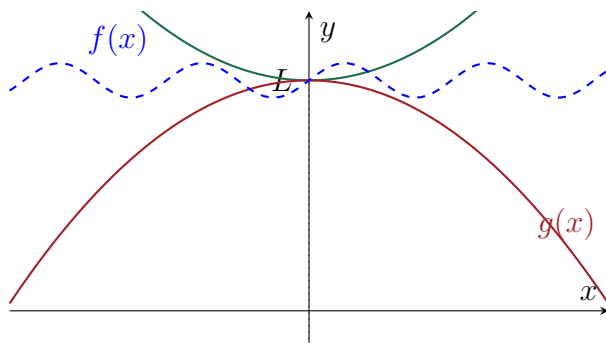
1.8 Determining Limits Using the Squeeze Theorem

Theorem 1.8.1: Squeeze (Sandwich) Theorem

Suppose $g(x) \leq f(x) \leq h(x)$ for all x near c (except possibly at c), and

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L.$$

Then $\lim_{x \rightarrow c} f(x) = L$ as well.



Example 1.8.1: Squeeze Theorem: $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$

Solution. For all $x \neq 0$, $-1 \leq \sin(1/x) \leq 1$, so

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2.$$

Since $\lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} x^2 = 0$, the Squeeze Theorem gives $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.

Remark

The squeeze theorem is also the standard justification for two foundational limits used constantly in Unit 2:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$$

1.9 Connecting Multiple Representations of Limits

The same limit statement can be expressed graphically, numerically, analytically, or verbally, and fluently translating between these is itself an AP skill (Skill 2.C).

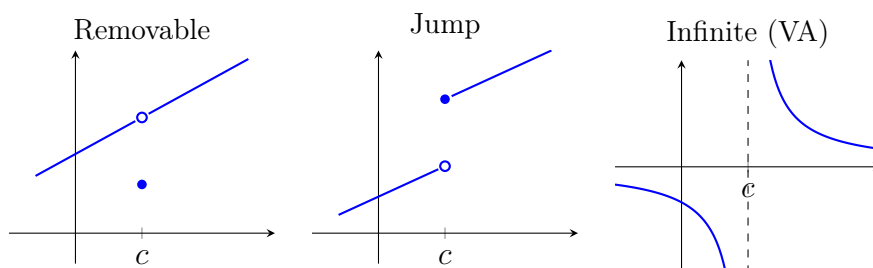
Representation	Example expressing $\lim_{x \rightarrow 2} f(x) = 5$
Analytical	$\lim_{x \rightarrow 2} f(x) = 5$
Verbal	“As x approaches 2, $f(x)$ approaches 5.”
Graphical	the curve has an open circle at height 5 above $x = 2$
Numerical	a table of $f(x)$ for x near 2 shows values converging to 5

An AP question may give one representation and ask for a conclusion normally drawn from another — e.g., a table of values used to justify a graphical claim, or a verbal description translated into limit notation.

1.10 Exploring Types of Discontinuities

Definition 1.10.1: Discontinuity

f has a **discontinuity** at $x = c$ if f fails to be continuous there (Topic 1.11 makes this precise). Three types appear on the AP Exam:



- **Removable:** the two-sided limit exists at c , but either $f(c)$ is undefined or $f(c) \neq \lim_{x \rightarrow c} f(x)$. Can be “fixed” by redefining $f(c)$ (Topic 1.13).
- **Jump:** both one-sided limits exist but are unequal, so the two-sided limit fails to exist.
- **Infinite:** at least one one-sided limit is $\pm\infty$; corresponds to a vertical asymptote (Topic 1.14).

1.11 Defining Continuity at a Point

Definition 1.11.1: Continuity at a Point

f is **continuous at** $x = c$ if all three conditions hold:

- $f(c)$ exists,
- $\lim_{x \rightarrow c} f(x)$ exists (equivalently, the one-sided limits agree),
- $\lim_{x \rightarrow c} f(x) = f(c)$.

AP Exam Tip

On free-response questions, *all three conditions must be stated and verified explicitly* to earn full justification credit — stating the conclusion alone (“it’s continuous”) without checking each hypothesis is a common way points are lost.

Example 1.11.1: Verifying Continuity at a Point

Let $f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$. Is f continuous at $x = 2$?

Solution. (i) $f(2) = 4$ exists. (ii) For $x \neq 2$, $\frac{x^2 - 4}{x - 2} = x + 2$, so $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x + 2) = 4$, which exists. (iii) $\lim_{x \rightarrow 2} f(x) = 4 = f(2)$. All three conditions hold, so f is continuous at $x = 2$.

1.12 Confirming Continuity over an Interval

Definition 1.12.1: Continuity on an Interval

f is **continuous on an interval** I if it is continuous at every point of I (at endpoints, only the one-sided limit into the interval is required).

Theorem 1.12.1: Continuity of Elementary Functions

Polynomial, rational, power, exponential, logarithmic, and trigonometric functions are continuous at every point of their domains.

Example 1.12.1: Interval of Continuity

On what interval(s) is $f(x) = \frac{\sqrt{x+3}}{x-1}$ continuous?

Solution. $\sqrt{x+3}$ requires $x \geq -3$; the quotient additionally requires $x \neq 1$. Both the numerator and denominator are continuous on their domains (root and polynomial, respectively), so f is continuous on $[-3, 1) \cup (1, \infty)$.

1.13 Removing Discontinuities

Theorem 1.13.1: Removing a Discontinuity

If $\lim_{x \rightarrow c} f(x) = L$ exists but f is undefined (or has the wrong value) at c , define $g(x) = f(x)$ for $x \neq c$ and $g(c) = L$. Then g is continuous at c . For a piecewise function to be continuous at a boundary point $x = b$ separating two pieces, all of the following must be equal: the limit from the left, the limit from the right, and the function's value at b .

Example 1.13.1: Choosing a Parameter for Continuity

Find k so that $f(x) = \begin{cases} 3x + k, & x \leq 2 \\ x^2, & x > 2 \end{cases}$ is continuous at $x = 2$.

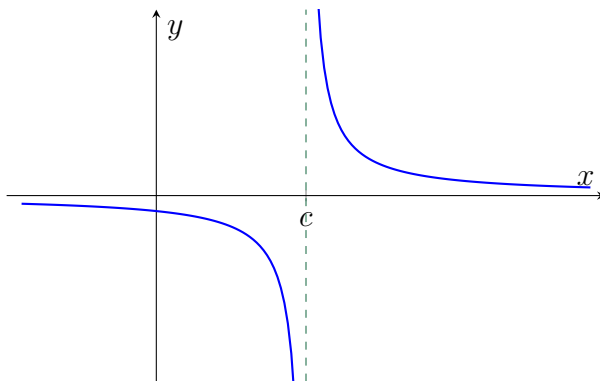
Solution. Need $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$:

$$3(2) + k = 2^2 \implies 6 + k = 4 \implies k = -2.$$

1.14 Connecting Infinite Limits and Vertical Asymptotes

Definition 1.14.1: Infinite Limit / Vertical Asymptote

$\lim_{x \rightarrow c} f(x) = \infty$ means $f(x)$ grows without bound as $x \rightarrow c$ (similarly for $-\infty$). If $\lim_{x \rightarrow c^-} f(x) = \pm\infty$ or $\lim_{x \rightarrow c^+} f(x) = \pm\infty$, the line $x = c$ is a **vertical asymptote** of f .



Example 1.14.1: Finding a Vertical Asymptote

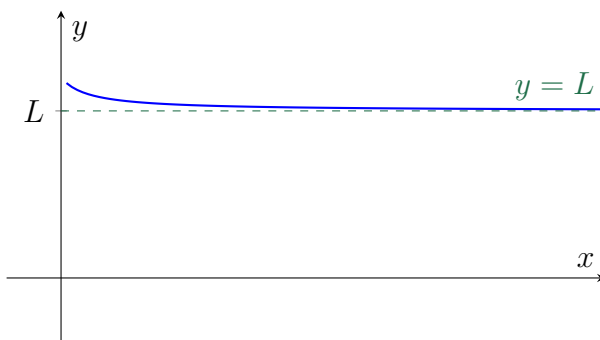
Analyze $\lim_{x \rightarrow 3^\pm} \frac{2}{x-3}$.

Solution. As $x \rightarrow 3^+$, $x-3 \rightarrow 0^+$, so $\frac{2}{x-3} \rightarrow +\infty$. As $x \rightarrow 3^-$, $x-3 \rightarrow 0^-$, so $\frac{2}{x-3} \rightarrow -\infty$. Hence $x=3$ is a vertical asymptote, and the two-sided limit does not exist (the one-sided limits are not even both finite, let alone equal).

1.15 Connecting Limits at Infinity and Horizontal Asymptotes

Definition 1.15.1: Limit at Infinity / Horizontal Asymptote

$\lim_{x \rightarrow \infty} f(x) = L$ means $f(x)$ gets arbitrarily close to L as x grows without bound (similarly for $x \rightarrow -\infty$). If either limit at infinity equals a finite L , then $y=L$ is a **horizontal asymptote** of f .



Theorem 1.15.1: End Behavior of Rational Functions

For $f(x) = \frac{a_n x^n + \dots}{b_m x^m + \dots}$,

$$\lim_{x \rightarrow \pm\infty} f(x) = \begin{cases} 0 & n < m, \\ a_n/b_m & n = m, \\ \pm\infty & n > m. \end{cases}$$

Example 1.15.1: Horizontal Asymptote of a Rational Function

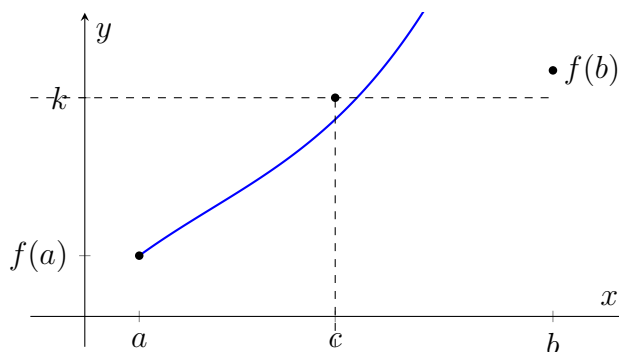
Find $\lim_{x \rightarrow \infty} \frac{5x + 3}{2x + 1}$.

Solution. Numerator and denominator both have degree 1, so the limit is the ratio of leading coefficients: $\lim_{x \rightarrow \infty} \frac{5x + 3}{2x + 1} = \frac{5}{2}$. Thus $y = \frac{5}{2}$ is a horizontal asymptote.

1.16 Working with the Intermediate Value Theorem (IVT)

Theorem 1.16.1: Intermediate Value Theorem

If f is continuous on $[a, b]$ and k is any value between $f(a)$ and $f(b)$, then there exists at least one $c \in (a, b)$ with $f(c) = k$.

**AP Exam Tip**

IVT is an *existence* theorem: it proves a c exists but never tells you what c equals. On the AP Exam, correct use always cites (a) continuity of f on the closed interval, and (b) that k lies between $f(a)$ and $f(b)$, before concluding a c exists.

Example 1.16.1: Applying the IVT to Show a Root Exists

Show that $f(x) = x^3 - x - 1$ has a root in $(1, 2)$.

Solution. f is a polynomial, hence continuous on $[1, 2]$. Compute $f(1) = 1 - 1 - 1 = -1 < 0$ and $f(2) = 8 - 2 - 1 = 5 > 0$. Since 0 is between $f(1)$ and $f(2)$, the IVT

guarantees some $c \in (1, 2)$ with $f(c) = 0$ — that is, a root of f in $(1, 2)$.

UNIT 2

Differentiation: Definition and Fundamental Properties

Class Periods	~13–14 (AB)	~9–10 (BC)
Big Ideas	Change (CHA)	Analysis of Functions (FUN)

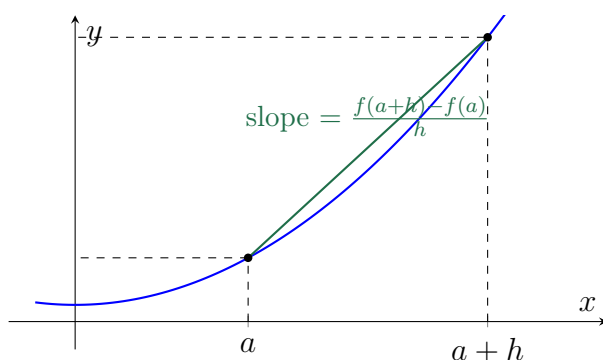
Unit 1 used limits to make sense of instantaneous change informally. Unit 2 makes that idea precise: the **derivative** is defined as a limit of average rates of change, and from that single definition we build an entire toolkit — the power rule, sum/difference rules, and the derivatives of the basic transcendental functions — that lets us differentiate without returning to the limit definition every time.

2.1 Defining Average and Instantaneous Rates of Change at a Point

Definition 2.1.1: Difference Quotient

For a function f and a fixed point a , the **average rate of change** of f over $[a, a+h]$ ($h \neq 0$) is the **difference quotient**:

$$\frac{f(a+h) - f(a)}{h}.$$



Definition 2.1.2: Instantaneous Rate of Change

The **instantaneous rate of change** of f at $x = a$ is

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

provided the limit exists — the limiting slope of the difference quotient as the

second point slides into a .

Example 2.1.1: Average vs. Instantaneous Velocity

A particle's position is $s(t) = t^2 + 1$ (meters, seconds). Find its average velocity on $[1, 3]$ and its instantaneous velocity at $t = 1$.

Solution. Average velocity: $\frac{s(3) - s(1)}{3 - 1} = \frac{10 - 2}{2} = 4$ m/s. Instantaneous velocity:

$$\lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} = \lim_{h \rightarrow 0} \frac{(1+h)^2 + 1 - 2}{h} = \lim_{h \rightarrow 0} \frac{2h + h^2}{h} = \lim_{h \rightarrow 0} (2 + h) = 2 \text{ m/s.}$$

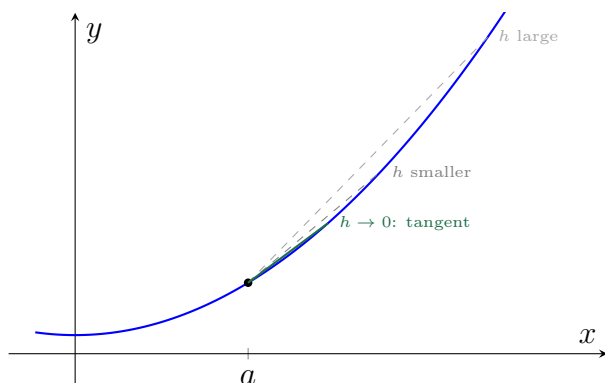
2.2 Defining the Derivative of a Function and Using Derivative Notation

Definition 2.2.1: The Derivative

The **derivative** of f at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists. If $f'(a)$ exists, f is **differentiable** at a , and $f'(a)$ is the slope of the tangent line to $y = f(x)$ at $x = a$.



Remark

As $h \rightarrow 0$, the secant lines through $(a, f(a))$ and $(a+h, f(a+h))$ rotate toward the tangent line at a . This is the geometric content of the derivative definition.

Theorem 2.2.1: Notation for the Derivative

Common equivalent notations for the derivative of $y = f(x)$:

$$f'(x), \quad y', \quad \frac{dy}{dx}, \quad \frac{d}{dx}[f(x)], \quad Df(x).$$

The value of the derivative at $x = a$ is written $f'(a)$ or $\left. \frac{dy}{dx} \right|_{x=a}$.

Example 2.2.1: Computing a Derivative from the Definition

Use the limit definition to find $f'(x)$ for $f(x) = x^2$.

Solution.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x.$$

2.3 Estimating Derivatives of a Function at a Point

When f is known only graphically or numerically, $f'(a)$ can be *estimated* using a nearby secant slope — most accurately, a **central** (symmetric) difference quotient.

Theorem 2.3.1: Difference Quotient Approximations

$$f'(a) \approx \frac{f(a+h) - f(a)}{h} \quad (\text{forward}), \quad f'(a) \approx \frac{f(a) - f(a-h)}{h} \quad (\text{backward}),$$

$$f'(a) \approx \frac{f(a+h) - f(a-h)}{2h} \quad (\text{central; generally most accurate}).$$

Example 2.3.1: Estimating a Derivative from a Table

x	1.9	2.0	2.1	2.2
$f(x)$	3.24	3.60	3.98	4.38

Estimate $f'(2.0)$.

Solution. Using the central difference quotient with $h = 0.1$:

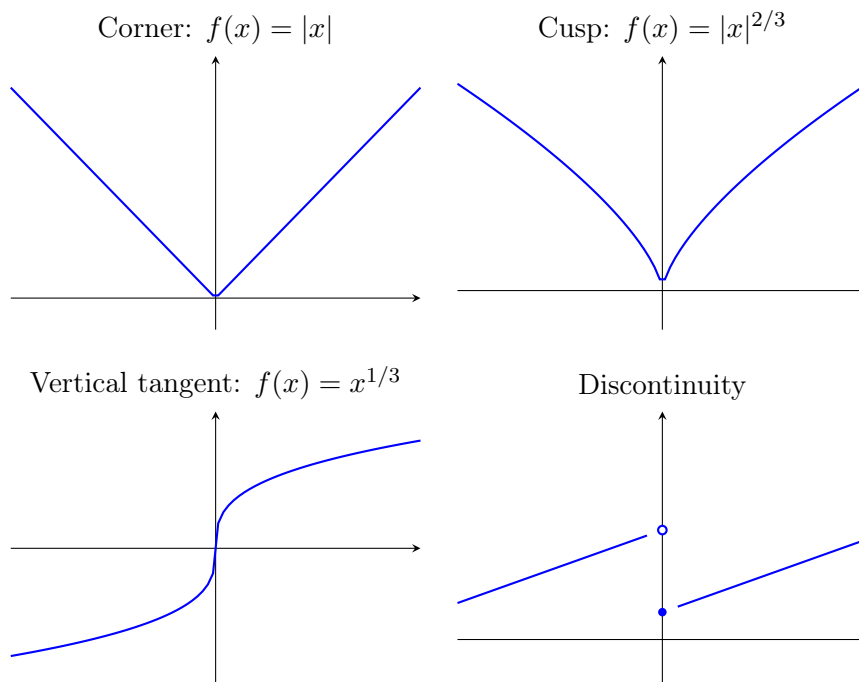
$$f'(2.0) \approx \frac{f(2.1) - f(1.9)}{0.2} = \frac{3.98 - 3.24}{0.2} = 3.7.$$

2.4 Connecting Differentiability and Continuity: Determining When Derivatives Do and Do Not Exist

Theorem 2.4.1: Differentiability Implies Continuity

If f is differentiable at a , then f is continuous at a .

The converse is false: a function can be continuous at a without being differentiable there.

**Remark**

A derivative fails to exist at $x = a$ for exactly these reasons: a **corner** (one-sided derivatives disagree), a **cusp** (one-sided derivatives both go to $+\infty$ or both to $-\infty$, but the tangent line would be vertical from both sides in an unmatched way), a **vertical tangent** (the difference quotient itself diverges to $\pm\infty$), or a **discontinuity** (differentiability requires continuity, so any discontinuity rules out a derivative).

Example 2.4.1: $f(x) = |x|$ Is Not Differentiable at 0

Show that $f(x) = |x|$ is continuous but not differentiable at $x = 0$.

Solution. Continuity is clear: $\lim_{x \rightarrow 0} |x| = 0 = f(0)$. For differentiability, compare one-sided difference quotients:

$$\lim_{h \rightarrow 0^+} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1, \quad \lim_{h \rightarrow 0^-} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1.$$

The one-sided limits disagree, so $f'(0)$ does not exist.

2.5 Applying the Power Rule

Theorem 2.5.1: Power Rule

For any real number n ,

$$\frac{d}{dx} [x^n] = nx^{n-1}.$$

Example 2.5.1: Power Rule with Negative and Fractional Exponents

Differentiate $g(x) = \frac{1}{x^3} + \sqrt{x}$.

Solution. Rewrite: $g(x) = x^{-3} + x^{1/2}$. Then

$$g'(x) = -3x^{-4} + \frac{1}{2}x^{-1/2} = -\frac{3}{x^4} + \frac{1}{2\sqrt{x}}.$$

2.6 Derivative Rules: Constant, Sum, Difference, and Constant Multiple

Theorem 2.6.1: Constant, Sum, Difference, and Constant Multiple Rules

Let f, g be differentiable and k a constant.

$$\frac{d}{dx}[k] = 0, \quad \frac{d}{dx}[k f(x)] = k f'(x), \quad \frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x).$$

Example 2.6.1: Combining Rules

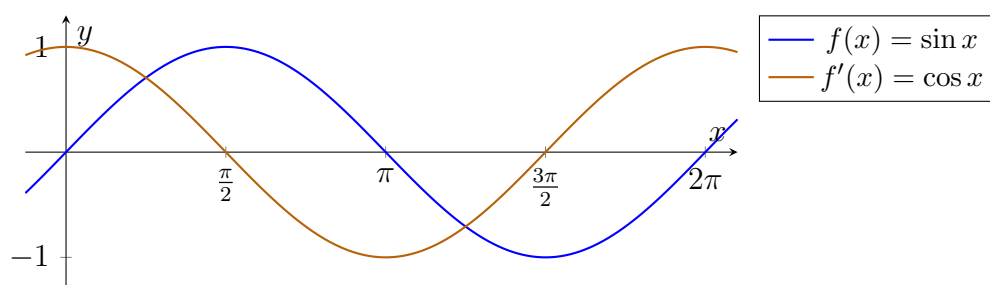
Differentiate $h(x) = 5x^4 - 3x^2 + 7x - 2$.

Solution. Apply the power rule term by term: $h'(x) = 20x^3 - 6x + 7$.

2.7 Derivatives of $\cos x$, $\sin x$, e^x , and $\ln x$

Theorem 2.7.1: Basic Transcendental Derivatives

$$\frac{d}{dx}[\sin x] = \cos x, \quad \frac{d}{dx}[\cos x] = -\sin x, \quad \frac{d}{dx}[e^x] = e^x, \quad \frac{d}{dx}[\ln x] = \frac{1}{x} \ (x > 0).$$

**Remark**

The graph makes the derivative relationship visible: $\sin x$ has horizontal tangents (slope 0) exactly where $\cos x = 0$, and $\sin x$ is increasing fastest (largest positive slope) exactly where $\cos x = 1$.

Example 2.7.1: Differentiating a Combination

Differentiate $y = 3e^x - 4 \ln x + \cos x$.

Solution. $y' = 3e^x - \frac{4}{x} - \sin x$.

2.8 The Product Rule

Theorem 2.8.1: Product Rule

If f and g are differentiable,

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

Example 2.8.1: Applying the Product Rule

Differentiate $y = x^2 \sin x$.

Solution. With $f = x^2$ ($f' = 2x$) and $g = \sin x$ ($g' = \cos x$):

$$y' = 2x \sin x + x^2 \cos x.$$

2.9 The Quotient Rule

Theorem 2.9.1: Quotient Rule

If f and g are differentiable and $g(x) \neq 0$,

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}.$$

AP Exam Tip

A common error is reversing the order of terms in the numerator. The mnemonic "low d-high minus high d-low, over low squared" (using f = high, g = low) helps students keep the sign correct.

Example 2.9.1: Applying the Quotient Rule

Differentiate $y = \frac{e^x}{x^2 + 1}$.

Solution. With $f = e^x$ ($f' = e^x$) and $g = x^2 + 1$ ($g' = 2x$):

$$y' = \frac{e^x(x^2 + 1) - e^x(2x)}{(x^2 + 1)^2} = \frac{e^x(x^2 - 2x + 1)}{(x^2 + 1)^2} = \frac{e^x(x - 1)^2}{(x^2 + 1)^2}.$$

2.10 Finding the Derivatives of Tangent, Cotangent, Secant, and/or Cosecant Functions

Theorem 2.10.1: Derivatives of the Remaining Trigonometric Functions

$$\begin{aligned}\frac{d}{dx}[\tan x] &= \sec^2 x, & \frac{d}{dx}[\cot x] &= -\csc^2 x, \\ \frac{d}{dx}[\sec x] &= \sec x \tan x, & \frac{d}{dx}[\csc x] &= -\csc x \cot x.\end{aligned}$$

Example 2.10.1: Deriving $\frac{d}{dx}[\tan x]$ from the Quotient Rule

Derive the formula for $\frac{d}{dx}[\tan x]$.

Solution. Write $\tan x = \frac{\sin x}{\cos x}$ and apply the quotient rule:

$$\frac{d}{dx}[\tan x] = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

Example 2.10.2: Using the New Rules

Differentiate $y = x \sec x$.

Solution. By the product rule: $y' = \sec x + x \sec x \tan x$.

UNIT 3

Differentiation: Composite, Implicit, and Inverse Functions

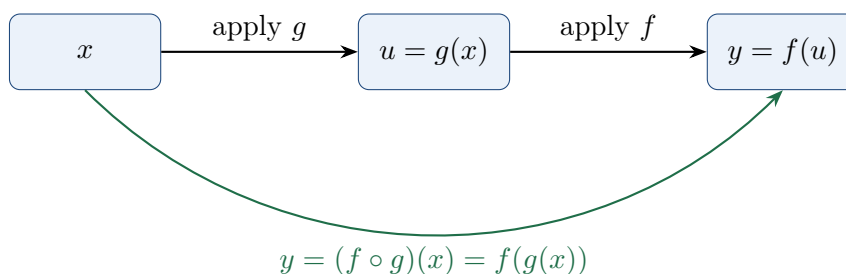
Class Periods	~10–11 (AB)	~8–9 (BC)
Big Ideas	Change (CHA)	Analysis of Functions (FUN)

Unit 2 differentiated functions built directly from x by sums, products, and quotients. Unit 3 extends the toolkit to functions built by *composition* (the chain rule), relations where y is not solved explicitly in terms of x (implicit differentiation), and inverse functions — including the inverse trigonometric functions.

3.1 The Chain Rule

Definition 3.1.1: Composite Function

The **composition** of f and g is $(f \circ g)(x) = f(g(x))$: first apply g , then apply f to the result.



Theorem 3.1.1: Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, then $f \circ g$ is differentiable at x , and

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$,

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

AP Exam Tip

Think of the chain rule as “derivative of the outside (leaving the inside alone) times derivative of the inside.” Forgetting the inner-derivative factor is one of the most common AP scoring errors.

Example 3.1.1: Applying the Chain Rule

Differentiate $y = \sin(x^2)$ and $y = (3x + 1)^5$.

Solution. For $y = \sin(x^2)$: outer = $\sin(\cdot)$, inner = x^2 , so $y' = \cos(x^2) \cdot 2x$. For $y = (3x + 1)^5$: outer = $(\cdot)^5$, inner = $3x + 1$, so $y' = 5(3x + 1)^4 \cdot 3 = 15(3x + 1)^4$.

Example 3.1.2: Chain Rule with Multiple Layers

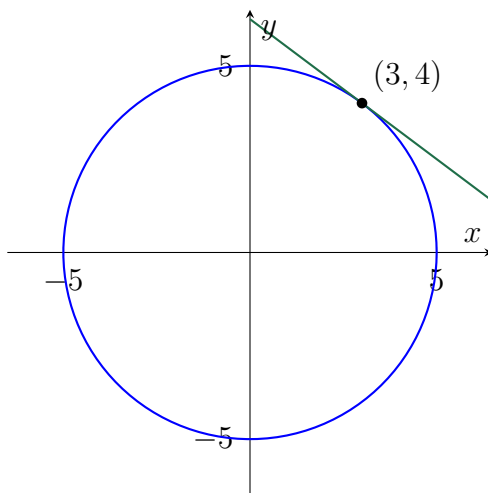
Differentiate $y = e^{\cos(2x)}$.

Solution. Working from the outside in: $\frac{d}{dx}[e^u] = e^u u'$ with $u = \cos(2x)$, and $u' = -\sin(2x) \cdot 2$. So

$$y' = e^{\cos(2x)} \cdot (-2 \sin(2x)) = -2 \sin(2x) e^{\cos(2x)}.$$

3.2 Implicit Differentiation

Many curves — such as $x^2 + y^2 = 25$ — are not graphs of a single function $y = f(x)$, yet still have well-defined tangent lines. **Implicit differentiation** finds $\frac{dy}{dx}$ directly from such an equation by differentiating both sides with respect to x , treating y as an unknown differentiable function of x and applying the chain rule to every y -term.

**Example 3.2.1: Implicit Differentiation of a Circle**

Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$, and evaluate the slope at $(3, 4)$.

Solution. Differentiate both sides with respect to x , treating y as a function of x :

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

At $(3, 4)$: $\frac{dy}{dx} = -\frac{3}{4}$, matching the tangent line drawn above.

Example 3.2.2: Implicit Differentiation with a Product Term

Find $\frac{dy}{dx}$ for $xy + y^2 = 6$.

Solution. Differentiate, using the product rule on xy :

$$\left(y + x \frac{dy}{dx}\right) + 2y \frac{dy}{dx} = 0.$$

Collect all $\frac{dy}{dx}$ terms:

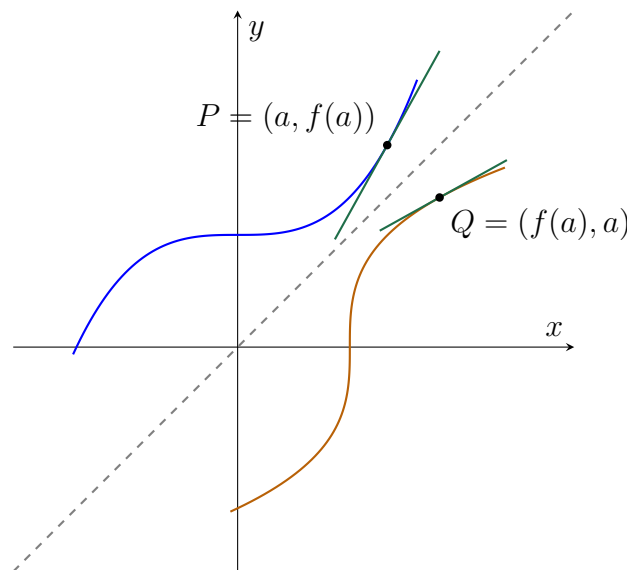
$$\frac{dy}{dx}(x + 2y) = -y \implies \frac{dy}{dx} = \frac{-y}{x + 2y}.$$

3.3 Differentiating Inverse Functions

Theorem 3.3.1: Derivative of an Inverse Function

If f is differentiable and has an inverse f^{-1} that is differentiable at x , then

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}, \quad \text{provided } f'(f^{-1}(x)) \neq 0.$$



Remark

Because the graphs of f and f^{-1} are reflections across $y = x$, the tangent line at $Q = (f(a), a)$ on f^{-1} is the reflection of the tangent line at $P = (a, f(a))$ on f — so its slope is the *reciprocal* of $f'(a)$, exactly as the theorem states.

Example 3.3.1: Using the Inverse Function Derivative Formula

Suppose $f(2) = 5$ and $f'(2) = 3$. Find $(f^{-1})'(5)$.

Solution. Since $f(2) = 5$, we have $f^{-1}(5) = 2$, so

$$(f^{-1})'(5) = \frac{1}{f'(f^{-1}(5))} = \frac{1}{f'(2)} = \frac{1}{3}.$$

3.4 Differentiating Inverse Trigonometric Functions

Theorem 3.4.1: Derivatives of the Inverse Trigonometric Functions

$$\frac{d}{dx} [\arcsin x] = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx} [\arccos x] = \frac{-1}{\sqrt{1-x^2}}, \quad \frac{d}{dx} [\arctan x] = \frac{1}{1+x^2}.$$

Remark

$\arcsin x$ and $\arccos x$ are defined for $|x| < 1$ on the interval of differentiability; $\arctan x$ is differentiable for all real x . (The derivatives of arcsec , arccsc , and arccot follow the same pattern up to sign and are rarely needed explicitly on the AP Exam.)

Example 3.4.1: Chain Rule with an Inverse Trig Function

Differentiate $y = \arctan(3x)$.

Solution. With inner function $u = 3x$:

$$y' = \frac{1}{1+(3x)^2} \cdot 3 = \frac{3}{1+9x^2}.$$

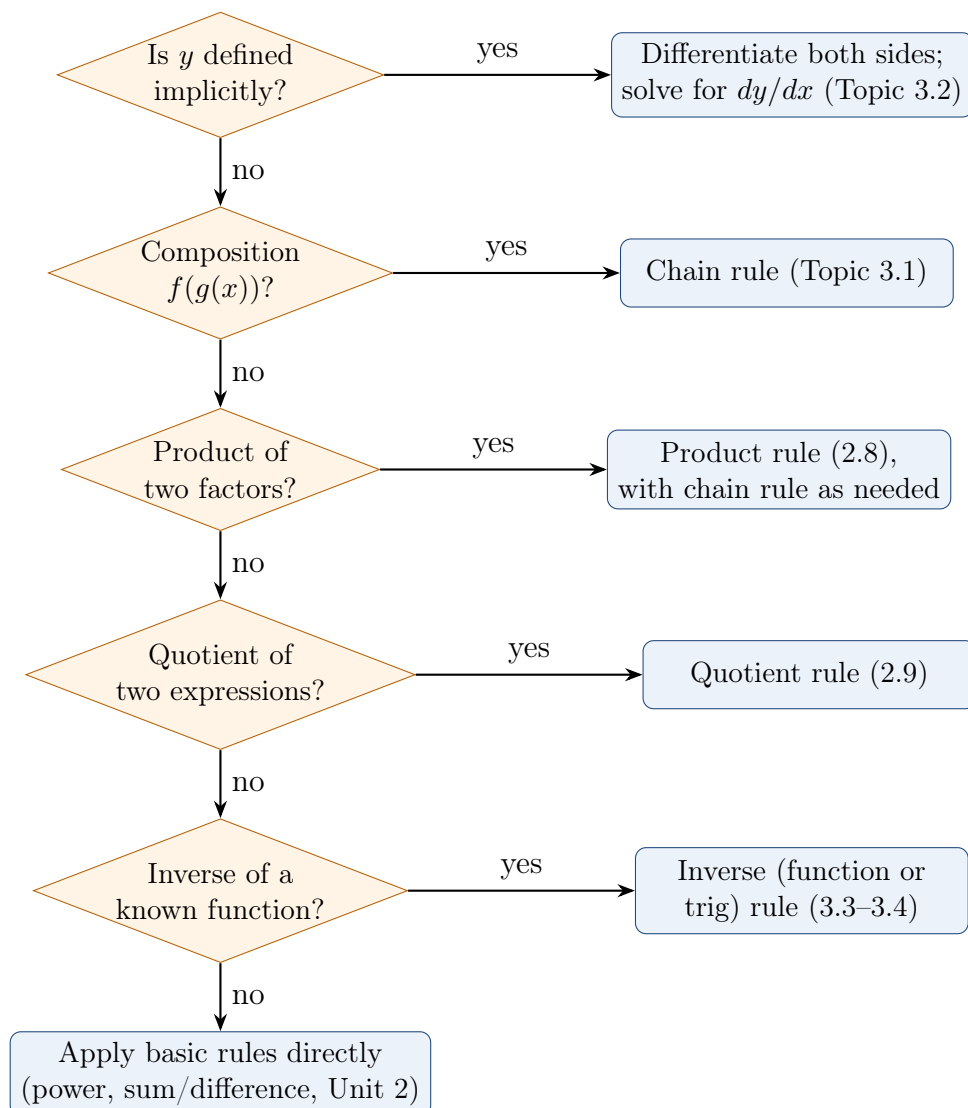
Example 3.4.2: Inverse Trig Function of a Nonlinear Inner Function

Differentiate $y = \arcsin(x^2)$.

Solution.

$$y' = \frac{1}{\sqrt{1-(x^2)^2}} \cdot 2x = \frac{2x}{\sqrt{1-x^4}}.$$

3.5 Selecting Procedures for Calculating Derivatives



Remark

Most AP-level derivatives require *combining* several rules (e.g. quotient rule where one piece also needs the chain rule). Treat the diagram above as identifying the *outermost* structure first, then recurse into each piece.

Example 3.5.1: Combining Multiple Rules

Differentiate $y = \frac{\sin^3(x^2)}{x+1}$.

Solution. Outermost structure: a *quotient*. Let $f(x) = \sin^3(x^2) = [\sin(x^2)]^3$ and $g(x) = x + 1$.

For f , apply the chain rule twice: $f'(x) = 3[\sin(x^2)]^2 \cdot \cos(x^2) \cdot 2x = 6x \sin^2(x^2) \cos(x^2)$.

Now apply the quotient rule:

$$y' = \frac{f'(x)(x+1) - \sin^3(x^2) \cdot 1}{(x+1)^2} = \frac{6x(x+1)\sin^2(x^2)\cos(x^2) - \sin^3(x^2)}{(x+1)^2}.$$

3.6 Calculating Higher-Order Derivatives

Definition 3.6.1: Higher-Order Derivatives

The **second derivative** $f''(x)$ is the derivative of $f'(x)$; the **third derivative** $f'''(x)$ is the derivative of $f''(x)$; and so on. Common notations for the n th derivative:

$$f^{(n)}(x), \quad y^{(n)}, \quad \frac{d^n y}{dx^n}.$$

Example 3.6.1: Computing Higher-Order Derivatives

Find $f''(x)$ for $f(x) = x^4 - 3x^2 + 5x$.

Solution. $f'(x) = 4x^3 - 6x + 5$, so $f''(x) = 12x^2 - 6$.

Example 3.6.2: Position, Velocity, and Acceleration

A particle's position is $s(t) = t^3 - 6t^2 + 9t$. Find its velocity and acceleration functions.

Solution. Velocity is the first derivative, acceleration the second:

$$v(t) = s'(t) = 3t^2 - 12t + 9, \quad a(t) = v'(t) = s''(t) = 6t - 12.$$

Remark

Higher-order derivatives reappear throughout the course: the second derivative drives concavity analysis in Unit 5, and position/velocity/acceleration relationships are revisited with integrals in Unit 8.

UNIT 4

Contextual Applications of Differentiation

Class Periods	~10–11 (AB)	~6–7 (BC)
Big Ideas	Change (CHA)	Limits (LIM)

Units 2 and 3 were about *computation*: given a formula for f , produce a formula for f' . That mechanical fluency is necessary, but it is not the point of calculus — the point is what a derivative *tells us*. Unit 4 turns the toolkit outward. We will read derivatives as rates embedded in real settings (position and velocity, volume and flow, cost and revenue), use the chain rule to relate several changing quantities to one another at once (related rates), use the tangent line as a cheap stand-in for a complicated function near a point (linearization), and finally circle back to Unit 1 with a new weapon for limits that algebra alone cannot resolve (L'Hospital's Rule). Every topic here answers a version of the same question that opened the course in Topic 1.1: what does it mean for change to happen, and how do we quantify it at a single instant?

4.1 Interpreting the Meaning of the Derivative in Context

Before doing anything computational, it is worth pausing on a habit of mind. Every derivative $f'(a)$ is a limit of average rates of change, and average rates of change are always ratios: *change in output over change in input*. That ratio structure survives the limiting process, which means $f'(a)$ always carries units, and those units always tell you how to put the number into a sentence.

Theorem 4.1.1: Units of a Derivative

If $y = f(x)$, the units of $f'(x)$ are

$$\frac{\text{units of } y}{\text{units of } x}.$$

This is a small statement with a big consequence: it gives a template for translating symbols into language. If $C(t)$ is the cost, in dollars, of producing t units, then $C'(t)$ is measured in dollars *per unit*, and a statement like $C'(50) = 3.2$ should always be read as a complete sentence: “When 50 units have been produced, cost is increasing at a rate of \$3.20 per additional unit.” On the AP Exam, this sentence — a value, a direction (increasing/decreasing), units, and the point at which it holds — is exactly what full-credit justification requires. A bare number with no context earns little.

Example 4.1.1: Interpreting a Derivative in Context

The volume of water in a tank, in gallons, after t minutes is $V(t)$. Interpret the meaning of $V'(12) = -8$.

Solution. At $t = 12$ minutes, the volume of water in the tank is decreasing at a rate of 8 gallons per minute.

With the language of interpretation established, we now apply it to the single richest source of derivative contexts on the AP Exam: motion along a line.

4.2 Straight-Line Motion: Connecting Position, Velocity, and Acceleration

Motion is the original motivating example for the derivative (recall Topic 1.1's question, "can change occur at an instant?"), and it remains the clearest place to practice reading derivatives as rates, because everyone has direct physical intuition for position, velocity, and acceleration.

Definition 4.2.1: Velocity and Acceleration

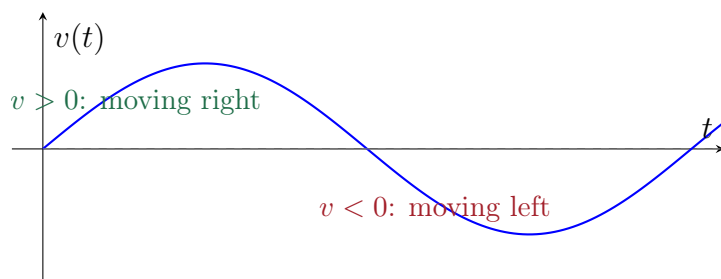
If $s(t)$ is the position of a particle moving along a line at time t , then its **velocity** is $v(t) = s'(t)$ and its **acceleration** is $a(t) = v'(t) = s''(t)$. Its **speed** is $|v(t)|$.

Velocity carries a sign that position alone does not: $v(t) > 0$ means the particle moves in the positive direction, $v(t) < 0$ means it moves in the negative direction, and $v(t) = 0$ means it is momentarily at rest. Speed discards that sign and only measures how fast the particle is moving, regardless of direction — a distinction that matters constantly on the exam, since "find the velocity" and "find the speed" are different questions.

Theorem 4.2.1: Speeding Up and Slowing Down

A particle is **speeding up** when $v(t)$ and $a(t)$ have the *same* sign, and **slowing down** when they have *opposite* signs.

It is worth pausing on *why* this is true rather than memorizing it: speed is $|v(t)|$, and a quantity's absolute value increases exactly when the quantity and its own rate of change point the same way. If the particle is already moving in the positive direction ($v > 0$) and accelerating in that same direction ($a > 0$), its speed climbs; if it is moving in the positive direction but accelerating in the negative direction ($a < 0$), the acceleration is fighting the motion and the speed falls. The graph below organizes this logic at a glance.



Example 4.2.1: Direction and Speed from a Position Function

A particle moves along a line with position $s(t) = t^3 - 6t^2 + 9t$ (meters, seconds). Determine whether the particle is speeding up or slowing down at $t = 1$.

Solution. From Example 3.6.2, $v(t) = 3t^2 - 12t + 9$ and $a(t) = 6t - 12$. At $t = 1$: $v(1) = 3 - 12 + 9 = 0$ and $a(1) = 6 - 12 = -6$. Since $v(1) = 0$, the particle is instantaneously at rest — the question of speeding up or slowing down does not apply at this exact instant, but for t slightly greater than 1, $v(t) < 0$ (check $v(1.1) = 3(1.21) - 13.2 + 9 \approx -0.57$) while $a(t) < 0$ throughout, so the particle is speeding up in the negative direction just after $t = 1$.

Motion is only one setting among many where a rate of change carries physical meaning. The next section generalizes the same reasoning to contexts with no motion at all.

4.3 Rates of Change in Applied Contexts Other Than Motion

Nothing about the derivative's construction — a limit of a difference quotient — refers to motion specifically. Position and velocity are simply the most familiar instance of a much more general pattern: whenever one quantity is a differentiable function of another, the derivative measures how fast the first responds to changes in the second. Economics gives marginal cost and marginal revenue; biology gives population growth rates; physics beyond motion gives rates of heating, draining, and expanding. Recognizing the pattern is more valuable than memorizing any one context.

Example 4.3.1: A Non-Motion Rate of Change

Air is pumped into a spherical balloon so that its volume is $V(r) = \frac{4}{3}\pi r^3$, where r is the radius in centimeters. Find the rate of change of volume with respect to radius when $r = 3$ cm, and interpret the result.

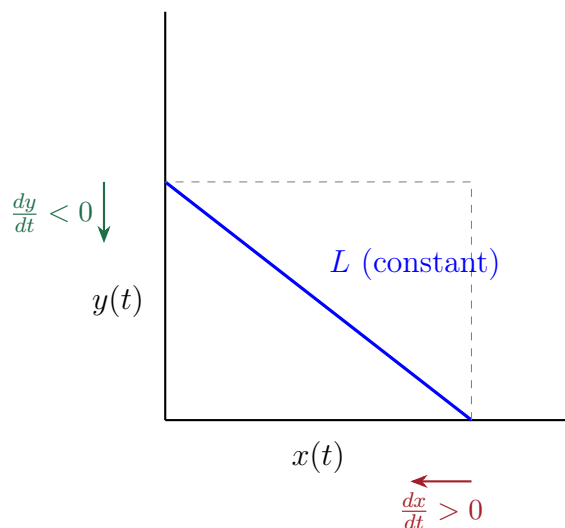
Solution. $V'(r) = 4\pi r^2$, so $V'(3) = 36\pi \approx 113.1$. When the radius is 3 cm, the volume is increasing at a rate of approximately 113.1 cubic centimeters per centimeter of radius.

That example quietly did something new: it related the rate of change of *one* quantity (V) to *another* quantity (r) that is itself changing, rather than to time directly. When two or more such quantities are linked by an equation and are *all* changing with time simultaneously, we need a systematic way to relate their rates — which is exactly the subject of related rates.

4.4 Introduction to Related Rates

Picture a ladder sliding down a wall. At every instant, the base of the ladder, the height it reaches on the wall, and the (fixed) length of the ladder satisfy the Pythagorean relationship. The base's distance from the wall and the height on the wall are both changing with time — and because they are tied together by an equation, their rates of

change are tied together too. **Related rates** problems are implicit differentiation (Topic 3.2) applied with respect to time.



Theorem 4.4.1: General Strategy for Related Rates

- (1) Draw and label a diagram; identify which quantities change with time and assign variables to them.
- (2) Write an equation relating the variables, valid at *every* instant of time (not just the instant in question).
- (3) Differentiate both sides with respect to t , using the chain rule on every variable that depends on t .
- (4) Substitute the known values — *only after* differentiating — and solve for the desired rate.

The order of steps 3 and 4 is not a formality. Substituting numbers before differentiating silently freezes the relationship at one instant and destroys the very information calculus was supposed to extract; the equation from step 2 must stay in general symbolic form until every derivative in it has already been taken.

4.5 Solving Related Rates Problems

With the strategy in hand, the remaining work in a related rates problem is executing it carefully on a specific configuration. The two examples below are the two archetypes almost every AP related-rates question is built from: a rigid length constraint (the ladder) and a similar-triangles constraint (a draining or filling container).

Example 4.5.1: The Sliding Ladder

A 10-foot ladder leans against a vertical wall. The base slides away from the wall at 2 ft/s. How fast is the top of the ladder sliding down the wall when the base is 6 feet from the wall?

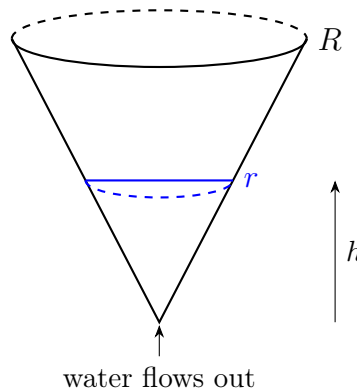
Solution. Let x be the distance from the wall to the base, and y the height of the top on the wall. The length is fixed, so at every instant $x^2 + y^2 = 100$. Differentiate with respect to t :

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0.$$

At the instant in question, $x = 6$, so $y = \sqrt{100 - 36} = 8$, and $\frac{dx}{dt} = 2$. Substitute now, after differentiating:

$$2(6)(2) + 2(8) \frac{dy}{dt} = 0 \implies 24 + 16 \frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{3}{2}.$$

The top of the ladder is sliding down the wall at 1.5 ft/s (the negative sign confirms y is decreasing).



Example 4.5.2: The Draining Conical Tank

Water drains from an inverted conical tank with radius $R = 2$ m and height $H = 5$ m at the top. At the instant the water depth is $h = 3$ m, the depth is decreasing at 0.5 m/min. How fast is the volume decreasing at that instant?

Solution. By similar triangles, the water's radius and depth are proportional: $\frac{r}{h} = \frac{R}{H} = \frac{2}{5}$, so $r = \frac{2}{5}h$. The volume of the water is $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{2}{5}h\right)^2 h = \frac{4\pi}{75}h^3$ — a relation with a single variable h , obtained by using the constraint to eliminate r before differentiating. Differentiate with respect to t :

$$\frac{dV}{dt} = \frac{4\pi}{75} \cdot 3h^2 \frac{dh}{dt} = \frac{4\pi h^2}{25} \frac{dh}{dt}.$$

At $h = 3$, $\frac{dh}{dt} = -0.5$:

$$\frac{dV}{dt} = \frac{4\pi(9)}{25}(-0.5) = -\frac{18\pi}{25} \approx -2.26 \text{ m}^3/\text{min}.$$

The volume is decreasing at approximately 2.26 cubic meters per minute.

Remark

Notice the extra step in the conical tank example: because the equation $V = \frac{1}{3}\pi r^2 h$ involves *two* time-dependent variables, r and h , we first used the geometric constraint between them to rewrite V in terms of h alone. Reducing to a single independent variable before differentiating is often what separates a related-rates problem that is merely tedious from one that is unsolvable as stated.

4.6 Approximating Values of a Function Using Local Linearity and Linearization

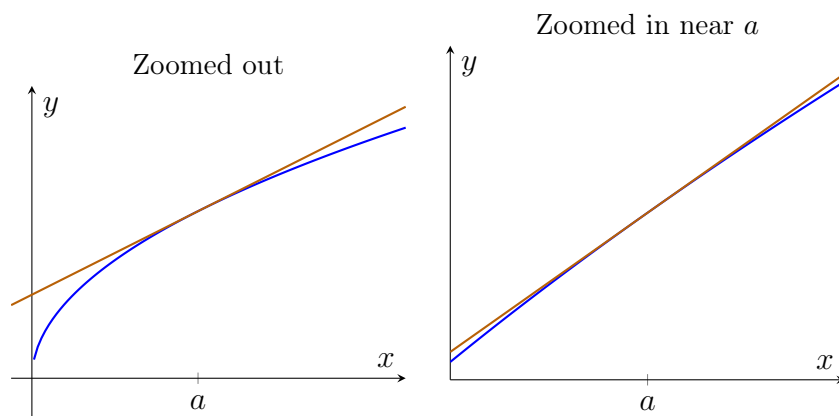
Related rates used the derivative to link *simultaneous* rates of change. We now use it for a different purpose: producing a fast, simple estimate of f near a point where f itself may be inconvenient to evaluate exactly. The underlying idea, **local linearity**, is that every differentiable function looks like a straight line if you zoom in far enough on its graph — and that line is precisely the tangent line.

Definition 4.6.1: Linearization

The **linearization** of f at $x = a$ is the tangent line function

$$L(x) = f(a) + f'(a)(x - a).$$

For x near a , $f(x) \approx L(x)$.



The two panels above show the same two curves — $f(x) = \sqrt{x}$ and its tangent line at $a = 4$ — at two different scales. Zoomed out, they visibly separate; zoomed in near a , they are nearly indistinguishable. That visual convergence *is* the content of local linearity, and it is what licenses replacing a hard computation ($\sqrt{4.1}$, say) with an easy one (evaluating a line).

Example 4.6.1: Linear Approximation

Use linearization to approximate $\sqrt{4.1}$.

Solution. Let $f(x) = \sqrt{x}$, $a = 4$. Then $f(4) = 2$ and $f'(x) = \frac{1}{2\sqrt{x}}$, so $f'(4) = \frac{1}{4}$.

The linearization is $L(x) = 2 + \frac{1}{4}(x - 4)$, so

$$\sqrt{4.1} \approx L(4.1) = 2 + \frac{1}{4}(0.1) = 2.025.$$

(For comparison, $\sqrt{4.1} = 2.02485\dots$ — the approximation is accurate to three decimal places.)

Remark

Because a tangent line only ever touches a curve at one point, a linearization is exact only at $x = a$ and drifts from the true curve as x moves away from a — a lesson Unit 5 sharpens using concavity: the tangent line lies *above* the curve where f is concave down, and *below* it where f is concave up, which tells you in advance whether a linear approximation over- or under-estimates.

4.7 Using L'Hospital's Rule for Determining Limits of Indeterminate Forms

We close this unit by returning to where the course began: limits. Topic 1.6 handled indeterminate forms like $\frac{0}{0}$ using algebra — factoring, rationalizing, trigonometric identities — and Topic 1.8 handled a narrow class using the Squeeze Theorem. Both approaches require finding the *right trick* for each expression. Now that derivatives are available, we have a single method that resolves a much broader family of indeterminate limits directly.

Theorem 4.7.1: L'Hospital's Rule

Suppose f and g are differentiable near a (except possibly at a itself), $g'(x) \neq 0$ near a , and

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

is of indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided the limit on the right exists (or is $\pm\infty$). The rule also applies as $x \rightarrow \pm\infty$.

AP Exam Tip

L'Hospital's Rule may only be applied to a limit that is *already* in indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ at the moment it is applied — check this explicitly before differentiating, and re-check it before every additional application. Differentiating a fraction that is not indeterminate (using the rule instead of the quotient rule, or when direct substitution already gives an answer) produces a wrong answer, and is one of the most common errors on the AP Exam.

Example 4.7.1: A Basic Application

Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

Solution. Direct substitution gives $\frac{0}{0}$, so L'Hospital's Rule applies:

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x}.$$

This is again $\frac{0}{0}$, so apply the rule a second time:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}.$$

Example 4.7.2: An Indeterminate Form at Infinity

Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$.

Solution. As $x \rightarrow \infty$, this has the form $\frac{\infty}{\infty}$, so

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

4.7.1 A Rigorous Proof of L'Hospital's Rule

Everything so far has *used* L'Hospital's Rule; we now *prove* it. The argument below is standard real-analysis material, well beyond what is expected of an AP student — but as the instructor, you should be able to justify every tool you hand your students, and this proof is included for that purpose. It requires no machinery beyond single-variable differentiation, but it does lean on two classical results ordinarily proved in a first analysis course: Rolle's Theorem and its generalization, the Cauchy Mean Value Theorem.

Theorem 4.7.2: Rolle's Theorem

Let f be continuous on $[p, q]$ and differentiable on (p, q) , with $f(p) = f(q)$. Then there exists $\xi \in (p, q)$ with $f'(\xi) = 0$.

Proof. If f is constant on $[p, q]$, then $f' \equiv 0$ on (p, q) and any ξ works. Otherwise, since f is continuous on the closed, bounded interval $[p, q]$, the Extreme Value Theorem guarantees that f attains both a maximum and a minimum value on $[p, q]$. Because f is non-constant while $f(p) = f(q)$, at least one of these extreme values differs from $f(p)$, and so is attained at some interior point $\xi \in (p, q)$, not at an endpoint. At this interior extremum, examine the difference quotient $\frac{f(\xi + h) - f(\xi)}{h}$: if ξ is a maximum, this quotient is ≤ 0 for $h > 0$ and ≥ 0 for $h < 0$. Since $f'(\xi)$ exists, it is the limit of this quotient as $h \rightarrow 0$ from *either* side, so both one-sided limits must agree — forcing $f'(\xi) = 0$ (the minimum case is identical with inequalities reversed). ■

Theorem 4.7.3: Cauchy's Mean Value Theorem

Let f, g be continuous on $[p, q]$ and differentiable on (p, q) . Then there exists $\xi \in (p, q)$ such that

$$[f(q) - f(p)]g'(\xi) = [g(q) - g(p)]f'(\xi).$$

Proof. Define $h(x) = [g(q) - g(p)]f(x) - [f(q) - f(p)]g(x)$ for $x \in [p, q]$. As a linear combination of f and g , h is continuous on $[p, q]$ and differentiable on (p, q) . A direct computation gives

$$h(p) = g(q)f(p) - f(q)g(p), \quad h(q) = f(p)g(q) - g(p)f(q),$$

so $h(p) = h(q)$. By Rolle's Theorem, there is $\xi \in (p, q)$ with $h'(\xi) = 0$; since $h'(x) = [g(q) - g(p)]f'(x) - [f(q) - f(p)]g'(x)$, this is exactly the claimed identity. ■

Theorem 4.7.4: Corollary: A Nonvanishing Derivative Separates End-point Values

If h is continuous on $[p, q]$, differentiable on (p, q) , and $h'(t) \neq 0$ for every $t \in (p, q)$, then $h(p) \neq h(q)$.

Proof. If $h(p) = h(q)$, Rolle's Theorem produces $\xi \in (p, q)$ with $h'(\xi) = 0$, contradicting the hypothesis. ■

With these tools available, we can prove the $\frac{0}{0}$ form of L'Hospital's Rule precisely.

Theorem 4.7.5: L'Hospital's Rule — the $\frac{0}{0}$ Case, Proved

Let $a \in \mathbb{R}$ and let f, g be differentiable on an open interval (a, b) , with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose

$$\lim_{x \rightarrow a^+} f(x) = 0, \quad \lim_{x \rightarrow a^+} g(x) = 0, \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}$$

Then $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$.

Proof. f and g need not originally be defined at a ; extend them by declaring $f(a) := 0$ and $g(a) := 0$. Since $\lim_{x \rightarrow a^+} f(x) = 0 = f(a)$ (and likewise for g), this extension makes f, g continuous on $[a, x]$ for every $x \in (a, b)$: continuous at every interior point, since differentiable there, and right-continuous at a by construction.

Fix $x \in (a, b)$. By the Corollary applied to g on $[a, x]$, $g(x) \neq g(a) = 0$. By Cauchy's Mean Value Theorem applied to f, g on $[a, x]$, there exists $c_x \in (a, x)$ with

$$[f(x) - f(a)]g'(c_x) = [g(x) - g(a)]f'(c_x) \implies f(x)g'(c_x) = g(x)f'(c_x)$$

(using $f(a) = g(a) = 0$). Since $g'(c_x) \neq 0$ and $g(x) \neq 0$, divide to obtain

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}, \quad a < c_x < x. \quad (*)$$

Now let $\varepsilon > 0$. Because $\lim_{t \rightarrow a^+} \frac{f'(t)}{g'(t)} = L$, there is $\delta > 0$ such that

$$0 < t - a < \delta \implies \left| \frac{f'(t)}{g'(t)} - L \right| < \varepsilon. \quad (**)$$

Let x satisfy $0 < x - a < \delta$. Since $a < c_x < x$, also $0 < c_x - a < \delta$, so c_x satisfies the hypothesis of (**), giving $\left| \frac{f'(c_x)}{g'(c_x)} - L \right| < \varepsilon$. By (*), this says $\left| \frac{f(x)}{g(x)} - L \right| < \varepsilon$.

We have shown: for every $\varepsilon > 0$ there is $\delta > 0$ such that $0 < x - a < \delta$ implies $\left| \frac{f(x)}{g(x)} - L \right| < \varepsilon$ — precisely the statement $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$. $\blacksquare$

Remark

The two-sided limit $x \rightarrow a$ follows by applying this argument separately on (a, b) (giving $x \rightarrow a^+$) and on an interval (a', a) (giving $x \rightarrow a^-$); if both one-sided limits equal L , so does the two-sided limit. An infinite limit $L = \pm\infty$ follows the identical argument, with the ε -neighborhood of (**) replaced by “eventually larger than any bound M .”

Theorem 4.7.6: Corollary: Limits as $x \rightarrow \infty$

If f, g are differentiable for $x > M$, $g'(x) \neq 0$ for $x > M$, $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = 0$, and $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = L$, then $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$.

Proof. Substitute $t = \frac{1}{x}$, so $x \rightarrow \infty$ corresponds to $t \rightarrow 0^+$. Define $F(t) = f(1/t)$, $G(t) = g(1/t)$ for $0 < t < 1/M$. By the chain rule, $F'(t) = -\frac{1}{t^2}f'(1/t)$ and $G'(t) = -\frac{1}{t^2}g'(1/t)$ (both well-defined, and $G'(t) \neq 0$ since $g'(1/t) \neq 0$), so

$$\frac{F'(t)}{G'(t)} = \frac{f'(1/t)}{g'(1/t)} \xrightarrow{t \rightarrow 0^+} L,$$

since $1/t \rightarrow \infty$. Also $F(t) \rightarrow 0$, $G(t) \rightarrow 0$ as $t \rightarrow 0^+$. The $\frac{0}{0}$ case just proved (applied at the finite point $a = 0$) gives $\lim_{t \rightarrow 0^+} \frac{F(t)}{G(t)} = L$. Since $F(t)/G(t) = f(1/t)/g(1/t) = f(x)/g(x)$

with $x = 1/t \rightarrow \infty$, this is exactly $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$. $\blacksquare$

Finally, we prove the $\frac{\infty}{\infty}$ form used in Example 4.7.2. Its proof is more delicate, because we can no longer extend f, g to equal 0 at a — instead, we compare f, g at a nearby *fixed* point rather than at a itself.

Theorem 4.7.7: L'Hospital's Rule — the $\frac{\infty}{\infty}$ Case, Proved

Let f, g be differentiable on (a, b) with $g'(x) \neq 0$ on (a, b) . Suppose

$$\lim_{x \rightarrow a^+} g(x) = +\infty \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}.$$

Then $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$.

Proof. Step 1 (a fixed comparison point). Let $\varepsilon > 0$. Choose $\delta_1 > 0$ (with $a + \delta_1 < b$) such that

$$a < t < a + \delta_1 \implies \left| \frac{f'(t)}{g'(t)} - L \right| < \frac{\varepsilon}{4}.$$

Fix $x_0 := a + \frac{1}{2}\delta_1$. Because $x_0 \in (a, b)$ is now *fixed*, $f(x_0)$ and $g(x_0)$ are fixed finite real numbers.

Step 2 (Cauchy's Mean Value Theorem on $[x, x_0]$). For any $x \in (a, x_0)$, f, g are continuous on $[x, x_0]$ and differentiable on $(x, x_0) \subset (a, a + \delta_1)$. By the Corollary, $g(x_0) \neq g(x)$; by Cauchy's Mean Value Theorem, there is $c = c_x \in (x, x_0) \subset (a, a + \delta_1)$ with

$$[f(x_0) - f(x)]g'(c) = [g(x_0) - g(x)]f'(c) \implies f(x) - f(x_0) = [g(x) - g(x_0)] \frac{f'(c)}{g'(c)}.$$

Since $c \in (a, a + \delta_1)$, Step 1 gives $\left| \frac{f'(c)}{g'(c)} - L \right| < \frac{\varepsilon}{4}$.

Step 3 (divide by $g(x)$). Since $g(x) \rightarrow +\infty$ as $x \rightarrow a^+$, we may assume x is close enough to a that $g(x) \neq 0$; dividing the boxed identity by $g(x)$:

$$\frac{f(x)}{g(x)} = \frac{f(x_0)}{g(x)} + \left(1 - \frac{g(x_0)}{g(x)}\right) \frac{f'(c)}{g'(c)}.$$

Step 4 (the fixed terms vanish). Because $x_0, f(x_0), g(x_0)$ are fixed while $g(x) \rightarrow +\infty$ as $x \rightarrow a^+$, both $\frac{f(x_0)}{g(x)} \rightarrow 0$ and $\frac{g(x_0)}{g(x)} \rightarrow 0$. Choose $\delta_2 \in (0, \delta_1)$ so that for $0 < x - a < \delta_2$ (hence $x < x_0$),

$$\left| \frac{f(x_0)}{g(x)} \right| < \frac{\varepsilon}{4} \quad \text{and} \quad \left| \frac{g(x_0)}{g(x)} \right| < \min\left(1, \frac{\varepsilon}{4(|L| + 1)}\right).$$

Step 5 (assemble the estimate). Write $\frac{f'(c)}{g'(c)} = L + \eta$ with $|\eta| < \frac{\varepsilon}{4}$ (Step 2). Then

$$\frac{f(x)}{g(x)} - L = \frac{f(x_0)}{g(x)} - L \cdot \frac{g(x_0)}{g(x)} + \eta \left(1 - \frac{g(x_0)}{g(x)}\right).$$

By Step 4, $\left| \frac{f(x_0)}{g(x)} \right| < \frac{\varepsilon}{4}$, and $\left| L \cdot \frac{g(x_0)}{g(x)} \right| < |L| \cdot \frac{\varepsilon}{4(|L| + 1)} < \frac{\varepsilon}{4}$. Also $\left| 1 - \frac{g(x_0)}{g(x)} \right| < 1 + 1 = 2$ (Step 4 bound < 1), so $\left| \eta \left(1 - \frac{g(x_0)}{g(x)}\right) \right| < \frac{\varepsilon}{4} \cdot 2 = \frac{\varepsilon}{2}$. Altogether, for $0 < x - a < \delta_2$,

$$\left| \frac{f(x)}{g(x)} - L \right| < \frac{\varepsilon}{4} + \frac{\varepsilon}{4} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$. ■

Remark

Both proofs share the same engine: the Cauchy Mean Value Theorem converts a statement about the *ratio of derivatives* at some unknown intermediate point into a statement about the *ratio of the functions themselves*. Everything else — the extension trick for $\frac{0}{0}$, the fixed comparison point for $\frac{\infty}{\infty}$ — is bookkeeping needed

to control that intermediate point as $x \rightarrow a$.

Having learned to interpret derivatives contextually, relate simultaneous rates, approximate with tangent lines, and resolve indeterminate limits (now on fully rigorous footing), we are equipped for Unit 5, where the derivative turns from a tool for answering isolated questions into a tool for understanding the global shape of a function's graph.

UNIT 5

Analytical Applications of Differentiation

Class Periods	~15–16 (AB)	~10–11 (BC)	
Big Ideas	Change (CHA)	Limits (LIM)	Analysis of Functions (FUN)

Unit 4 used derivatives to answer local, isolated questions: what is happening *at this instant, at this point*? Unit 5 asks a more global question: what does the derivative tell us about the entire shape of a function’s graph — where it rises and falls, where it bends, and where its most extreme values occur? The theoretical engine for all of this is a single theorem, the Mean Value Theorem, which is not new to us: it is the special case $g(x) = x$ of the Cauchy Mean Value Theorem we already proved in Unit 4 to justify L’Hospital’s Rule. Watching one carefully chosen theorem generate an entire unit’s worth of consequences — monotonicity, concavity, optimization — is one of the most satisfying patterns in calculus, and it is the throughline of everything that follows.

5.1 Using the Mean Value Theorem

Every average rate of change is a number you can compute without calculus — just a slope between two points. The Mean Value Theorem (MVT) makes a much stronger claim: that number is not just an abstraction, it is *actually achieved* as an instantaneous rate of change at some specific point in between.

Theorem 5.1.1: The Mean Value Theorem

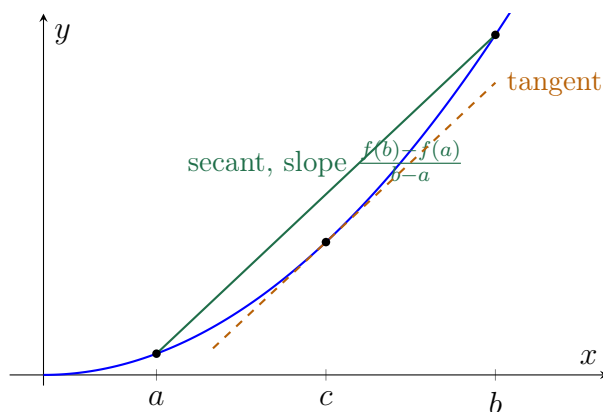
If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. This is precisely Cauchy’s Mean Value Theorem (Theorem 4.7.3) in the special case $g(x) = x$: since $g'(x) = 1$ everywhere, the conclusion $[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c)$ becomes $f(b) - f(a) = (b - a)f'(c)$, which rearranges to the statement above. ■

Remark

Rolle’s Theorem (Theorem 4.7.2) is, in turn, the special case of the MVT where $f(a) = f(b)$: the guaranteed point c then satisfies $f'(c) = \frac{f(b) - f(a)}{b - a} = 0$. The three theorems — Rolle’s, Cauchy’s, and the Mean Value Theorem — are really one idea viewed from three angles, and all of Unit 5 is built on top of them.



The tangent line at c is drawn *parallel* to the secant line — that parallelism, geometrically, is exactly the statement $f'(c)$ equals the secant slope.

Example 5.1.1: Verifying the Mean Value Theorem

Verify that $f(x) = x^2$ satisfies the hypotheses of the MVT on $[1, 4]$, and find the value(s) of c it guarantees.

Solution. f is a polynomial, hence continuous on $[1, 4]$ and differentiable on $(1, 4)$; the hypotheses hold. The secant slope is $\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = 5$. Since $f'(x) = 2x$, solve $2c = 5$: $c = 2.5$, which indeed lies in $(1, 4)$.

AP Exam Tip

Always verify the hypotheses (continuity on the closed interval, differentiability on the open interval) explicitly before invoking the MVT — this is exactly the habit flagged in Unit 1's discussion of theorem justification, and it is graded explicitly on AP free-response questions.

5.2 Extreme Value Theorem, Global Versus Local Extrema, and Critical Points

Before derivatives can help us map a function's peaks and valleys, we need precise vocabulary for what a "peak" or "valley" even is — and a guarantee that, on a closed interval, such things actually exist to be found.

Theorem 5.2.1: Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value and an absolute minimum value on $[a, b]$.

This is the same Extreme Value Theorem invoked, without proof, inside our proof of Rolle's Theorem last unit — there, it guaranteed a maximum or minimum existed *somewhere* on the interval; here, it is the star of the show.

Definition 5.2.1: Global and Local Extrema

f has an **absolute (global) maximum** at c if $f(c) \geq f(x)$ for every x in the domain; an **absolute (global) minimum** is defined with $\leq$. f has a **local (relative) maximum** at c if $f(c) \geq f(x)$ for all x near c (in some open interval around c); a **local minimum** is defined with $\leq$.

Definition 5.2.2: Critical Point

An interior point c in the domain of f is a **critical point** if $f'(c) = 0$ or $f'(c)$ does not exist.

Theorem 5.2.2: Fermat's Theorem: Local Extrema Occur at Critical Points

If f has a local maximum or local minimum at an interior point c , and $f'(c)$ exists, then $f'(c) = 0$.

We already proved this fact — it was embedded, unnamed, inside the proof of Rolle's Theorem: the argument that a difference quotient at an interior extremum is ≤ 0 from one side and ≥ 0 from the other, forcing the derivative to be 0, is exactly Fermat's Theorem.

AP Exam Tip

The converse of Fermat's Theorem is false, and this is one of the most commonly tested misconceptions on the AP Exam: a critical point need not be a local extremum. For instance $f(x) = x^3$ has $f'(0) = 0$, but 0 is neither a local max nor a local min — f is increasing right through it. Critical points are only *candidates*.

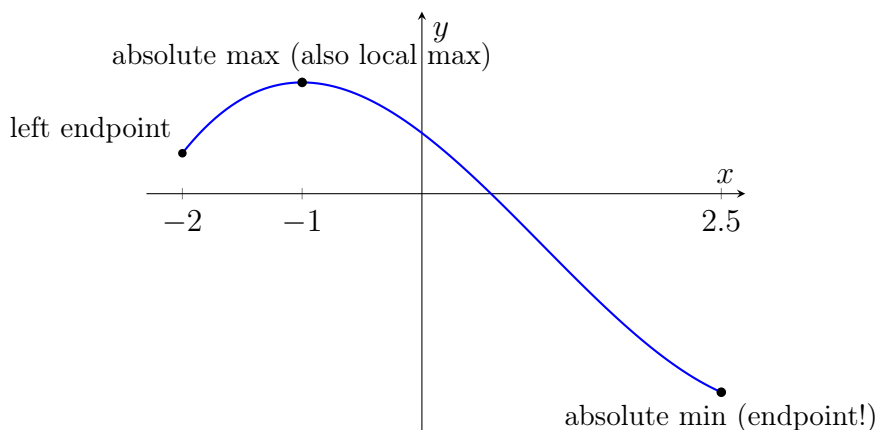
Throughout the rest of this unit, we will repeatedly analyze one running example, so that each new tool can be checked against the same concrete function:

$$g(x) = \frac{1}{3}x^3 - x^2 - 3x + 2.$$

Example 5.2.1: Finding Critical Points

Find the critical points of $g(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$.

Solution. $g'(x) = x^2 - 2x - 3 = (x - 3)(x + 1)$, which is defined for all x and equals 0 at $x = -1$ and $x = 3$. These are the critical points; $g(-1) = \frac{11}{3}$ and $g(3) = -7$.



On the closed interval $[-2, 2.5]$ shown above, g 's absolute maximum happens to coincide with its only interior critical point — but its absolute minimum occurs at the right *endpoint*, $x = 2.5$, which is not a critical point at all. Endpoints are always legitimate candidates for global extrema, a fact Topic 5.5 turns into a formal procedure.

5.3 Determining Intervals on Which a Function is Increasing or Decreasing

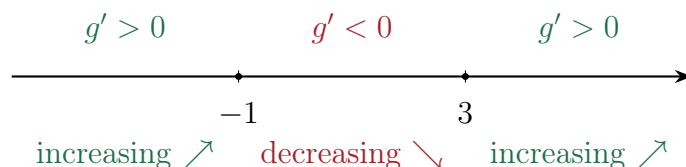
Since Unit 2, we have used the intuitive fact that a positive derivative means a function is increasing. With the Mean Value Theorem available, we can finally prove it.

Theorem 5.3.1: Increasing/Decreasing Function Theorem

Let f be continuous on an interval I and differentiable on its interior. If $f'(x) > 0$ for every interior x , then f is increasing on I . If $f'(x) < 0$ for every interior x , then f is decreasing on I .

Proof. Let $x_1 < x_2$ be any two points of I . By the MVT applied to f on $[x_1, x_2]$, there is $c \in (x_1, x_2)$ with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$. If $f'(x) > 0$ throughout I , then $f'(c) > 0$, and since $x_2 - x_1 > 0$, the product $f'(c)(x_2 - x_1) > 0$, so $f(x_2) > f(x_1)$. As $x_1 < x_2$ were arbitrary, f is increasing. The decreasing case is identical with inequalities reversed. ■

This gives a completely mechanical procedure: find where f' is zero or undefined (the critical points), use them to partition the domain into intervals, and test the sign of f' on each.



Example 5.3.1: Intervals of Increase and Decrease

Determine where $g(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$ is increasing and decreasing.

Solution. From Example 5.2.1, $g'(x) = (x - 3)(x + 1)$, zero at $x = -1, 3$. Testing

a point in each of the three resulting intervals (as diagrammed above): $g'(-2) = 5 > 0$, $g'(0) = -3 < 0$, $g'(4) = 5 > 0$. So g is increasing on $(-\infty, -1)$, decreasing on $(-1, 3)$, and increasing on $(3, \infty)$.

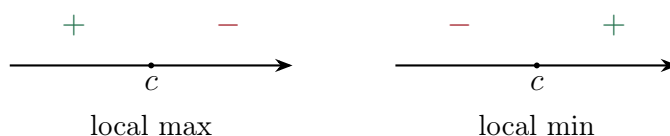
5.4 Using the First Derivative Test to Determine Relative (Local) Extrema

The sign chart from Topic 5.3 already contains everything needed to classify each critical point — we simply have not yet stated the conclusion as a theorem.

Theorem 5.4.1: First Derivative Test

Let c be a critical point of a continuous function f .

- If f' changes from positive to negative at c , f has a **local maximum** at c .
- If f' changes from negative to positive at c , f has a **local minimum** at c .
- If f' does not change sign at c , f has **neither**.



Example 5.4.1: Applying the First Derivative Test

Classify the critical points of $g(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$.

Solution. From Example 5.3.1, g' changes from $+$ to $-$ at $x = -1$, so g has a **local maximum** there, $g(-1) = \frac{11}{3}$. g' changes from $-$ to $+$ at $x = 3$, so g has a **local minimum** there, $g(3) = -7$.

5.5 Using the Candidates Test to Determine Absolute (Global) Extrema

The First Derivative Test classifies *local* behavior at a critical point, using only information near that point. Finding *global* extrema on a closed interval is a different, and easier, task — the Extreme Value Theorem already guarantees global extrema exist, so we only need to check a short, finite list of candidates.

Theorem 5.5.1: Candidates Test

If f is continuous on $[a, b]$, its absolute extrema occur either at a critical point in (a, b) or at an endpoint a or b . To find them: compute f at every critical point in (a, b) and at both endpoints, then compare.

Remark

This is a direct consequence of the Extreme Value Theorem and Fermat's Theorem: the EVT guarantees a global max and min exist *somewhere* on $[a, b]$; Fermat's Theorem guarantees that if one occurs at an interior point, that point must be a critical point. The only other possibility left is an endpoint. Hence the full list of candidates is exactly: interior critical points, together with the two endpoints.

Example 5.5.1: Candidates Test

Find the absolute extrema of $g(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$ on $[-2, 4]$.

Solution. Critical points in $(-2, 4)$: $x = -1, 3$ (Example 5.2.1). Evaluate g at these and at both endpoints:

x	-2	-1	3	4
$g(x)$	1.333	3.667	-7	-4.667

The absolute maximum is $g(-1) = \frac{11}{3}$; the absolute minimum is $g(3) = -7$. (Note that on this interval, both global extrema happen to occur at interior critical points — contrast this with Example 5.2.2, where the domain $[-2, 2.5]$ instead put the absolute minimum at an endpoint.)

5.6 Determining Concavity of Functions over Their Domains

Monotonicity (Topic 5.3) describes whether a graph rises or falls; **concavity** describes how it *bends* while doing so — and, just as the sign of f' controlled monotonicity of f , the sign of f'' (the derivative of f') controls monotonicity of f' , which is exactly concavity.

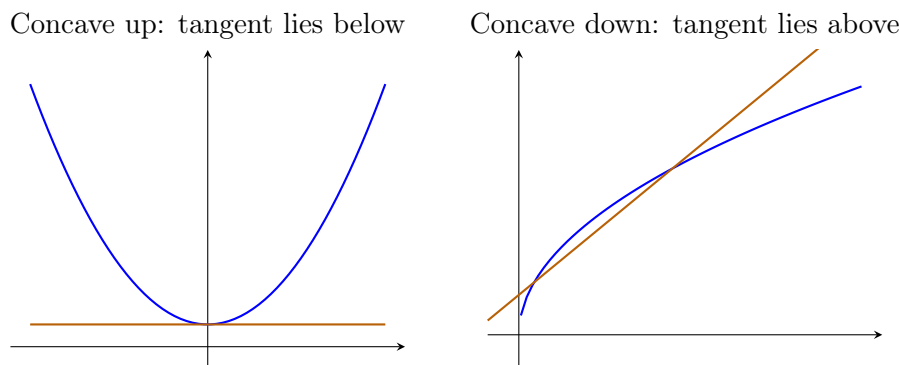
Definition 5.6.1: Concavity

f is **concave up** on an interval if f' is increasing there, and **concave down** if f' is decreasing there. A point where concavity changes is an **inflection point**.

Theorem 5.6.1: Concavity Test

If $f''(x) > 0$ on an interval, f is concave up there. If $f''(x) < 0$, f is concave down there.

This theorem requires no new proof: it is simply the Increasing/Decreasing Function Theorem (Topic 5.3) applied to f' instead of f , since f'' is the derivative of f' .

**Remark**

This is exactly the geometric fact behind Unit 4's remark on linearization error: a tangent line under-approximates a concave-up curve and over-approximates a concave-down curve, because the curve bends away from (concave up) or toward (concave down) the straight tangent line on either side of the point of tangency.

Example 5.6.1: Determining Concavity

Determine the concavity of $g(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$.

Solution. $g'(x) = x^2 - 2x - 3$, so $g''(x) = 2x - 2$, zero at $x = 1$. Testing: $g''(0) = -2 < 0$ and $g''(2) = 2 > 0$. So g is concave down on $(-\infty, 1)$ and concave up on $(1, \infty)$; $x = 1$ is an inflection point, with $g(1) = -\frac{5}{3}$.

5.7 Using the Second Derivative Test to Determine Extrema

Concavity gives a second, often faster, way to classify a critical point: if the graph is concave down at a critical point, the point “looks like” a peak; if concave up, a valley.

Theorem 5.7.1: Second Derivative Test

Let $f'(c) = 0$.

- If $f''(c) < 0$, f has a **local maximum** at c .
- If $f''(c) > 0$, f has a **local minimum** at c .
- If $f''(c) = 0$, the test is **inconclusive** — fall back on the First Derivative Test.

Example 5.7.1: Applying the Second Derivative Test

Reclassify the critical points of $g(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$ using $g''(x) = 2x - 2$.

Solution. $g''(-1) = -2 - 2 = -4 < 0$, confirming a **local maximum** at $x = -1$. $g''(3) = 6 - 2 = 4 > 0$, confirming a **local minimum** at $x = 3$ — agreeing with the First Derivative Test result of Example 5.4.1.

AP Exam Tip

When $f''(c) = 0$, students sometimes wrongly conclude c is an inflection point. It might be — but it might also be a local extremum with a “flatter than usual” turn, or neither. $f''(c) = 0$ only rules out the Second Derivative Test; it does not, by itself, determine anything.

That warning cuts both ways: just as $f''(c) = 0$ does not guarantee the Second Derivative Test’s failure matters, it also does not guarantee an inflection point. Points where $f'' = 0$ (or is undefined) deserve exactly the same careful treatment we gave critical points in Topic 5.2 — they are *candidates*, not conclusions.

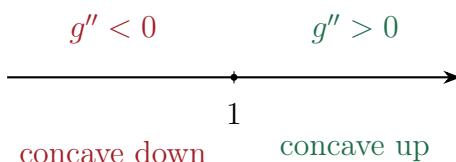
Definition 5.7.1: Candidates for Inflection Points

A point c in the domain of f is a **candidate** for an inflection point if $f''(c) = 0$ or $f''(c)$ does not exist. (Compare this to the definition of a critical point in Topic 5.2 — the same logic, one derivative order higher.)

Theorem 5.7.2: Test for Inflection Points

Let c be a candidate for an inflection point. If f'' changes sign at c , then f has an **inflection point** at c . If f'' does **not** change sign at c , then f does not have an inflection point there, even though $f''(c) = 0$.

This, once again, requires no new proof: it is the Increasing/Decreasing Function Theorem (Topic 5.3) applied to f' instead of f — a sign change in f'' is exactly a change in the monotonicity of f' , which is exactly a change in the concavity of f . Locating and testing candidates for inflection points is thus a sign-chart procedure identical in spirit to the one built for critical points in Topics 5.3 and 5.4, just applied one derivative higher.

**Example 5.7.2: Confirming an Inflection Point**

Confirm that $x = 1$ is an inflection point of $g(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$.

Solution. From Example 5.6.1, $g''(x) = 2x - 2$, zero at $x = 1$ — a candidate. Testing on either side, as diagrammed above: $g''(0) = -2 < 0$ and $g''(2) = 2 > 0$. Since g'' changes sign at $x = 1$, the candidate is confirmed to be an inflection point.

Example 5.7.3: A Candidate That Is Not an Inflection Point

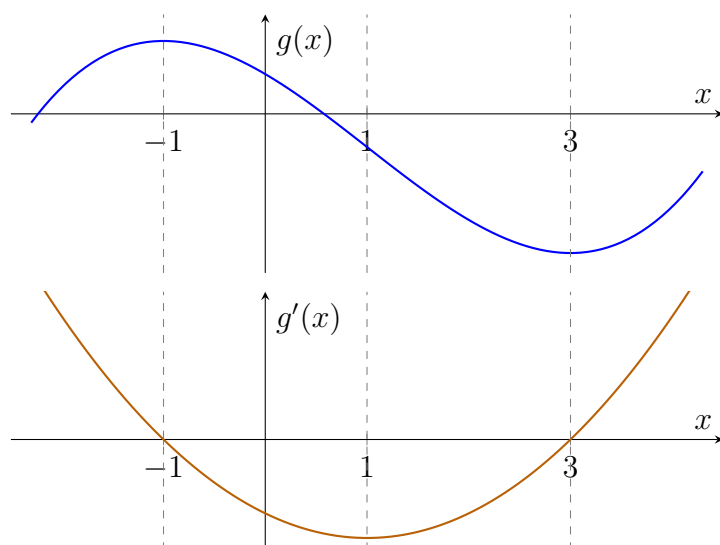
Show that $x = 0$ is *not* an inflection point of $f(x) = x^4$, even though $f''(0) = 0$.

Solution. $f'(x) = 4x^3$, so $f''(x) = 12x^2$, which is indeed 0 at $x = 0$ — a candidate. But $f''(x) = 12x^2 \geq 0$ for *every* x : it never changes sign, so f is concave up on both sides of 0 alike. There is no inflection point at $x = 0$. This is the exact concavity analogue of $f(x) = x^3$ having a critical point at $x = 0$ that is not an extremum.

(Topic 5.2's AP Exam Tip) — a candidate that fails to deliver.

5.8 Sketching Graphs of Functions and Their Derivatives

We now have every tool needed to translate between a function and its derivatives graphically — arguably the single most tested skill across Units 2 through 5. The two panels below show g and g' stacked so that every feature of one has a visible counterpart in the other.



Remark

Read the dashed guide lines across both panels: at $x = -1$ and $x = 3$, g has horizontal tangents *exactly* where g' crosses zero. At $x = 1$, g has its inflection point (its steepest downhill moment) *exactly* where g' reaches its own minimum — because an inflection point of g is a critical point of g' .

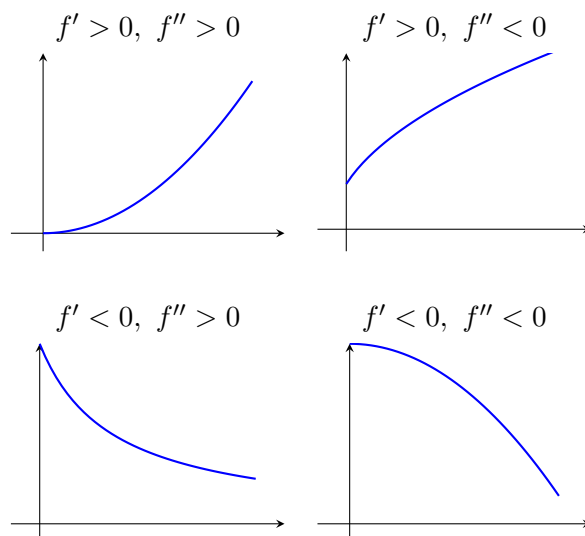
Example 5.8.1: From a Derivative Graph to the Original Function

Suppose h' is positive on $(-\infty, 2)$, negative on $(2, 5)$, and positive on $(5, \infty)$, with $h''(x) < 0$ on $(-\infty, 2) \cup (2, 3.5)$ and $h''(x) > 0$ on $(3.5, \infty)$. Describe the shape of h .

Solution. From the sign of h' : h increases on $(-\infty, 2)$, decreases on $(2, 5)$, increases on $(5, \infty)$ — a local max at $x = 2$ and a local min at $x = 5$ (First Derivative Test). From the sign of h'' : h is concave down on $(-\infty, 3.5)$ and concave up on $(3.5, \infty)$, with an inflection point at $x = 3.5$ (which falls inside the decreasing interval, so the graph is decreasing but changing from bending-down to bending-up there).

5.9 Connecting a Function, Its First Derivative, and Its Second Derivative

Every local piece of a graph's shape is determined by exactly two signs — that of f' and that of f'' — combined. There are only four possible combinations, and it is worth internalizing all four as shapes, not just as sign tables.



Example 5.9.1: Sign Analysis Without a Formula

A function f satisfies $f'(x) < 0$ and $f''(x) < 0$ on the interval $(2, 5)$. Describe the graph of f there, and sketch its general shape.

Solution. $f' < 0$: f is decreasing on $(2, 5)$. $f'' < 0$: f is concave down on $(2, 5)$. Matching the bottom-right icon above, the graph falls, and it falls *more and more steeply* as x increases through the interval (concave down means the slope itself is decreasing, i.e. becoming more negative).

5.10 Introduction to Optimization Problems

Optimization problems ask a very concrete question — what dimensions minimize cost, what shape maximizes area, what production level maximizes profit — and answer it with a strategy that is the direct descendant of the Candidates Test (Topic 5.5), in exactly the way Topic 4.5's related rates problems were the direct descendant of implicit differentiation.

Theorem 5.10.1: General Strategy for Optimization

- (1) Identify the quantity to be optimized, and express it as a function of the relevant variable(s). A diagram often clarifies the relationship.
- (2) If there is more than one variable, use a given constraint to eliminate all but one, exactly as in related rates (Topic 4.4).
- (3) Determine the domain of the remaining variable dictated by the physical

context (lengths and areas cannot be negative, for instance).

- (4) Apply the Candidates Test on that domain — or, if the domain is open or unbounded, the First or Second Derivative Test — to find the optimal value.

5.11 Solving Optimization Problems

Example 5.11.1: Minimizing the Surface Area of a Can

A cylindrical can must hold a fixed volume $V = 1000 \text{ cm}^3$. Find the radius and height that minimize the amount of material used (the total surface area), and show that at the optimum, $h = 2r$.

Solution. Let r be the radius and h the height. The surface area (top, bottom, and lateral surface) is $S = 2\pi r^2 + 2\pi rh$, and the constraint is $V = \pi r^2 h = 1000$, so $h = \frac{1000}{\pi r^2}$. Substitute to eliminate h :

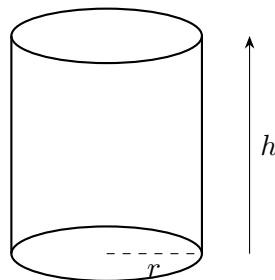
$$S(r) = 2\pi r^2 + 2\pi r \cdot \frac{1000}{\pi r^2} = 2\pi r^2 + \frac{2000}{r}, \quad r > 0.$$

Differentiate and set equal to zero:

$$S'(r) = 4\pi r - \frac{2000}{r^2} = 0 \implies r^3 = \frac{2000}{4\pi} = \frac{500}{\pi} \implies r = \left(\frac{500}{\pi}\right)^{1/3} \approx 5.42 \text{ cm}.$$

Since $S''(r) = 4\pi + \frac{4000}{r^3} > 0$ for all $r > 0$, this critical point is a minimum (Second Derivative Test) — and since it is the *only* critical point on the open domain $(0, \infty)$, it is the absolute minimum. Then $h = \frac{1000}{\pi r^2} \approx 10.83 \text{ cm}$.

For the general relationship: from $r^3 = \frac{V}{2\pi}$ we get $V = 2\pi r^3$, so $h = \frac{V}{\pi r^2} = \frac{2\pi r^3}{\pi r^2} = 2r$ — confirming $h = 2r \approx 10.83 \text{ cm}$ independent of the specific volume.



Example 5.11.2: Maximizing the Area of an Inscribed Rectangle

Find the maximum area of a rectangle with its base on the x -axis and its upper two corners on the parabola $y = 4 - x^2$.

Solution. By symmetry, a corner at $(x, 4 - x^2)$ for $x \in (0, 2)$ determines the whole

rectangle: width $2x$, height $4 - x^2$, so

$$A(x) = 2x(4 - x^2) = 8x - 2x^3, \quad x \in (0, 2).$$

$A'(x) = 8 - 6x^2 = 0 \Rightarrow x^2 = \frac{4}{3} \Rightarrow x = \frac{2}{\sqrt{3}}$. Since $A''(x) = -12x < 0$ for $x > 0$, this is a maximum. The maximum area is

$$A\left(\frac{2}{\sqrt{3}}\right) = 8 \cdot \frac{2}{\sqrt{3}} - 2\left(\frac{2}{\sqrt{3}}\right)^3 = \frac{16}{\sqrt{3}} - \frac{16}{3\sqrt{3}} = \frac{32}{3\sqrt{3}} = \frac{32\sqrt{3}}{9} \approx 6.16.$$

5.12 Exploring Behaviors of Implicit Relations

Every tool built in this unit — increasing/decreasing, concavity, extrema — was stated for functions $y = f(x)$. Implicit differentiation (Topic 3.2) lets us extend all of it to curves defined implicitly, by working with $\frac{dy}{dx}$ as an expression in both x and y .

Theorem 5.12.1: Horizontal and Vertical Tangents on Implicit Curves

On a curve defined implicitly, a **horizontal tangent** occurs where $\frac{dy}{dx} = 0$ — that is, where the numerator of $\frac{dy}{dx}$ (fully reduced) is zero while the denominator is not.

A **vertical tangent** occurs where $\frac{dy}{dx}$ is undefined because its denominator is zero while the numerator is not.

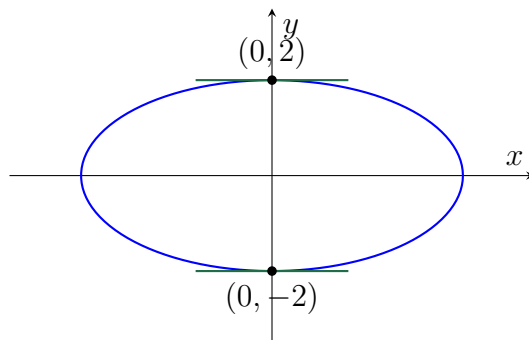
Example 5.12.1: Horizontal Tangents on an Ellipse

Find the points on $x^2 + 4y^2 = 16$ where the tangent line is horizontal.

Solution. Differentiate implicitly:

$$2x + 8y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{4y}.$$

This is zero exactly when $x = 0$ (and $y \neq 0$, so the denominator does not also vanish). Substituting $x = 0$ into the original equation: $4y^2 = 16 \Rightarrow y = \pm 2$. The horizontal tangents occur at $(0, 2)$ and $(0, -2)$.



Remark

By the same reasoning, vertical tangents on this ellipse occur where the denominator $4y = 0$ (i.e. $y = 0$) while $x \neq 0$: substituting $y = 0$ gives $x^2 = 16$, so $x = \pm 4$ — the points $(\pm 4, 0)$, visible in the figure as the leftmost and rightmost points of the curve.

With monotonicity, concavity, and extrema now fully in hand — for explicit *and* implicit curves alike — Unit 6 turns the derivative's logic in reverse: given a rate of change, how do we recover the total amount of change it produces? That question is the definite integral.

UNIT 6

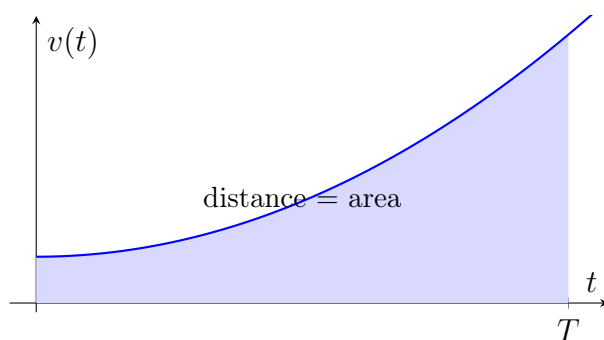
Integration and Accumulation of Change

Class Periods	~18–20 (AB)	~15–16 (BC)	
Big Ideas	Change (CHA)	Limits (LIM)	Analysis of Functions (FUN)

Every unit since Unit 2 has asked: given a function, what is its rate of change? Unit 6 turns that question around: given a *rate* of change, how much total change accumulates? This is not a new limiting process — it is the same limiting idea as the derivative, run in reverse, and once again built out of a familiar object: the sum of many small pieces. By the end of this unit, the two questions will turn out to be inverse operations of one another, tied together by the single most important theorem in the course: the Fundamental Theorem of Calculus.

6.1 Exploring Accumulations of Change

If a car travels at a constant velocity of 60 mph for 2 hours, it covers $60 \times 2 = 120$ miles — velocity times time. But velocity is rarely constant. If $v(t)$ varies continuously, no single multiplication computes the distance traveled; instead, the natural idea is to chop the time interval into short pieces so brief that velocity is *nearly* constant on each one, multiply each piece's (approximately constant) velocity by its short duration, and add the results.



That sum — velocity samples times a small time step, added up — is an approximation to the total distance traveled, and geometrically it is an approximation to the **area between the velocity graph and the t -axis**. As the time step shrinks, the approximation improves; the exact distance is whatever that sum approaches in the limit. This informal idea — accumulation as area, and area as a limit of sums — is made precise over the next two topics.

Example 6.1.1: Estimating Accumulated Change from a Table

A pump's flow rate is given below. Estimate the total volume pumped over $[0, 4]$ minutes using the rate at the *start* of each 1-minute interval.

t (min)	0	1	2	3	4
$r(t)$ (L/min)	2.0	3.5	4.0	3.0	2.5

Solution. Using the rate at the left endpoint of each interval, each multiplied by the 1-minute width:

$$2.0(1) + 3.5(1) + 4.0(1) + 3.0(1) = 12.5 \text{ liters.}$$

6.2 Approximating Areas with Riemann Sums

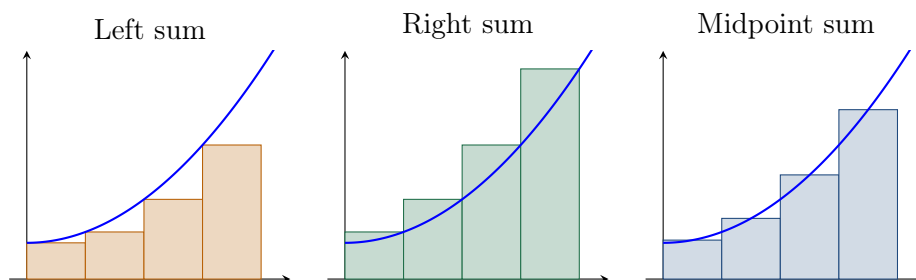
The estimate above used only the left endpoint of each subinterval — one of several reasonable choices. Formalizing this choice-of-sample-point idea produces the **Riemann sum**.

Definition 6.2.1: Riemann Sum

Partition $[a, b]$ into n subintervals, for simplicity of equal width $\Delta x = \frac{b-a}{n}$, with endpoints $a = x_0 < x_1 < \cdots < x_n = b$. Choose a sample point x_i^* in each subinterval $[x_{i-1}, x_i]$. The corresponding Riemann sum is

$$\sum_{i=1}^n f(x_i^*) \Delta x.$$

A **left Riemann sum** uses $x_i^* = x_{i-1}$ (the left endpoint); a **right Riemann sum** uses $x_i^* = x_i$; a **midpoint Riemann sum** uses the midpoint of each subinterval.

**Remark**

For $f(x) = 0.3x^2 + 1$ above, which is increasing throughout $[0, 4]$: the left sum consistently uses the smaller value on each subinterval, so it **underestimates** the true area; the right sum consistently uses the larger value, so it **overestimates**. If f were decreasing instead, these roles would swap. This connects directly to Unit 5: monotonicity of f (not concavity) determines whether left/right sums over- or under-estimate.

Example 6.2.1: Computing a Left and Right Riemann Sum

Using $f(x) = 0.3x^2 + 1$ on $[0, 4]$ with $n = 4$ (as pictured above), compute the left and right Riemann sums.

Solution. With $\Delta x = 1$: Left sum $= 1(1) + 1(1.3) + 1(2.2) + 1(3.7) = 8.2$. Right sum $= 1(1.3) + 1(2.2) + 1(3.7) + 1(5.8) = 13.0$. The true area (found exactly in Example 6.7.1) lies between these two estimates, at 10.4.

6.3 Riemann Sums, Summation Notation, and Definite Integral Notation

As $n \rightarrow \infty$ (equivalently, as $\Delta x \rightarrow 0$), a Riemann sum's approximation becomes exact — regardless of which sample points were chosen, *provided* f is continuous. This limit is given its own name and notation.

Definition 6.3.1: The Definite Integral

If f is continuous on $[a, b]$, the **definite integral** of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x, \quad \Delta x = \frac{b-a}{n},$$

the common limit of all Riemann sums, regardless of the sample points x_i^* chosen. Geometrically, when $f(x) \geq 0$ on $[a, b]$, this equals the area between the graph of f and the x -axis on $[a, b]$.

Remark

The integral sign $\int$ is a stretched S , for “sum,” and dx is the continuous echo of the discrete Δx — the notation is built to remind you that an integral *is* a limit of sums, every time you write one.

Example 6.3.1: From a Riemann Sum Limit to Integral Notation

Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(1 + \frac{3i}{n}\right)^2 \cdot \frac{3}{n}$ as a definite integral.

Solution. Matching the pattern $\Delta x = \frac{3}{n}$ suggests $b - a = 3$; matching the sample point $x_i^* = 1 + \frac{3i}{n}$ (a right endpoint starting from $a = 1$) with $f(x) = x^2$ gives

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(1 + \frac{3i}{n}\right)^2 \cdot \frac{3}{n} = \int_1^4 x^2 dx.$$

6.4 The Fundamental Theorem of Calculus and Accumulation Functions

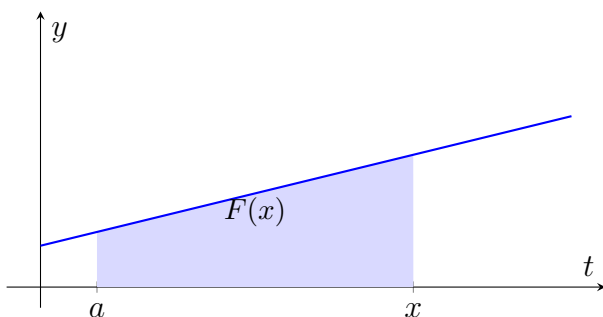
Fix a continuous function f and a starting point a . Letting the *upper* limit of integration vary turns the definite integral into a function in its own right.

Definition 6.4.1: Accumulation Function

For continuous f , the function

$$F(x) = \int_a^x f(t) dt$$

is the **accumulation function** of f based at a : $F(x)$ is the net signed area under f from a to x .



Theorem 6.4.1: Fundamental Theorem of Calculus, Part 1

If f is continuous on an interval containing a , and $F(x) = \int_a^x f(t) dt$, then F is differentiable and

$$F'(x) = f(x).$$

Remark

This is the precise sense in which integration reverses differentiation: the derivative of the accumulated area function gives back the original rate function. Intuitively, over a tiny interval of width h near x , $F(x+h) - F(x)$ is (approximately) a thin rectangle of area $f(x) \cdot h$, so $\frac{F(x+h) - F(x)}{h} \approx f(x)$ — and this approximation becomes exact in the limit $h \rightarrow 0$ defining $F'(x)$.

Example 6.4.1: Differentiating an Accumulation Function

Let $F(x) = \int_2^x \sqrt{t^2 + 1} dt$. Find $F'(x)$.

Solution. By FTC Part 1, $F'(x) = \sqrt{x^2 + 1}$ directly — no antidifferentiation required.

Example 6.4.2: Accumulation Function with a Chain Rule Twist

Let $G(x) = \int_0^{x^3} \sin(t) dt$. Find $G'(x)$.

Solution. The upper limit is x^3 , not x , so G is a composition of the accumulation function with $u = x^3$. By FTC Part 1 together with the chain rule (Topic 3.1):

$$G'(x) = \sin(x^3) \cdot \frac{d}{dx}[x^3] = 3x^2 \sin(x^3).$$

6.5 Interpreting the Behavior of Accumulation Functions Involving Area

Because $F'(x) = f(x)$, everything Unit 5 built to analyze a function from its derivative applies immediately to accumulation functions — with f playing the role of “the derivative” throughout.

Theorem 6.5.1: Reading F from f

Let $F(x) = \int_a^x f(t) dt$.

- F is increasing where $f > 0$, decreasing where $f < 0$ (Topic 5.3, applied to $F' = f$).
- F has a local max where f changes $+$ $\rightarrow$ $-$, a local min where f changes $-$ $\rightarrow$ $+$ (First Derivative Test, applied to F).
- Since $F'' = f'$, F is concave up where f is increasing and concave down where f is decreasing.

Example 6.5.1: Analyzing an Accumulation Function from a Graph of f

Suppose f is positive and decreasing on $(0, 3)$, and negative and decreasing on $(3, 6)$. Describe $F(x) = \int_0^x f(t) dt$ on $(0, 6)$.

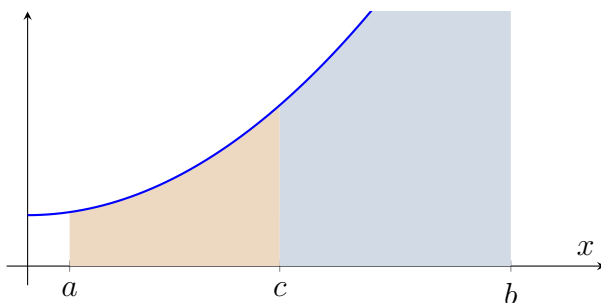
Solution. Since $f > 0$ on $(0, 3)$, F is increasing there; since $f < 0$ on $(3, 6)$, F is decreasing there — so F has a local **maximum** at $x = 3$ (Topic 5.4, First Derivative Test). Since f is decreasing on the entire interval $(0, 6)$, $F'' = f' < 0$ throughout, so F is concave down on all of $(0, 6)$ — with no inflection point, despite the local max in the middle.

6.6 Applying Properties of Definite Integrals

A short list of algebraic properties, all provable directly from the Riemann sum definition, make definite integrals far more flexible to work with than the definition alone would suggest.

Theorem 6.6.1: Properties of Definite Integrals

$$\begin{aligned}
\int_a^b [f(x) \pm g(x)] dx &= \int_a^b f(x) dx \pm \int_a^b g(x) dx, \\
\int_a^b k f(x) dx &= k \int_a^b f(x) dx \quad (k \in \mathbb{R}), \\
\int_a^a f(x) dx &= 0, \quad \int_b^a f(x) dx = - \int_a^b f(x) dx, \\
\int_a^b f(x) dx &= \int_a^c f(x) dx + \int_c^b f(x) dx \quad (\text{additivity}).
\end{aligned}$$

**Example 6.6.1: Using Additivity**

Given $\int_1^5 f(x) dx = 12$ and $\int_1^3 f(x) dx = 5$, find $\int_3^5 f(x) dx$.

Solution. By additivity, $\int_1^5 f dx = \int_1^3 f dx + \int_3^5 f dx$, so $\int_3^5 f(x) dx = 12 - 5 = 7$.

6.7 The Fundamental Theorem of Calculus and Definite Integrals

FTC Part 1 told us that accumulation functions are antiderivatives. FTC Part 2 turns this around into the single most useful computational tool in the course: it lets us evaluate a definite integral *without* ever taking a limit of Riemann sums, using only *any* antiderivative.

Theorem 6.7.1: Fundamental Theorem of Calculus, Part 2

If f is continuous on $[a, b]$ and F is *any* antiderivative of f (i.e. $F' = f$), then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Why any antiderivative works. If F and G are both antiderivatives of f , then $(F - G)' = f - f = 0$, so by the Increasing/Decreasing Function Theorem (Topic 5.3, applied in both directions) $F - G$ is constant. Hence $F(b) - F(a) = G(b) - G(a)$ for *any* two

antiderivatives — the difference $F(b) - F(a)$ does not depend on which antiderivative is used.

Example 6.7.1: Evaluating a Definite Integral Exactly

Evaluate $\int_0^4 (0.3x^2 + 1) dx$, and compare to the Riemann sum estimates of Example 6.2.1.

Solution. An antiderivative of $0.3x^2 + 1$ is $F(x) = 0.1x^3 + x$.

$$\int_0^4 (0.3x^2 + 1) dx = F(4) - F(0) = (6.4 + 4) - (0) = 10.4.$$

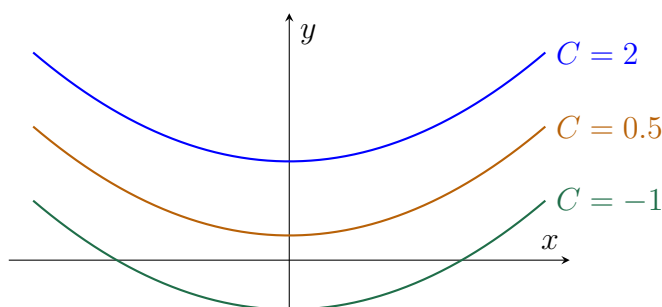
This lies between the left sum (8.2) and right sum (13.0) found earlier, as it must.

6.8 Finding Antiderivatives and Indefinite Integrals: Basic Rules and Notation

FTC Part 2 makes antiderivatives the central computational object of this unit. Since differentiation rules were built in Units 2–3, most basic antidifferentiation is simply reading those rules backward.

Definition 6.8.1: Antiderivative and Indefinite Integral

F is an **antiderivative** of f if $F' = f$. If F is one antiderivative, every antiderivative has the form $F(x) + C$ for a constant C (by the same argument used in Topic 6.7). The **indefinite integral** notation $\int f(x) dx = F(x) + C$ denotes this entire family.



Remark

All members of the family $\int x dx = \frac{1}{2}x^2 + C$ share the same derivative at every x (the vertical spacing between the curves is constant), which is exactly why they are all valid antiderivatives of the same function — this figure is the direct converse of Topic 4.6's picture of a single tangent line, now showing an entire stack of curves that all share one slope-field pattern (a preview of Unit 7).

Theorem 6.8.1: Basic Antiderivative Rules

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \quad \int \frac{1}{x} dx = \ln|x| + C, \quad \int e^x dx = e^x + C,$$

$$\int \sin x dx = -\cos x + C, \quad \int \cos x dx = \sin x + C, \quad \int \sec^2 x dx = \tan x + C.$$

Example 6.8.1: Basic Antidifferentiation

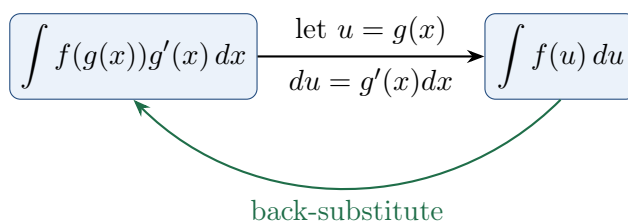
Find $\int \left(4x^3 - \frac{2}{x} + 3 \sin x\right) dx$.

Solution. Antidifferentiate term by term:

$$\int \left(4x^3 - \frac{2}{x} + 3 \sin x\right) dx = x^4 - 2 \ln|x| - 3 \cos x + C.$$

6.9 Integrating Using Substitution

Just as the chain rule was needed to differentiate composite functions, its reverse — ***u*-substitution** — is needed to antidifferentiate them.

**Theorem 6.9.1: *u*-Substitution**

If $u = g(x)$ is differentiable and f is continuous on the range of g , then

$$\int f(g(x)) g'(x) dx = \int f(u) du.$$

Example 6.9.1: Substitution in an Indefinite Integral

Find $\int 2x \cos(x^2) dx$.

Solution. Let $u = x^2$, so $du = 2x dx$ — which is exactly the remaining factor in the integrand:

$$\int 2x \cos(x^2) dx = \int \cos u du = \sin u + C = \sin(x^2) + C.$$

Example 6.9.2: Substitution in a Definite Integral

Evaluate $\int_0^2 x e^{x^2} dx$.

Solution. Let $u = x^2$, $du = 2x dx$, so $x dx = \frac{1}{2} du$. Converting the bounds: $x = 0 \Rightarrow u = 0$; $x = 2 \Rightarrow u = 4$.

$$\int_0^2 x e^{x^2} dx = \frac{1}{2} \int_0^4 e^u du = \frac{1}{2} [e^u]_0^4 = \frac{1}{2} (e^4 - 1).$$

6.9.1 Products of Trigonometric Powers

Integrands built from products like $\sin^m x \cos^n x$ or $\tan^m x \sec^n x$ look intimidating, but nearly all of them reduce to ordinary substitution once a single Pythagorean identity peels off exactly one factor to serve as du .

Theorem 6.9.2: Strategy for $\int \sin^m x \cos^n x dx$

- If m is **odd**, save one factor of $\sin x$ for du , and convert the remaining (even) power of sine to cosines using $\sin^2 x = 1 - \cos^2 x$; substitute $u = \cos x$.
- If n is **odd**, symmetrically save one factor of $\cos x$, convert the remaining even power of cosine to sines using $\cos^2 x = 1 - \sin^2 x$; substitute $u = \sin x$.
- If m and n are **both even**, no factor can be saved; instead apply the power-reduction identities from Topic 0.1, $\sin^2 x = \frac{1 - \cos 2x}{2}$ and $\cos^2 x = \frac{1 + \cos 2x}{2}$, to halve the total degree, and repeat if necessary.

Example 6.9.3: Odd Power of Sine

Find $\int \sin^3 x \cos^4 x dx$.

Solution. $m = 3$ is odd. Save one $\sin x$ and rewrite the rest: $\sin^3 x = \sin x(1 - \cos^2 x)$. Let $u = \cos x$, $du = -\sin x dx$:

$$\begin{aligned} \int \sin^3 x \cos^4 x dx &= \int (1 - \cos^2 x) \cos^4 x \sin x dx = - \int (1 - u^2) u^4 du = - \int (u^4 - u^6) du \\ &= -\frac{u^5}{5} + \frac{u^7}{7} + C = -\frac{\cos^5 x}{5} + \frac{\cos^7 x}{7} + C. \end{aligned}$$

Example 6.9.4: Both Powers Even

Find $\int \sin^2 x \cos^2 x dx$.

Solution. Neither power is odd, so use power reduction. Since $\sin x \cos x = \frac{1}{2} \sin 2x$, we have $\sin^2 x \cos^2 x = \frac{1}{4} \sin^2 2x$; applying power reduction once more to $\sin^2 2x$:

$$\int \sin^2 x \cos^2 x dx = \frac{1}{4} \int \sin^2 2x dx = \frac{1}{4} \int \frac{1 - \cos 4x}{2} dx = \frac{x}{8} - \frac{\sin 4x}{32} + C.$$

Theorem 6.9.3: Strategy for $\int \tan^m x \sec^n x dx$

- If n is **even**, save a factor of $\sec^2 x$ for du , and convert the remaining even power of secant to tangents using $\sec^2 x = 1 + \tan^2 x$; substitute $u = \tan x$.
- If m is **odd**, save one factor of $\sec x \tan x$ for du , and convert the remaining even power of tangent to secants using $\tan^2 x = \sec^2 x - 1$; substitute $u = \sec x$.

Example 6.9.5: Even Power of Secant

Find $\int \tan^4 x \sec^2 x dx$.

Solution. $n = 2$ is even; let $u = \tan x$, $du = \sec^2 x dx$ directly:

$$\int \tan^4 x \sec^2 x dx = \int u^4 du = \frac{u^5}{5} + C = \frac{\tan^5 x}{5} + C.$$

Example 6.9.6: Odd Power of Tangent

Find $\int \tan^3 x \sec^3 x dx$.

Solution. $m = 3$ is odd. Write $\tan^3 x \sec^3 x = \tan^2 x \sec^2 x \cdot (\sec x \tan x) = (\sec^2 x - 1) \sec^2 x \cdot (\sec x \tan x)$. Let $u = \sec x$, $du = \sec x \tan x dx$:

$$\int \tan^3 x \sec^3 x dx = \int (u^2 - 1)u^2 du = \frac{u^5}{5} - \frac{u^3}{3} + C = \frac{\sec^5 x}{5} - \frac{\sec^3 x}{3} + C.$$

6.9.2 Trigonometric Substitution

Some integrands contain $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$ in a way that no algebraic substitution removes. Because each expression matches a Pythagorean identity, substituting x itself as a trigonometric function of a new variable θ eliminates the radical entirely.

Theorem 6.9.4: Trigonometric Substitutions

Radical form	Substitution	Result
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$\sqrt{a^2 - x^2} = a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$\sqrt{a^2 + x^2} = a \sec \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sqrt{x^2 - a^2} = a \tan \theta$

After integrating in θ , a right-triangle diagram (built from the substitution equation) converts the answer back to x .

Example 6.9.7: Trigonometric Substitution with $\sqrt{a^2 - x^2}$

Find $\int \sqrt{9 - x^2} dx$.

Solution. Let $x = 3 \sin \theta$, $dx = 3 \cos \theta d\theta$, so $\sqrt{9 - x^2} = 3 \cos \theta$:

$$\int \sqrt{9 - x^2} dx = \int 3 \cos \theta \cdot 3 \cos \theta d\theta = 9 \int \cos^2 \theta d\theta = 9 \left(\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right) + C.$$

Since $\sin 2\theta = 2 \sin \theta \cos \theta$, this is $\frac{9}{2}\theta + \frac{9}{2} \sin \theta \cos \theta + C$. With $\sin \theta = \frac{x}{3}$, $\cos \theta = \frac{\sqrt{9-x^2}}{3}$, and $\theta = \arcsin \frac{x}{3}$:

$$\int \sqrt{9 - x^2} dx = \frac{9}{2} \arcsin\left(\frac{x}{3}\right) + \frac{x\sqrt{9 - x^2}}{2} + C.$$

Example 6.9.8: Trigonometric Substitution with $\sqrt{a^2 + x^2}$

Find $\int \frac{dx}{(x^2 + 4)^{3/2}}$.

Solution. Let $x = 2 \tan \theta$, $dx = 2 \sec^2 \theta d\theta$, so $x^2 + 4 = 4 \sec^2 \theta$ and $(x^2 + 4)^{3/2} = 8 \sec^3 \theta$:

$$\int \frac{dx}{(x^2 + 4)^{3/2}} = \int \frac{2 \sec^2 \theta d\theta}{8 \sec^3 \theta} = \frac{1}{4} \int \cos \theta d\theta = \frac{1}{4} \sin \theta + C.$$

Since $\tan \theta = \frac{x}{2}$, a right triangle with opposite x and adjacent 2 has hypotenuse $\sqrt{x^2 + 4}$, so $\sin \theta = \frac{x}{\sqrt{x^2 + 4}}$:

$$\int \frac{dx}{(x^2 + 4)^{3/2}} = \frac{x}{4\sqrt{x^2 + 4}} + C.$$

6.9.3 Antiderivatives Involving Inverse Trigonometric Functions

Reversing Topic 3.4's inverse trigonometric derivatives produces three recognizable integral patterns — spotting them (sometimes after a small algebraic adjustment) avoids trigonometric substitution altogether.

Theorem 6.9.5: Inverse Trigonometric Antiderivative Forms

$$\begin{aligned} \int \frac{dx}{\sqrt{a^2 - x^2}} &= \arcsin\left(\frac{x}{a}\right) + C, & \int \frac{dx}{a^2 + x^2} &= \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \\ \int \frac{dx}{x\sqrt{x^2 - a^2}} &= \frac{1}{a} \operatorname{arcsec}\left(\frac{|x|}{a}\right) + C. \end{aligned}$$

Example 6.9.9: Direct Pattern Matching

Find $\int \frac{dx}{\sqrt{16 - x^2}}$ and $\int \frac{dx}{x^2 + 9}$.

Solution. These match the patterns above with $a = 4$ and $a = 3$ respectively:

$$\int \frac{dx}{\sqrt{16-x^2}} = \arcsin\left(\frac{x}{4}\right) + C, \quad \int \frac{dx}{x^2+9} = \frac{1}{3} \arctan\left(\frac{x}{3}\right) + C.$$

Example 6.9.10: Pattern Matching After a Substitution

Find $\int \frac{dx}{\sqrt{4-9x^2}}$.

Solution. The coefficient on x^2 prevents direct matching. Let $u = 3x$, $du = 3 dx$:

$$\int \frac{dx}{\sqrt{4-9x^2}} = \int \frac{\frac{1}{3} du}{\sqrt{4-u^2}} = \frac{1}{3} \arcsin\left(\frac{u}{2}\right) + C = \frac{1}{3} \arcsin\left(\frac{3x}{2}\right) + C.$$

(Check: differentiating the right side with the chain rule returns $1/\sqrt{4-9x^2}$, confirming the antiderivative.)

6.10 Integrating Functions Using Long Division and Completing the Square

Not every rational integrand is ready for substitution. Two algebraic preprocessing techniques from precalculus — long division and completing the square — often convert an unfamiliar integrand into one built from Topic 6.8's basic rules.

Example 6.10.1: Long Division Before Integrating

Find $\int \frac{x^2+1}{x-1} dx$.

Solution. The integrand is an improper rational expression; dividing first:

$$\frac{x^2+1}{x-1} = x+1 + \frac{2}{x-1}.$$

Now each term is directly integrable:

$$\int \left(x+1 + \frac{2}{x-1}\right) dx = \frac{x^2}{2} + x + 2 \ln|x-1| + C.$$

Example 6.10.2: Completing the Square Before Integrating

Find $\int \frac{dx}{x^2+6x+13}$.

Solution. Complete the square in the denominator: $x^2+6x+13 = (x+3)^2+4$. This matches the arctangent pattern $\int \frac{du}{u^2+a^2} = \frac{1}{a} \arctan \frac{u}{a} + C$ (Topic 3.4, reversed)

with $u = x + 3$, $a = 2$:

$$\int \frac{dx}{(x+3)^2 + 4} = \frac{1}{2} \arctan\left(\frac{x+3}{2}\right) + C.$$

6.11 Integrating Using Integration by Parts

BC Only

Topic 6.11 is assessed on the BC Exam only.

Substitution reverses the chain rule; **integration by parts** reverses the product rule (Topic 2.8).

Theorem 6.11.1: Integration by Parts

If u and v are differentiable functions of x ,

$$\int u \, dv = uv - \int v \, du.$$

Justification. By the product rule, $(uv)' = u'v + uv'$, i.e. $uv' = (uv)' - u'v$. Integrating both sides and writing $dv = v'dx$, $du = u'dx$: $\int u \, dv = uv - \int v \, du$.

AP Exam Tip

Choose u to be the factor that *simplifies* upon differentiation (polynomials are ideal), and dv to be the factor that is easy to antidifferentiate. The mnemonic “LIPET” (Logarithmic, Inverse trigonometric, Polynomial, Exponential, Trigonometric) ranks typical priority for choosing u .

Example 6.11.1: Integration by Parts

Find $\int x e^x \, dx$.

Solution. Let $u = x$ (simplifies to $du = dx$) and $dv = e^x \, dx$ (so $v = e^x$):

$$\int x e^x \, dx = x e^x - \int e^x \, dx = x e^x - e^x + C.$$

6.12 Integrating Using Linear Partial Fractions

BC Only

Topic 6.12 is assessed on the BC Exam only.

When a rational integrand's denominator factors into distinct linear factors, it can be split into simpler fractions, each directly integrable using the $\ln|\cdot|$ rule.

Theorem 6.12.1: Partial Fraction Decomposition (Distinct Linear Factors)

If $\deg(\text{numerator}) < \deg(\text{denominator})$ and the denominator factors as $(x - r_1)(x - r_2)$ with $r_1 \neq r_2$, then

$$\frac{P(x)}{(x - r_1)(x - r_2)} = \frac{A}{x - r_1} + \frac{B}{x - r_2}$$

for constants A, B determined by clearing denominators and matching coefficients (or substituting convenient values of x).

Example 6.12.1: Partial Fractions

Find $\int \frac{5x - 1}{(x - 1)(x + 2)} dx$.

Solution. Write $\frac{5x - 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$. Clearing denominators: $5x - 1 = A(x + 2) + B(x - 1)$. Setting $x = 1$: $4 = 3A \Rightarrow A = \frac{4}{3}$. Setting $x = -2$: $-11 = -3B \Rightarrow B = \frac{11}{3}$. So

$$\int \frac{5x - 1}{(x - 1)(x + 2)} dx = \frac{4}{3} \ln|x - 1| + \frac{11}{3} \ln|x + 2| + C.$$

6.13 Evaluating Improper Integrals

BC Only

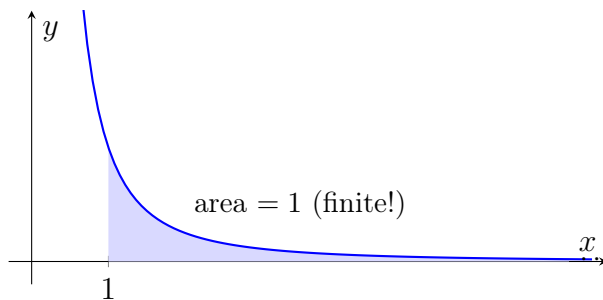
Topic 6.13 is assessed on the BC Exam only.

The definite integral was built from Riemann sums over a *finite* interval with a *continuous, bounded* integrand. An **improper integral** violates one of those conditions — an infinite interval, or an unbounded integrand — and is defined as a limit of proper integrals.

Definition 6.13.1: Improper Integral over an Infinite Interval

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

If this limit exists (is finite), the improper integral **converges**; otherwise it **diverges**.

**Example 6.13.1: A Convergent Improper Integral**

Evaluate $\int_1^\infty \frac{1}{x^2} dx$.

Solution.

$$\int_1^\infty \frac{dx}{x^2} = \lim_{b \rightarrow \infty} \int_1^b x^{-2} dx = \lim_{b \rightarrow \infty} \left[-x^{-1} \right]_1^b = \lim_{b \rightarrow \infty} \left(1 - \frac{1}{b} \right) = 1.$$

The region under $y = 1/x^2$ from $x = 1$ onward extends infinitely far right, yet encloses a *finite* area — exactly 1 — because the curve decays fast enough.

Example 6.13.2: A Divergent Improper Integral

Evaluate $\int_1^\infty \frac{1}{x} dx$.

Solution.

$$\int_1^\infty \frac{dx}{x} = \lim_{b \rightarrow \infty} \left[\ln x \right]_1^b = \lim_{b \rightarrow \infty} \ln b = \infty.$$

The integral diverges: $1/x$ decays too slowly for the area to stay finite (this exact contrast — $1/x^2$ converges, $1/x$ diverges — reappears in Unit 10 as the p -series test, made precise below).

6.13.1 Improper Integrals with a Discontinuous Integrand

The infinite interval is not the only way an integral can fail to be “proper.” If f is unbounded near a point in $[a, b]$ — typically a vertical asymptote — the same limiting idea applies, now taken as the troublesome endpoint is approached rather than as $b \rightarrow \infty$.

Definition 6.13.2: Improper Integral with an Infinite Discontinuity

If f is continuous on $[a, b)$ with $\lim_{x \rightarrow b^-} f(x) = \pm\infty$, then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx,$$

provided the limit exists. (If the discontinuity is instead at $x = a$, take $\lim_{t \rightarrow a^+} \int_t^b f(x) dx$ instead.)

Example 6.13.3: A Convergent Improper Integral with a Discontinuous Integrand

Evaluate $\int_0^1 \frac{dx}{\sqrt{x}}$.

Solution. $f(x) = x^{-1/2}$ is unbounded as $x \rightarrow 0^+$, so

$$\int_0^1 x^{-1/2} dx = \lim_{t \rightarrow 0^+} \int_t^1 x^{-1/2} dx = \lim_{t \rightarrow 0^+} \left[2\sqrt{x} \right]_t^1 = \lim_{t \rightarrow 0^+} (2 - 2\sqrt{t}) = 2.$$

Even though the graph shoots up without bound as $x \rightarrow 0^+$, the region still encloses a finite area of 2.

Example 6.13.4: A Divergent Improper Integral with a Discontinuous Integrand

Evaluate $\int_0^1 \frac{dx}{x}$.

Solution.

$$\int_0^1 \frac{dx}{x} = \lim_{t \rightarrow 0^+} \left[\ln x \right]_t^1 = \lim_{t \rightarrow 0^+} (0 - \ln t) = \infty.$$

The integral diverges. Contrast this with the previous example: both $1/\sqrt{x}$ and $1/x$ blow up at $x = 0$, but only $1/\sqrt{x}$ does so slowly enough for the enclosed area to stay finite.

6.13.2 The p -Integral: A Convergence Test We Can Prove Completely

The last four examples are not isolated curiosities: $\int_1^\infty x^{-p} dx$ has a complete answer for every real p , and it is worth proving in full — this exact result becomes the p -series test in Unit 10.

Theorem 6.13.1: Convergence of the p -Integral

$$\int_1^{\infty} \frac{dx}{x^p} \text{ converges if } p > 1, \quad \text{and diverges if } p \leq 1,$$

$$\text{and when it converges, } \int_1^{\infty} x^{-p} dx = \frac{1}{p-1}.$$

Proof. The case $p = 1$ is exactly Example 6.13.2: the integral diverges. Now let $p \neq 1$. Then

$$\int_1^b x^{-p} dx = \left[\frac{x^{1-p}}{1-p} \right]_1^b = \frac{b^{1-p} - 1}{1-p}.$$

To evaluate $\lim_{b \rightarrow \infty} b^{1-p}$, write $b^{1-p} = e^{(1-p)\ln b}$. Since $\ln b \rightarrow \infty$ as $b \rightarrow \infty$, the *sign* of the exponent $1-p$ entirely controls this limit:

- If $p > 1$, then $1-p < 0$, so $(1-p)\ln b \rightarrow -\infty$, giving $b^{1-p} = e^{(1-p)\ln b} \rightarrow 0$. Then

$$\lim_{b \rightarrow \infty} \frac{b^{1-p} - 1}{1-p} = \frac{0 - 1}{1-p} = \frac{1}{p-1},$$

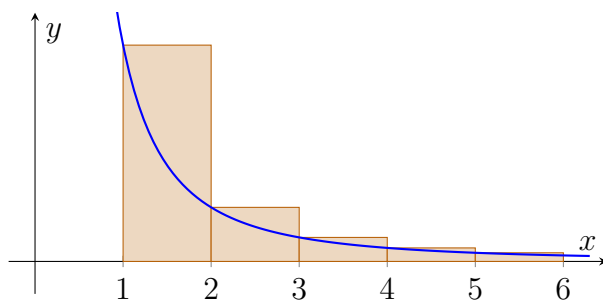
a finite value — the integral **converges** to $\frac{1}{p-1}$.

- If $p < 1$, then $1-p > 0$, so $(1-p)\ln b \rightarrow +\infty$, giving $b^{1-p} \rightarrow \infty$. Then $\frac{b^{1-p} - 1}{1-p} \rightarrow \infty$ as well (an unboundedly growing numerator divided by the fixed positive constant $1-p$) — the integral **diverges**.

Combining this with the $p = 1$ case: the integral converges precisely when $p > 1$. ■

Remark

This is exactly why Example 6.13.1 ($p = 2$) converged, to $\frac{1}{2-1} = 1$ as computed directly, while Example 6.13.2 ($p = 1$) diverged — both are special cases of the theorem just proved.

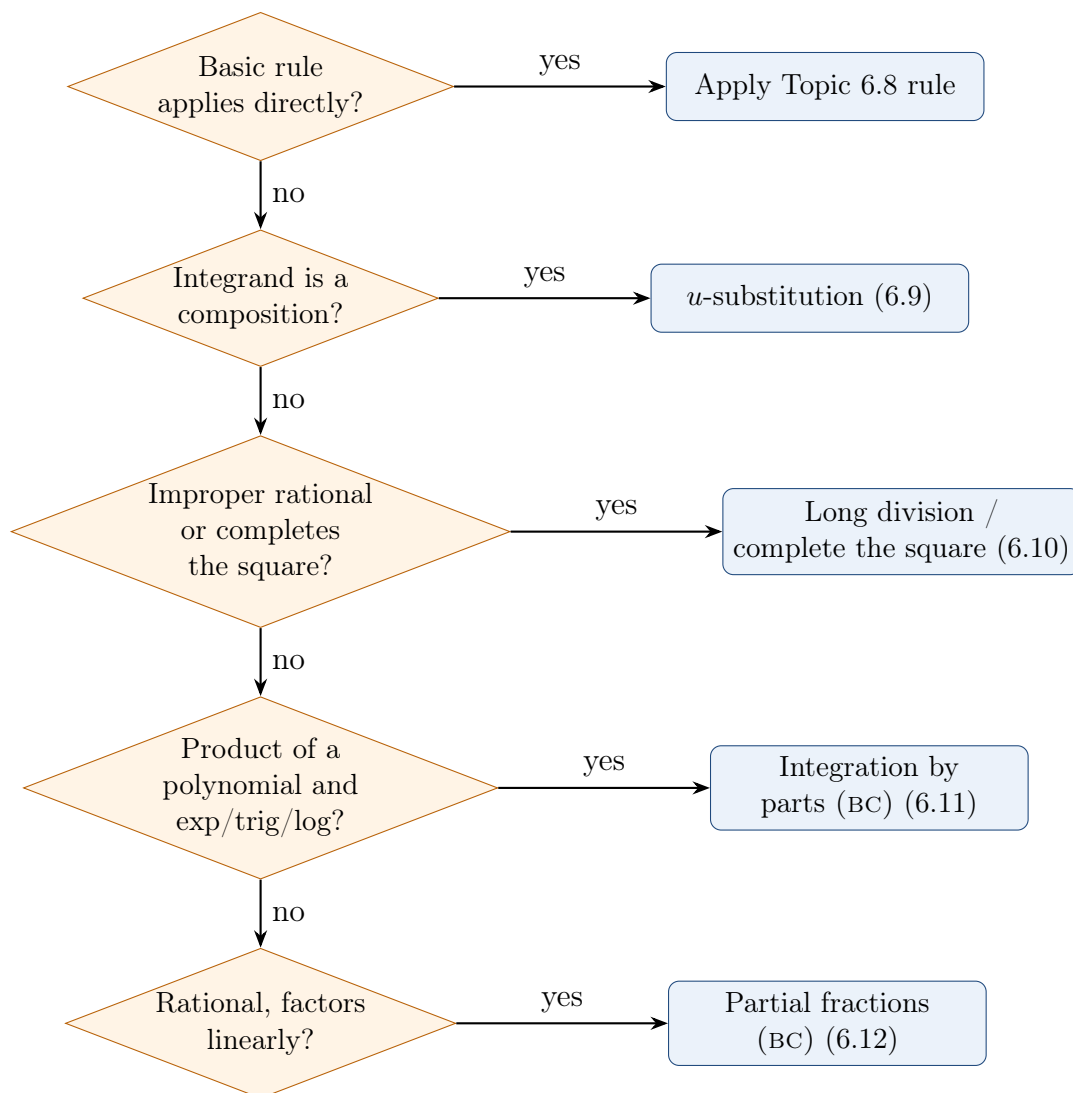
**Looking Ahead to Unit 10**

The picture above is exactly a **left Riemann sum** of the decreasing function $f(x) = 1/x^2$ — and we already know (Topic 6.2) that a left sum of a decreasing function *overestimates* the true area, so $\sum_{n=1}^5 f(n) \geq \int_1^6 f(x) dx$, with the same pattern holding as the picture extends indefinitely to the right. This is the entire

idea behind the **Integral Test** (Topic 10.4): because rectangles built from a positive, decreasing f trap its values against $\int_1^\infty f(x) dx$, the infinite series $\sum_{n=1}^\infty f(n)$ and the improper integral $\int_1^\infty f(x) dx$ must converge or diverge together. Applied to $f(x) = 1/x^p$, this is precisely how Topic 10.5 will prove that the p -series $\sum_{n=1}^\infty \frac{1}{n^p}$ converges if and only if $p > 1$ — the same threshold established above, one unit early.

6.14 Selecting Techniques for Antidifferentiation

With every technique now available, the practical skill is choosing among them quickly — the same kind of decision fluency Topics 1.7 and 3.5 built for limits and derivatives.



Remark

As with limits (Topic 1.7) and derivatives (Topic 3.5), this chart reflects the *most common* first move, not a rigid algorithm — some integrands need more than one technique in sequence (e.g. a substitution that produces a new integral requiring integration by parts).

6.14.1 Summary: A Complete Table of Common Antiderivatives

Every antiderivative introduced in this unit — whether read directly off a derivative rule from Units 2–3 or produced by one of Topics 6.9–6.13’s techniques — is collected here for quick reference.

Power and Rational Forms

$$\int x^n dx \quad \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} dx \quad \ln |x| + C$$

Exponential and Logarithmic

$$\int e^x dx \quad e^x + C$$

$$\int a^x dx \quad \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1)$$

Trigonometric

$$\int \sin x dx \quad -\cos x + C$$

$$\int \cos x dx \quad \sin x + C$$

$$\int \sec^2 x dx \quad \tan x + C$$

$$\int \csc^2 x dx \quad -\cot x + C$$

$$\int \sec x \tan x dx \quad \sec x + C$$

$$\int \csc x \cot x dx \quad -\csc x + C$$

$$\int \tan x dx \quad \ln |\sec x| + C$$

$$\int \cot x dx \quad \ln |\sin x| + C$$

$$\int \sec x dx \quad \ln |\sec x + \tan x| + C$$

$$\int \csc x dx \quad -\ln |\csc x + \cot x| + C$$

Inverse Trigonometric Forms

$$\int \frac{dx}{\sqrt{a^2 - x^2}} \quad \arcsin\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{a^2 + x^2} \quad \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{x\sqrt{x^2 - a^2}} \quad \frac{1}{a} \operatorname{arcsec}\left(\frac{|x|}{a}\right) + C$$

Remark

Every trigonometric entry traces back to a derivative already established: the first six are the Topic 2.7/2.10 derivatives read in reverse; $\tan x$ and $\cot x$ follow from a one-step u -substitution ($u = \cos x$ or $u = \sin x$); and $\sec x$'s antiderivative follows from the classic (if unmotivated-looking) trick of multiplying the integrand by

$\frac{\sec x + \tan x}{\sec x + \tan x}$, which turns the numerator into exactly du for $u = \sec x + \tan x$:

$$\int \sec x \, dx = \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} \, dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx = \int \frac{du}{u} = \ln |u| + C.$$

Unit 6 established what a definite integral *is* (a limit of Riemann sums), what it *means* (accumulated, signed area), and how to *compute* it (antidifferentiate and apply FTC Part 2). Unit 7 puts antiderivatives to work solving equations that describe rates directly — differential equations — before Unit 8 returns to the geometric applications of integration itself: areas, volumes, and arc length.

UNIT 7

Differential Equations

Class Periods	~8–9 (AB)	~9–10 (BC)
Big Idea	Function Analysis (FUN)	

Every equation solved so far has had a function on one side and an explicit formula on the other: $f(x) = x^2 \sin x$, $g(x) = \ln(x^2 + 1)$, and so on. A **differential equation** looks different — it is an equation relating a function to its own derivative, such as $\frac{dy}{dx} = 2y$. No explicit formula for y appears anywhere; instead, the equation describes a *rule* that y 's rate of change must obey, and the task is to recover y itself from that rule. This is where the two halves of calculus meet in the most direct way possible: antidifferentiation, the reverse of differentiation built in Unit 6, is exactly the tool needed to undo a derivative and solve for the unknown function. Along the way, a powerful new picture emerges — the **slope field** — that lets us visualize every solution to a differential equation simultaneously, often before doing any algebra at all.

7.1 Modeling Situations with Differential Equations

Word problems throughout this course have often stated a rate directly: “the population grows at a rate proportional to its size,” “the radius shrinks at a rate proportional to the surface area,” “the temperature changes at a rate proportional to the difference between the object’s temperature and the room’s.” Each of these sentences is really a compact description of a derivative — it says how fast a quantity changes, in terms of the quantity’s own current value. Translating such a sentence into symbols produces a **differential equation**, and learning to read this translation fluently is the first skill this unit builds.

Definition 7.1.1: Differential Equation and Order

A **differential equation** is an equation involving an unknown function and one or more of its derivatives. The **order** of a differential equation is the order of the highest derivative that appears in it. An equation such as

$$\frac{dy}{dx} = ky$$

is a **first-order** differential equation, since only a first derivative of y appears. Every differential equation in this course is first-order.

Remark

Compare this to every prior unit: given a formula for y , Units 2–3 computed $\frac{dy}{dx}$.

A differential equation reverses the question — given a rule for $\frac{dy}{dx}$ in terms of x and/or y , the goal is to recover y . Because the unknown y can appear on *both* sides of the relationship (as in $\frac{dy}{dx} = ky$, where y appears on the right), differential equations describe families of functions more naturally than they describe a single explicit formula.

The translation step — verbal statement to symbolic equation — is worth practicing on its own, independent of solving anything.

Example 7.1.1: Translating a Verbal Statement

Translate: “The rate of change of a quantity y with respect to time t is proportional to y .”

Solution. “Proportional to y ” means the rate equals a constant multiple of y . Writing k for that constant of proportionality,

$$\frac{dy}{dt} = ky.$$

Example 7.1.2: Translating a Geometric Statement

Translate: “At every point (x, y) on a curve, the slope of the tangent line equals the product of the coordinates.”

Solution. The slope of the tangent line at (x, y) is $\frac{dy}{dx}$, so

$$\frac{dy}{dx} = xy.$$

Example 7.1.3: Translating a Physical Law

Newton’s Law of Cooling states that a warm object in a cooler room loses heat at a rate proportional to the difference between its own temperature T and the (constant) ambient temperature T_s . Translate this into a differential equation.

Solution. “Rate proportional to the difference” gives

$$\frac{dT}{dt} = k(T - T_s),$$

for some constant k . Since a warm object *cools* (its temperature *decreases*) when $T > T_s$, the right-hand side must be negative in that case, which forces $k < 0$. This model reappears with concrete numbers in Topic 7.8.

AP Exam Tip

Watch for the phrase “proportional to.” It always introduces an unknown constant k multiplying whatever quantity follows it, never an explicit numerical rate. “Proportional to the difference between A and B ” means $k(A - B)$ — keep the order of subtraction consistent with which quantity is decreasing.

Not every differential equation can be solved by hand, and this unit will not attempt to solve most of them. What it *will* do is build, topic by topic, a complete toolkit for the well-behaved case that the AP Exam tests: confirming that a proposed function is a solution (Topic 7.2), visualizing every solution at once with a slope field (Topics 7.3–7.4), approximating a solution numerically when no closed form is available (Topic 7.5, BC), and finally solving explicitly using antidifferentiation (Topics 7.6–7.9).

7.2 Verifying Solutions for Differential Equations

Before solving a differential equation, it is worth understanding precisely what counts as an answer. A single differential equation is typically satisfied by an entire *family* of functions, not one function alone — and checking whether a specific candidate belongs to that family requires nothing more than differentiation and substitution.

Definition 7.2.1: Solution to a Differential Equation

A function $y = f(x)$ is a **solution** to a differential equation on an interval I if, after substituting $f(x)$ for y and $f'(x)$ (and any higher derivatives) for the corresponding derivatives, the equation becomes an identity for every x in I .

Remark

There may be infinitely many solutions to a single differential equation. For instance, $y = Ce^{2x}$ solves $\frac{dy}{dx} = 2y$ for *every* choice of the constant C — Example 7.2.1 below confirms this. This entire family is called the equation’s **general solution**; Topic 7.7 shows how an initial condition singles out one member of the family, called a **particular solution**.

Verifying a proposed solution is mechanical: differentiate the candidate, substitute both the candidate and its derivative into the differential equation, and confirm both sides match.

Example 7.2.1: Verifying a Family of Solutions

Show that $y = Ce^{2x}$ solves $\frac{dy}{dx} = 2y$ for every constant C .

Solution. Differentiating, $\frac{dy}{dx} = 2Ce^{2x}$. Comparing to $2y = 2(Ce^{2x}) = 2Ce^{2x}$: the two sides match for every C , so $y = Ce^{2x}$ is a solution for any choice of C .

Example 7.2.2: Verifying an Implicitly Defined Solution

Show that $x^2 + y^2 = 25$ implicitly defines a solution to $y \frac{dy}{dx} = -x$.

Solution. Differentiate both sides of $x^2 + y^2 = 25$ implicitly with respect to x (Topic 3.2):

$$2x + 2y \frac{dy}{dx} = 0 \implies y \frac{dy}{dx} = -x,$$

which is exactly the given differential equation. Every point on the circle $x^2 + y^2 = 25$ (away from $y = 0$, where the tangent is vertical) satisfies the equation.

Example 7.2.3: Verifying a Nonlinear Solution

Show that $y = \frac{1}{1-x}$ solves $\frac{dy}{dx} = y^2$.

Solution. Differentiating with the power/chain rule, $\frac{dy}{dx} = \frac{1}{(1-x)^2}$. Squaring the proposed solution, $y^2 = \frac{1}{(1-x)^2}$. The two expressions agree, so y solves the equation (on any interval avoiding $x = 1$, where the function itself is undefined).

The next two topics turn away from checking individual candidates and toward the more geometric question: without solving anything at all, what can the differential equation itself reveal about the shape of its solutions?

7.3 Sketching Slope Fields

A first-order differential equation $\frac{dy}{dx} = f(x, y)$ makes a remarkably strong claim: it specifies the slope of *every* solution curve at *every* point of the plane, before a single solution has been found. At the point $(1, 2)$, for instance, any solution curve that happens to pass through $(1, 2)$ must have slope exactly $f(1, 2)$ there — this is true regardless of which particular solution is being examined. Plotting a short line segment of the correct slope at many grid points turns this observation into a picture.

Definition 7.3.1: Slope Field

The **slope field** (or **direction field**) of the differential equation $\frac{dy}{dx} = f(x, y)$ is the collection of short line segments, one centered at each point (x, y) of a grid, each drawn with slope $f(x, y)$. A solution curve of the differential equation is tangent to the slope-field segment at every point it passes through.

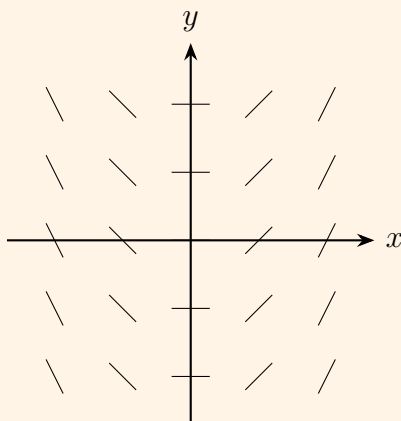
Constructing a slope field by hand is a matter of evaluating $f(x, y)$ at each grid point and drawing a short segment of that slope — tedious for a dense grid, but very fast for a coarse one.

Example 7.3.1: Building a Slope Field, $f(x, y) = x$

Sketch the slope field for $\frac{dy}{dx} = x$ on the grid $x, y \in \{-2, -1, 0, 1, 2\}$.

Solution. Here f does not depend on y at all, so every segment in a given *column* (fixed x) has the same slope, regardless of height:

x	-2	-1	0	1	2
slope = x	-2	-1	0	1	2



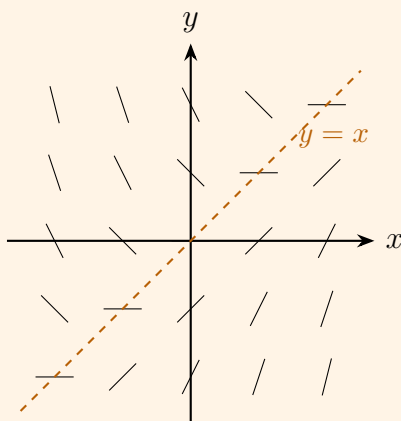
Every column is a stack of parallel segments — the picture already hints that solutions are horizontal shifts of a single shape, which matches the exact solution family $y = \frac{1}{2}x^2 + C$ found in Topic 7.6.

Example 7.3.2: Building a Slope Field, $f(x, y) = x - y$

Sketch the slope field for $\frac{dy}{dx} = x - y$ on the same grid.

Solution. Now the slope depends on both coordinates.

(x, y)	$(0, 0)$	$(1, 0)$	$(0, 1)$	$(2, 1)$	$(-1, 1)$
slope = $x - y$	0	1	-1	1	-2



Notice the diagonal line $y = x$, where every segment is flat (slope 0) — this is exactly where the defining expression $x - y$ vanishes. Every segment above that line points downward-left; every segment below it points upward-right. This kind of structural observation, made directly from the field, is the subject of Topic 7.4.

Remark

A slope field never needs the differential equation to be solvable in closed form — it can be constructed for *any* $f(x, y)$, purely by evaluation. This is exactly why it is so useful: it is a picture of the solutions that exists even when an explicit formula for them does not.

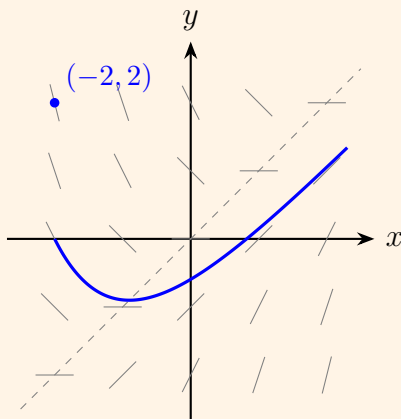
7.4 Reasoning Using Slope Fields

A slope field is only useful if it can be *read*. Given a slope field and a single starting point, a solution curve through that point can be sketched freehand by following the tangent directions continuously from segment to segment — the same way a hiker might follow a trail of arrows without ever seeing the destination in advance.

Example 7.4.1: Sketching a Solution Curve from a Slope Field

Using the slope field for $\frac{dy}{dx} = x - y$ from Example 7.3.2, sketch the solution curve passing through $(-2, 2)$.

Solution. Starting at $(-2, 2)$ (where the slope is $x - y = -2 - 2 = -4$, a steep downward direction), follow the local slope a short distance, re-evaluate, and continue. The curve descends steeply at first, flattens out as it approaches the line $y = x$ (where slopes are zero), and settles into the diagonal:



The curve shown is $y = x - 1 + 3e^{-x-2}$, the exact solution through $(-2, 2)$ (Topic 7.6 supplies the technique to find it algebraically); notice how closely a freehand sketch following the field would match it.

Slope fields also make it possible to *identify* a differential equation from its picture, by checking a few structural clues rather than plotting every point.

Example 7.4.2: Reading Structural Clues from a Field

Explain how to tell, from its slope field alone, that a differential equation has the form $\frac{dy}{dx} = ky$ for some constant $k > 0$ (exponential growth, Topic 7.8) rather than $\frac{dy}{dx} = x$.

Solution. For $\frac{dy}{dx} = ky$: along the horizontal line $y = 0$, every slope is $k(0) = 0$ — segments are flat only on that one line, and slopes grow steeper moving *vertically* away from it, symmetric under reflection through the origin (since $-y$ gives the negative slope). For $\frac{dy}{dx} = x$: slopes are constant along *vertical* lines and are completely independent of height, so the field looks identical when shifted up or down. Checking which direction (horizontal shifts vs. vertical shifts) leaves the field unchanged is often the fastest way to identify the dependency structure of $f(x, y)$.

Remark

A slope field can also reveal a **constant (equilibrium) solution** directly: any horizontal line $y = c$ along which every segment is flat corresponds to the constant solution $y = c$, since a horizontal line has slope 0 everywhere and the field confirms $f(x, c) = 0$. Topic 7.9's logistic model relies on exactly this observation.

7.5 Approximating Solutions Using Euler's Method

BC Only

Topic 7.5 is assessed on the BC Exam only.

Following a slope field freehand produces a qualitatively correct sketch, but not a number. **Euler's method** turns the same idea — walk in the direction the local slope indicates — into a precise numerical algorithm: starting from a known point on the solution curve, take a small step of length h in the x -direction along the *tangent line* at the current point, arrive at a new (approximate) point, recompute the slope there, and repeat.

Definition 7.5.1: Euler's Method

Given $\frac{dy}{dx} = f(x, y)$ with starting point (x_0, y_0) and step size h , Euler's method generates a sequence of approximations by

$$x_{n+1} = x_n + h, \quad y_{n+1} = y_n + h \cdot f(x_n, y_n).$$

Each (x_n, y_n) approximates a point on the true solution curve through (x_0, y_0) .

Remark

Euler's method is nothing more than repeated linear approximation (Topic 4.1, tangent line approximation): each step assumes the solution curve is well-approximated by its own tangent line over a short horizontal distance h , exactly as $L(x) = f(a) + f'(a)(x - a)$ approximated a function locally. Because the true curve is not perfectly straight, each step introduces a small error, and these errors accumulate — the approximation drifts further from the true solution the more steps are taken, and improves as h shrinks.

Example 7.5.1: Applying Euler's Method

Use Euler's method with step size $h = 0.5$ to estimate $y(1)$ for the solution of $\frac{dy}{dx} = x + y$, $y(0) = 1$.

Solution. Two steps of size 0.5 reach $x = 1$ from $x = 0$.

n	(x_n, y_n)	$f(x_n, y_n) = x_n + y_n$	$y_{n+1} = y_n + 0.5f(x_n, y_n)$
0	(0, 1)	1	$1 + 0.5(1) = 1.5$
1	(0.5, 1.5)	2	$1.5 + 0.5(2) = 2.5$

So $y(1) \approx 2.5$.

The linear differential equation in Example 7.5.1 happens to be exactly solvable (Topic 7.6's techniques do not directly apply since it is not separable, but the integrating-factor method from a differential equations course gives $y = 2e^x - x - 1$), so the Euler estimate can be checked against the truth: $y(1) = 2e - 2 \approx 3.437$, versus the Euler estimate of 2.5 — a sizeable error, because the curve bends upward much faster than its tangent line predicts.

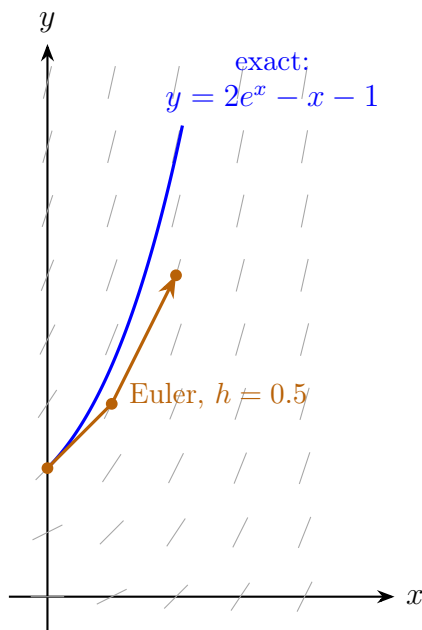
Example 7.5.2: Improving Accuracy with a Smaller Step

Repeat Example 7.5.1 with $h = 0.25$.

Solution. Four steps of size 0.25 reach $x = 1$.

n	x_n	y_n	$f(x_n, y_n)$	y_{n+1}
0	0	1	1	1.25
1	0.25	1.25	1.5	1.625
2	0.5	1.625	2.125	2.15625
3	0.75	2.15625	2.90625	2.88281

So $y(1) \approx 2.883$ — closer to the true value 3.437 than the $h = 0.5$ estimate of 2.5 was. Halving the step size roughly halved the error (from about 0.94 down to about 0.55), the hallmark of Euler's method's accuracy improving proportionally to h .



AP Exam Tip

On the AP Exam, Euler's-method questions are almost always scaffolded as a table to fill in one row at a time. Compute $f(x_n, y_n)$ first, then y_{n+1} , then increment x — organizing the work in a table like the ones above avoids arithmetic slips and matches how the answer is scored.

7.6 Finding General Solutions Using Separation of Variables

Euler's method approximates a solution numerically without ever finding a formula. When the differential equation has a special algebraic structure, however, an exact formula *can* be found — using nothing more than the antidifferentiation techniques already built in Unit 6.

Definition 7.6.1: Separable Differential Equation

A first-order differential equation is **separable** if it can be written as

$$\frac{dy}{dx} = f(x)g(y),$$

a product of a function of x alone and a function of y alone.

Theorem 7.6.1: Solving by Separation of Variables

To solve a separable equation $\frac{dy}{dx} = f(x)g(y)$ (where $g(y) \neq 0$), collect all y -terms with dy and all x -terms with dx , then antidifferentiate both sides:

$$\frac{dy}{g(y)} = f(x) dx \quad \implies \quad \int \frac{dy}{g(y)} = \int f(x) dx,$$

and add a single constant of integration to either side.

Remark

Both integrals technically produce their own constant, but the two constants can always be combined into one by subtracting: writing $F(y) + C_1 = G(x) + C_2$ is the same statement as $F(y) = G(x) + C$ with $C = C_2 - C_1$. Only one arbitrary constant ever needs to be written down.

Formally, treating $\frac{dy}{dx}$ as a ratio of infinitesimals dy and dx (rather than as a single symbol for the derivative) is not fully rigorous, but it is justified by the chain rule: if $\frac{dy}{dx} = f(x)g(y)$, dividing by $g(y)$ and antidifferentiating both sides with respect to x gives $\int \frac{1}{g(y)} \frac{dy}{dx} dx = \int f(x) dx$, and the left-hand side is exactly $\int \frac{dy}{g(y)}$ by u -substitution with $u = y(x)$ (Topic 6.9). The shortcut of formally “multiplying by dx ” is simply a fast way to arrive at the same substitution.

Example 7.6.1: General Solution by Separation

Find the general solution of $\frac{dy}{dx} = xy$.

Solution. Separate (assuming $y \neq 0$) and integrate both sides:

$$\frac{dy}{y} = x dx \implies \int \frac{dy}{y} = \int x dx \implies \ln |y| = \frac{x^2}{2} + C.$$

Exponentiating: $|y| = e^{x^2/2+C} = e^C e^{x^2/2}$. Since e^C ranges over all positive numbers as C varies, and y may be positive or negative, this can be written compactly as

$$y = Ae^{x^2/2}, \quad A \text{ an arbitrary constant.}$$

Example 7.6.2: A Lost Solution

Find the general solution of $\frac{dy}{dx} = y^2 \cos x$.

Solution. For $y \neq 0$, separate and integrate:

$$\frac{dy}{y^2} = \cos x dx \implies -\frac{1}{y} = \sin x + C \implies y = \frac{-1}{\sin x + C}.$$

AP Exam Tip

Dividing by $g(y)$ silently assumes $g(y) \neq 0$. Any constant value $y = c$ with $g(c) = 0$ solves the original equation trivially (both sides equal 0) but gets divided away and can vanish from the family found by separation. In Example 7.6.2, $y \equiv 0$ solves $\frac{dy}{dx} = y^2 \cos x$ (since $0 = 0 \cdot \cos x$) but is not of the form $\frac{-1}{\sin x + C}$ for any C — it must be reported as an additional solution. This exact issue reappears, deliberately,

in Topic 7.9: the logistic equation's two most important solutions are precisely its constant ones.

7.7 Finding Particular Solutions Using Initial Conditions and Separation of Variables

A general solution is a whole family, indexed by the constant C . An **initial condition** — one known point (x_0, y_0) on the desired solution curve — pins down exactly one member of that family, called a **particular solution**.

AP Exam Tip

The full procedure, worth automating through repetition: (1) separate variables; (2) antidifferentiate both sides, adding $+C$; (3) substitute the initial condition to solve for C ; (4) if possible, solve explicitly for y , choosing the correct sign/branch to match the initial condition; (5) state any domain restriction the solution requires.

Example 7.7.1: Particular Solution with Branch Selection

Solve $\frac{dy}{dx} = \frac{x}{y}$, $y(2) = -3$.

Solution. Separate and integrate:

$$y \, dy = x \, dx \quad \implies \quad \frac{y^2}{2} = \frac{x^2}{2} + C \quad \implies \quad y^2 = x^2 + K.$$

Apply the initial condition $y(2) = -3$: $(-3)^2 = 2^2 + K \Rightarrow 9 = 4 + K \Rightarrow K = 5$. So $y^2 = x^2 + 5$, giving $y = \pm\sqrt{x^2 + 5}$. Since $y(2) = -3 < 0$, the correct branch is the negative one:

$$y = -\sqrt{x^2 + 5}.$$

Because $x^2 + 5 > 0$ for every real x , this solution is valid on all of $\mathbb{R}$ — no domain restriction is needed here.

Example 7.7.2: Particular Solution with a Domain Restriction

Solve $\frac{dy}{dx} = -\frac{x}{y}$, $y(0) = 2$, and state the domain of the solution.

Solution. Separate and integrate:

$$y \, dy = -x \, dx \quad \implies \quad \frac{y^2}{2} = -\frac{x^2}{2} + C \quad \implies \quad x^2 + y^2 = K.$$

Apply $y(0) = 2$: $0 + 4 = K$, so $x^2 + y^2 = 4$ — a circle of radius 2. Solving for y and choosing the branch with $y(0) = 2 > 0$:

$$y = \sqrt{4 - x^2}.$$

This formula only makes sense, and only satisfies the *original* differential equation (which requires $y \neq 0$ in the denominator), for $4 - x^2 > 0$. At $x = \pm 2$ the solution

reaches $y = 0$, where $\frac{dy}{dx} = -x/y$ is undefined (a vertical tangent on the circle). The solution's domain must therefore be restricted to the open interval

$$-2 < x < 2.$$

Remark

Example 7.7.2 illustrates Essential Knowledge FUN-7.E.3 directly: solutions to differential equations are not automatically valid on all of $\mathbb{R}$ just because the algebraic formula produced by separation happens to make sense there. The differential equation itself — through points where a derivative is undefined — can force a smaller domain.

7.8 Exponential Models with Differential Equations

The single most common differential equation encountered in applications is the one translating the sentence “the rate of change of a quantity is proportional to the quantity itself” — already written down in Example 7.1.1. Its solution, worked out once in general here, becomes a reusable formula for every growth-and-decay context that fits this description.

Theorem 7.8.1: The Exponential Growth/Decay Model

The differential equation $\frac{dy}{dt} = ky$, with initial condition $y = y_0$ at $t = 0$, has solution

$$y = y_0 e^{kt}.$$

The quantity grows without bound if $k > 0$ and decays toward 0 if $k < 0$.

Derivation. This is a direct application of Topic 7.6. Separating variables (for $y \neq 0$),

$$\frac{dy}{y} = k dt \implies \ln |y| = kt + C \implies y = Ae^{kt}.$$

Applying $y(0) = y_0$ gives $y_0 = Ae^0 = A$, so $A = y_0$ and $y = y_0 e^{kt}$. ■

Remark

This theorem is worth memorizing as a pattern-matching shortcut: *any* time a context states or implies $\frac{dy}{dt} \propto y$, the solution is immediately $y = y_0 e^{kt}$ without repeating the separation-of-variables computation — exactly as Topics 6.10–6.13 built a library of standard forms to recognize on sight.

Example 7.8.1: Population Growth

A bacteria population grows at a rate proportional to its size. It starts at 500 and grows to 1500 after 3 hours. Find the population after 5 hours, and find the doubling time.

Solution. Here $P(t) = 500e^{kt}$. Using $P(3) = 1500$:

$$1500 = 500e^{3k} \implies e^{3k} = 3 \implies k = \frac{\ln 3}{3} \approx 0.3662.$$

Then $P(5) = 500e^{5k} = 500 \cdot 3^{5/3} \approx 500(6.24) \approx 3119$ bacteria. The doubling time solves $2 = e^{kt}$:

$$t = \frac{\ln 2}{k} = \frac{3 \ln 2}{\ln 3} \approx 1.89 \text{ hours.}$$

Example 7.8.2: Radioactive Decay

Carbon-14 has a half-life of 5730 years. A sample starts with 10 grams. How much remains after 2000 years?

Solution. A half-life of 5730 years means $A(5730) = \frac{1}{2}A_0$:

$$\frac{1}{2} = e^{5730k} \implies k = \frac{\ln(1/2)}{5730} \approx -0.0001209.$$

Then $A(2000) = 10e^{2000k} = 10e^{-0.2419} \approx 10(0.7852) \approx 7.85$ grams remain.

Example 7.8.3: Newton's Law of Cooling, Revisited

Recall Example 7.1.3: $\frac{dT}{dt} = k(T - T_s)$. A cup of coffee at 190°F is placed in a 68°F room; after 10 minutes it has cooled to 150°F . Find the temperature after 20 minutes and the time needed to reach 100°F .

Solution. Let $u = T - T_s = T - 68$. Then $\frac{du}{dt} = \frac{dT}{dt} = k(T - 68) = ku$, exactly the exponential model with u in place of y . So $u = u_0e^{kt}$, i.e.

$$T = 68 + Ae^{kt}, \quad A = T(0) - 68.$$

From $T(0) = 190$: $A = 190 - 68 = 122$. From $T(10) = 150$:

$$150 - 68 = 122e^{10k} \implies e^{10k} = \frac{82}{122} \approx 0.6721 \implies k \approx \frac{\ln(0.6721)}{10} \approx -0.03973.$$

Temperature at $t = 20$:

$$T(20) = 68 + 122e^{-0.03973(20)} = 68 + 122e^{-0.7946} \approx 68 + 55.1 \approx 123.1^\circ\text{F}.$$

Time to reach 100°F : solve $100 - 68 = 32 = 122e^{kt}$:

$$e^{kt} = \frac{32}{122} \approx 0.2623 \implies t = \frac{\ln(0.2623)}{k} \approx \frac{-1.338}{-0.03973} \approx 33.7 \text{ minutes.}$$

Remark

Every example in this section reduces, after at most a linear substitution like $u = T - T_s$, to the single model $y = y_0e^{kt}$. Learning to recognize this underlying pattern

beneath different variable names (P , A , T , and so on) is exactly the “ x ’s and y ’s to equivalent environments” skill the Course Framework emphasizes for this unit.

The exponential model assumes growth is unconstrained forever — realistic for radioactive decay, but not for a population with limited resources. The final topic replaces that assumption with a more realistic one.

7.9 Logistic Models with Differential Equations

BC Only

Topic 7.9 is assessed on the BC Exam only.

Unrestricted exponential growth $\frac{dP}{dt} = kP$ predicts a population that grows forever, doubling again and again at a fixed rate — but every real environment has a limited food supply, living space, or other resource, and population growth must slow as that limit is approached. The **logistic model** builds this leveling-off effect directly into the differential equation by making the growth rate proportional not just to the population, but also to the *remaining room to grow*.

Definition 7.9.1: The Logistic Differential Equation

Let $a > 0$ denote the **carrying capacity** — the maximum population the environment can sustain — and let $k > 0$ be a constant. The **logistic differential equation** is

$$\frac{dy}{dt} = ky(a - y).$$

Remark

The factor y makes the rate proportional to the current population (as in exponential growth); the new factor $(a - y)$ measures how much room remains before the carrying capacity is reached, and it *shrinks toward zero* as $y \rightarrow a$. The product of the two captures both effects: growth is fast when the population is sizable *and* far from capacity, and growth slows to a crawl as either factor shrinks.

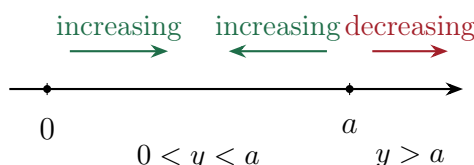
Remarkably, the Course Framework’s required knowledge about this model (FUN-7.H.1 through FUN-7.H.4) can all be extracted directly from the differential equation *without solving it* — by treating the right-hand side $g(y) = ky(a - y)$ as an ordinary function of y and analyzing it the way Unit 5 analyzed any function.

Theorem 7.9.1: Equilibrium Solutions

The constant functions $y \equiv 0$ and $y \equiv a$ are both solutions of $\frac{dy}{dt} = ky(a - y)$, since $g(0) = g(a) = 0$ makes $\frac{dy}{dt} = 0$ identically — a genuinely constant population never changes.

Theorem 7.9.2: Direction of Change Away from Equilibrium

For $0 < y < a$: both factors y and $(a - y)$ are positive, so $\frac{dy}{dt} = ky(a - y) > 0$ — the population **increases** toward the carrying capacity. For $y > a$: $y > 0$ but $(a - y) < 0$, so $\frac{dy}{dt} < 0$ — the population **decreases** back toward the carrying capacity. In both cases, $y \rightarrow a$ as $t \rightarrow \infty$: the carrying capacity is a stable limiting value approached from either side.



The last required fact — where the population is growing *fastest* — is a genuine optimization problem, solved with exactly the Unit 5 technique of maximizing a function via its critical points, except the function being maximized is the *rate* $g(y)$ itself, viewed as a function of y .

Theorem 7.9.3: The Point of Fastest Growth

The rate $\frac{dy}{dt} = g(y) = ky(a - y) = kay - ky^2$ is a downward-opening parabola in y , maximized where $g'(y) = ka - 2ky = 0$, i.e. at

$$y = \frac{a}{2}.$$

The population grows fastest when it equals exactly **half** the carrying capacity, at which point the maximum rate is $g(a/2) = k \cdot \frac{a}{2} \cdot \frac{a}{2} = \frac{ka^2}{4}$.

Remark

Geometrically, $y = a/2$ is the **inflection point** of the solution curve $y(t)$: since $\frac{d^2y}{dt^2} = g'(y) \cdot \frac{dy}{dt}$ by the chain rule, and $\frac{dy}{dt} \neq 0$ away from the equilibria, the second derivative changes sign exactly where $g'(y)$ does — at $y = a/2$. This is the same logic Topic 5.7 used to locate inflection points from a sign change in f'' , applied here one level removed.

Example 7.9.1: Interpreting a Logistic Model Without Solving

A population satisfies $\frac{dP}{dt} = 0.001P(500 - P)$, with $P(0) = 50$. Without solving the differential equation, find (a) the carrying capacity, (b) the population when it is growing fastest, and (c) the maximum growth rate.

Solution. Matching to $ky(a - y)$ gives $k = 0.001$ and $a = 500$.

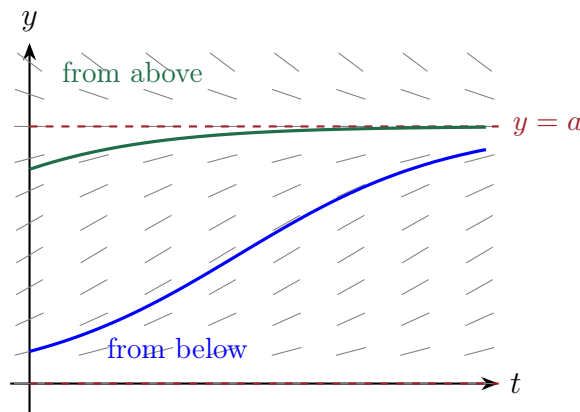
(a) The carrying capacity is $a = 500$.

(b) Fastest growth occurs at $P = a/2 = 250$.

(c) The maximum rate is $g(250) = 0.001(250)(500 - 250) = 0.001(250)(250) = 62.5$ (population units per unit time).

Every one of these facts follows from analyzing $g(P) = 0.001P(500 - P)$ directly — no antiderivative was needed.

Because the logistic equation depends on t only through y (the right-hand side $ky(a - y)$ has no explicit t), it is called **autonomous**, and its slope field has a distinctive signature: every segment along a fixed horizontal line $y = c$ has the identical slope $g(c)$, no matter how far left or right that point sits.



Remark

The picture above is the complete qualitative story of the logistic model in one image: every solution curve, regardless of starting value, bends toward the horizontal asymptote $y = a$ — climbing with an S-shaped (**sigmoid**) profile if it starts below the carrying capacity, or descending toward it if it starts above. The inflection point at $y = a/2$ is visible on the S-shaped curve as the point where concavity switches from up to down.

7.9.1 Beyond the AP Exam: Deriving the Explicit Logistic Solution

Everything established above — equilibria, direction of change, limiting value, point of fastest growth — is required knowledge for the BC Exam, and none of it needed an explicit formula for $y(t)$. The Course Framework explicitly notes that the logistic model “may be used or interpreted without performing the derivation.” Still, the equation *is* separable, and solving it in closed form is a satisfying capstone to this unit’s technique — as well as an excellent exercise in partial fractions (Topic 6.12). This subsection is enrichment beyond what the AP Exam requires.

Theorem 7.9.4: Explicit Solution of the Logistic Equation

The solution of $\frac{dy}{dt} = ky(a - y)$ with initial value $y(0) = y_0$ (where $0 < y_0 < a$) is

$$y(t) = \frac{a}{1 + Be^{-akt}}, \quad B = \frac{a - y_0}{y_0}.$$

Derivation. Separate variables:

$$\frac{dy}{y(a-y)} = k dt.$$

The left side is a rational function of y with distinct linear factors in the denominator, so it decomposes by partial fractions (Topic 6.12). Seeking constants M, N with $\frac{1}{y(a-y)} = \frac{M}{y} + \frac{N}{a-y}$, clearing denominators gives $1 = M(a-y) + Ny$; setting $y = 0$ gives $M = \frac{1}{a}$, and setting $y = a$ gives $N = \frac{1}{a}$. So

$$\frac{1}{y(a-y)} = \frac{1}{a} \left(\frac{1}{y} + \frac{1}{a-y} \right).$$

Substituting and integrating both sides with respect to t :

$$\frac{1}{a} \int \left(\frac{1}{y} + \frac{1}{a-y} \right) dy = \int k dt \implies \frac{1}{a} (\ln |y| - \ln |a-y|) = kt + C_1,$$

using $\int \frac{dy}{a-y} = -\ln |a-y| + C$ ($u = a-y$, $du = -dy$). Multiply by a and combine the logarithms:

$$\ln \left| \frac{y}{a-y} \right| = akt + C_2 \implies \frac{y}{a-y} = Ae^{akt}, \quad A = \pm e^{C_2}.$$

Solve for y : $y = Ae^{akt}(a-y) = Aae^{akt} - Ae^{akt}y$, so $y(1 + Ae^{akt}) = Aae^{akt}$, and dividing through by Ae^{akt} ,

$$y = \frac{a}{\frac{1}{A}e^{-akt} + 1} = \frac{a}{1 + Be^{-akt}}, \quad B := \frac{1}{A}.$$

Finally, impose $y(0) = y_0$: $y_0 = \frac{a}{1+B}$, so $1+B = \frac{a}{y_0}$ and

$$B = \frac{a - y_0}{y_0}. \quad \blacksquare$$

Remark

This formula deserves a direct check: differentiating $y = a(1 + Be^{-akt})^{-1}$ with the chain rule gives $\frac{dy}{dt} = a^2 B k e^{-akt} (1 + Be^{-akt})^{-2}$, while substituting the same solution into $ky(a-y)$ gives, after simplifying with $u = 1 + Be^{-akt}$, the identical expression $ka^2 B e^{-akt} u^{-2}$ — confirming the formula genuinely solves the differential equation, not merely resembles a solution.

Example 7.9.2: The Explicit Solution in Context

Continuing Example 7.9.1 ($k = 0.001$, $a = 500$, $P(0) = 50$), find the explicit formula for $P(t)$, and determine exactly when the population reaches 250 (half the carrying capacity).

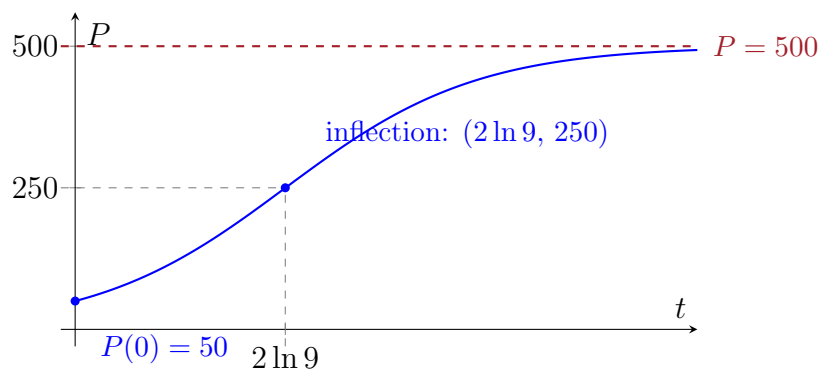
Solution. Here $ak = 0.001(500) = 0.5$ and $B = \frac{500 - 50}{50} = 9$, so

$$P(t) = \frac{500}{1 + 9e^{-0.5t}}.$$

Setting $P(t) = 250$:

$$250 = \frac{500}{1 + 9e^{-0.5t}} \implies 1 + 9e^{-0.5t} = 2 \implies e^{-0.5t} = \frac{1}{9} \implies t = 2 \ln 9 \approx 4.39.$$

Consistent with Theorem 7.9.3, this is exactly the moment the population passes through $a/2 = 250$, so it also marks the inflection point of this particular S -curve.



Unit 7 completed the second half of the promise made when antidifferentiation was introduced: not only can a rate be turned back into a total amount (Unit 6), a rate expressed only in terms of the quantity itself can be turned back into the quantity as a function of time. Unit 8 returns to integration's more geometric applications — areas, volumes, and (for BC) arc length — built once again from the same limit-of-Riemann-sums idea that has anchored every accumulation argument since Topic 6.1.

UNIT 8

Applications of Integration

Class Periods	~19–20 (AB)	~13–14 (BC)
Big Idea	Change (CHA)	

The definite integral was built in Unit 6 for one specific purpose: computing the area between a curve and the x -axis. But nothing in its construction — chop an interval into pieces, multiply a value by a small width, add, take a limit — actually depended on “area” at all. That construction computes a running total of *anything* that can be expressed as a rate times a small width: a rate of change accumulates into net change, a velocity accumulates into displacement, a cross-sectional area accumulates into volume, and (for BC) a small piece of arc length accumulates into total length. Unit 8 is a tour of exactly these reinterpretations. The computations will look different from anything in Unit 6, but every single one is, underneath, the same limit of a Riemann sum — set up by identifying what is being chopped into small pieces, and what is being multiplied by that small piece’s width.

8.1 Finding the Average Value of a Function on an Interval

The average of finitely many numbers is a familiar idea: add them and divide by how many there are. But what is the “average” of a continuously varying function, such as the temperature $T(t)$ over the course of a day — a quantity with infinitely many values, one at every instant? The definite integral answers this by treating the continuous case as a limit of the discrete one.

Partition $[a, b]$ into n equal subintervals of width $\Delta x = \frac{b-a}{n}$, and average the function’s value at the n sample points $x_1, \dots, x_n$:

$$\text{average} \approx \frac{f(x_1) + f(x_2) + \cdots + f(x_n)}{n} = \frac{1}{n} \sum_{i=1}^n f(x_i).$$

Multiplying and dividing by $\Delta x = \frac{b-a}{n}$ (so $\frac{1}{n} = \frac{\Delta x}{b-a}$) turns the sum into something recognizable:

$$\frac{1}{n} \sum_{i=1}^n f(x_i) = \frac{1}{b-a} \sum_{i=1}^n f(x_i) \Delta x.$$

The sum on the right is exactly a Riemann sum for $\int_a^b f(x) dx$. Letting $n \rightarrow \infty$ turns the discrete average into the definition below.

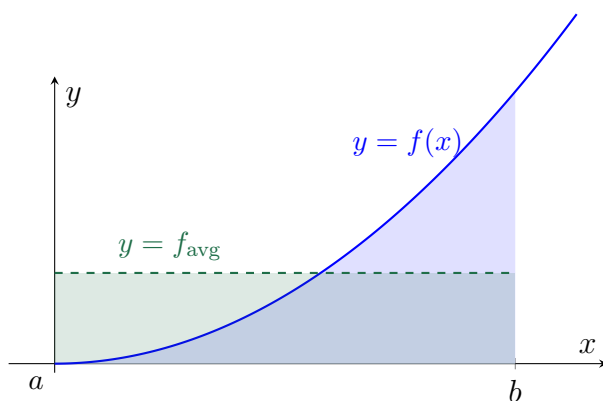
Definition 8.1.1: Average Value of a Function

The **average value** of a continuous function f on $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

Remark

This formula is often rearranged as $\int_a^b f(x) dx = f_{\text{avg}} \cdot (b-a)$ — the area under f equals the area of a rectangle of width $b-a$ and height f_{avg} . This gives a useful geometric picture: f_{avg} is the height at which a flat rectangle over $[a, b]$ encloses exactly the same area as the curve does.



Because f_{avg} is itself a value that f takes on somewhere (when f is continuous), the Extreme Value and Intermediate Value Theorems (Topic 1.15) guarantee a point where the function actually *equals* its own average — a continuous analogue of the Mean Value Theorem (Topic 5.2), but for integrals instead of derivatives.

Theorem 8.1.1: Mean Value Theorem for Integrals

If f is continuous on $[a, b]$, there exists at least one $c \in [a, b]$ such that

$$f(c) = f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

Proof. Let $F(x) = \int_a^x f(t) dt$, an antiderivative of f by the Fundamental Theorem of Calculus, Part 1 (Topic 6.4). Since F is differentiable on $[a, b]$, the Mean Value Theorem (Topic 5.2) guarantees a $c \in (a, b)$ with

$$F'(c) = \frac{F(b) - F(a)}{b-a}.$$

The left side is $F'(c) = f(c)$; the right side is $\frac{\int_a^b f(t) dt - 0}{b-a} = \frac{1}{b-a} \int_a^b f(x) dx$, which is exactly f_{avg} . So $f(c) = f_{\text{avg}}$. ■

Example 8.1.1: Computing an Average Value

Find the average value of $f(x) = x^2$ on $[0, 3]$, and find a value of c guaranteed by the Mean Value Theorem for Integrals.

Solution.

$$f_{\text{avg}} = \frac{1}{3-0} \int_0^3 x^2 dx = \frac{1}{3} \left[\frac{x^3}{3} \right]_0^3 = \frac{1}{3}(9) = 3.$$

Setting $f(c) = 3$: $c^2 = 3 \Rightarrow c = \sqrt{3} \approx 1.73$, which lies in $[0, 3]$, as guaranteed.

Example 8.1.2: Average Value from a Table

The rate at which cars pass a checkpoint is recorded every 10 minutes.

t (min)	0	10	20	30	40
$r(t)$ (cars/min)	4	7	9	6	5

Estimate the average rate over $[0, 40]$ using a left Riemann sum with the given data.

Solution. A left sum with $\Delta t = 10$ estimates $\int_0^{40} r(t) dt \approx 10(4 + 7 + 9 + 6) = 260$. Dividing by the interval length $40 - 0 = 40$:

$$r_{\text{avg}} \approx \frac{260}{40} = 6.5 \text{ cars/min.}$$

AP Exam Tip

Average value ($f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$, Topic 8.1) and **average rate of change** ($\frac{f(b)-f(a)}{b-a}$, Topic 2.1) are computed from completely different formulas and answer different questions — the first averages the *output values* of f , the second averages how fast f itself is changing. They are easy to confuse because of the shared word “average.” There is, however, a clean relationship between them:

Remark

If $f = g'$ for some function g , then by the Fundamental Theorem of Calculus, Part 2,

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b g'(x) dx = \frac{g(b) - g(a)}{b-a},$$

which is exactly the *average rate of change* of g on $[a, b]$. In words: *the average value of a derivative equals the average rate of change of the original function.* The two concepts are distinct in general, but coincide in this specific circumstance.

8.2 Connecting Position, Velocity, and Acceleration Using Integrals

Topic 4.2 differentiated position to get velocity, and velocity to get acceleration. Antidifferentiation reverses both steps, and the definite integral gives a precise meaning to the

result: the integral of velocity over a time interval accumulates displacement, exactly as Topic 6.1's opening example accumulated distance from a (constant-sign) velocity graph.

Theorem 8.2.1: Displacement and Total Distance

For a particle moving along a line with velocity $v(t)$ on $[a, b]$:

$$\text{displacement} = \int_a^b v(t) dt = s(b) - s(a), \quad \text{total distance traveled} = \int_a^b |v(t)| dt.$$

Remark

Displacement is a *signed* net change in position — rightward motion (where $v > 0$) and leftward motion (where $v < 0$) partially cancel. Total distance treats every moment of motion as a positive contribution, regardless of direction, which is why it requires integrating $|v(t)|$ rather than $v(t)$ directly. When v does not change sign on $[a, b]$, the two quantities agree.

AP Exam Tip

To integrate $|v(t)|$ by hand: find where $v(t) = 0$ to split $[a, b]$ into intervals of constant sign, integrate v on each piece separately, and add the *absolute values* of those pieces. Equivalently, $\int_a^b |v(t)| dt = \sum_i |V(t_{i+1}) - V(t_i)|$ where V is any antiderivative of v and t_i are the sign-change points (together with a and b).

Example 8.2.1: Displacement vs. Total Distance

A particle has acceleration $a(t) = 2$ and initial velocity $v(0) = -3$. Find its displacement and total distance traveled over $[0, 3]$.

Solution. Antidifferentiating the acceleration: $v(t) = -3 + 2t$ (using $v(0) = -3$). This is zero at $t = 1.5$: the particle moves left on $[0, 1.5)$ and right on $(1.5, 3]$. An antiderivative of v is $V(t) = t^2 - 3t$.

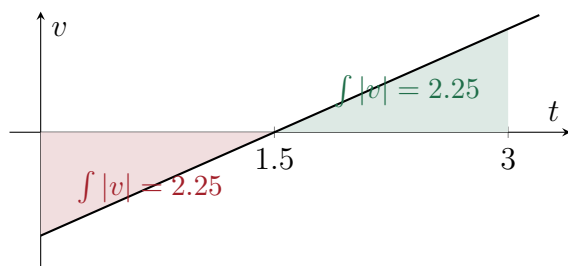
Displacement:

$$\int_0^3 v(t) dt = V(3) - V(0) = (9 - 9) - 0 = 0.$$

Total distance: split at $t = 1.5$ and add the magnitude of each piece:

$$|V(1.5) - V(0)| + |V(3) - V(1.5)| = |-2.25 - 0| + |0 - (-2.25)| = 2.25 + 2.25 = 4.5.$$

The particle returns to its starting position (displacement 0) but travels a total of 4.5 units getting there and back.

**Remark**

The picture makes the displacement-vs-distance distinction visible: the signed area below the axis (red) and above the axis (green) are equal in magnitude and opposite in sign, so they cancel in the displacement integral $\int v \, dt$ but add together in the distance integral $\int |v| \, dt$.

8.3 Using Accumulation Functions and Definite Integrals in Applied Contexts

Topic 6.4 introduced the accumulation function $F(x) = F(a) + \int_a^x f(t) \, dt$ in the abstract. In applied contexts — almost always phrased as a rate flowing *into* some quantity and another rate flowing *out* — this same idea becomes the single most common setup on the AP Free Response section.

Theorem 8.3.1: Net Change from Competing Rates

If a quantity W has an inflow rate $f(t)$ and outflow rate $g(t)$, both in the same units, then for $t \geq 0$,

$$W(t) = W(0) + \int_0^t (f(s) - g(s)) \, ds,$$

and W is increasing exactly where $f(t) > g(t)$ and decreasing exactly where $f(t) < g(t)$ — the First Derivative Test (Topic 5.3) applied to $W'(t) = f(t) - g(t)$.

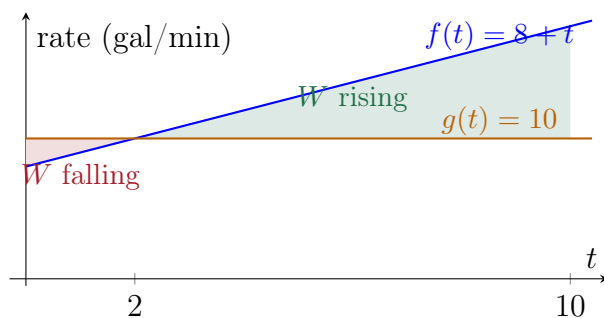
Example 8.3.1: A Complete Tank Problem

Water flows into a tank at rate $f(t) = 8 + t$ gallons/min and is pumped out at a constant rate $g(t) = 10$ gallons/min, for $0 \leq t \leq 10$. The tank starts with $W(0) = 50$ gallons. Find when the amount of water is decreasing, find the minimum amount of water, and find the amount at $t = 10$.

Solution. The tank's rate of change is $W'(t) = f(t) - g(t) = (8 + t) - 10 = t - 2$, which is negative for $t < 2$ and positive for $t > 2$: W **decreases** on $[0, 2)$ and **increases** on $(2, 10]$, so the minimum occurs at $t = 2$.

$$W(2) = 50 + \int_0^2 (t - 2) \, dt = 50 + \left[\frac{t^2}{2} - 2t \right]_0^2 = 50 + (2 - 4) = 48 \text{ gallons.}$$

$$W(10) = 50 + \int_0^{10} (t - 2) \, dt = 50 + \left[\frac{t^2}{2} - 2t \right]_0^{10} = 50 + (50 - 20) = 80 \text{ gallons.}$$

**AP Exam Tip**

When asked to *interpret* a definite integral in context (rather than compute it), the AP rubric expects a complete sentence naming the quantity, the time interval, and the correct units — for instance: “ $\int_2^5 r(t) dt$ is the net change in the number of gallons of water in the tank from $t = 2$ to $t = 5$ minutes.” A bare numerical answer with no units or context earns little to no credit for this skill.

Topics 8.1–8.3 reinterpreted the definite integral as an average, a displacement, and a net change — all one-dimensional accumulations. The remaining topics of this unit push the same idea into two and three dimensions: areas between curves, and volumes of solids.

8.4 Finding the Area Between Curves Expressed as Functions of x

Topic 6.1 computed area between a single curve and the x -axis — really just the special case of area between $y = f(x)$ and the constant function $y = 0$. Replacing that constant function with any second curve $y = g(x)$ generalizes the idea with almost no new work.

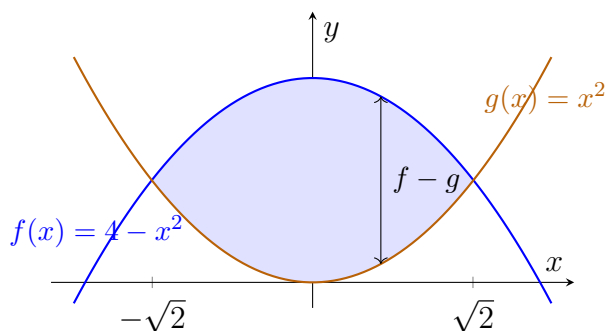
Theorem 8.4.1: Area Between Two Curves (Functions of x)

If $f(x) \geq g(x)$ for all $x \in [a, b]$, the area of the region between their graphs is

$$A = \int_a^b [f(x) - g(x)] dx.$$

Remark

This is the Riemann-sum idea one small step removed: chop $[a, b]$ into thin vertical strips of width Δx ; each strip is (approximately) a rectangle of height $f(x) - g(x)$ — the gap between the curves — and width Δx . Summing these thin rectangles and taking the limit produces exactly the integral above. The formula reduces to Topic 6.1’s when $g(x) \equiv 0$.

**Example 8.4.1: Area Between Two Parabolas**

Find the area of the region enclosed by $f(x) = 4 - x^2$ and $g(x) = x^2$.

Solution. Find intersections: $4 - x^2 = x^2 \Rightarrow 4 = 2x^2 \Rightarrow x = \pm\sqrt{2}$. On this interval, $f(x) \geq g(x)$ (checked at $x = 0$: $4 \geq 0$). So

$$A = \int_{-\sqrt{2}}^{\sqrt{2}} [(4 - x^2) - x^2] dx = \int_{-\sqrt{2}}^{\sqrt{2}} (4 - 2x^2) dx = \left[4x - \frac{2x^3}{3} \right]_{-\sqrt{2}}^{\sqrt{2}}.$$

By symmetry this is twice the value at $x = \sqrt{2}$:

$$A = 2 \left(4\sqrt{2} - \frac{2(2\sqrt{2})}{3} \right) = 2 \left(4\sqrt{2} - \frac{4\sqrt{2}}{3} \right) = 2 \cdot \frac{8\sqrt{2}}{3} = \frac{16\sqrt{2}}{3} \approx 7.54.$$

8.5 Finding the Area Between Curves Expressed as Functions of y

Some regions are far more naturally sliced with *horizontal* strips than vertical ones — typically when the region is bounded by curves that are easy to solve for x in terms of y , or when a curve fails the vertical line test but passes the horizontal one.

Theorem 8.5.1: Area Between Two Curves (Functions of y)

If $u(y) \geq v(y)$ for all $y \in [c, d]$ (i.e. the curve $x = u(y)$ lies to the right of $x = v(y)$), the area between them is

$$A = \int_c^d [u(y) - v(y)] dy.$$

AP Exam Tip

The choice between dx and dy is entirely about convenience, never about correctness — both compute the same area when set up correctly. Sketch the region first: if a horizontal strip has a single, simple description of its left and right ends in terms of y , integrate in dy ; if a vertical strip has a simpler top/bottom description in terms of x , integrate in dx .

Example 8.5.1: Area Using Horizontal Strips

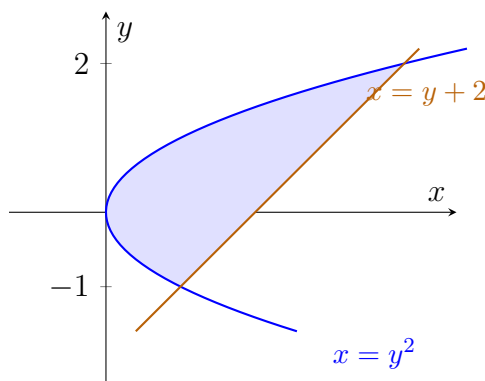
Find the area enclosed by $x = y^2$ and $x = y + 2$.

Solution. Solve for intersections: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y - 2)(y + 1) = 0 \Rightarrow y = -1, 2$. For $y \in [-1, 2]$, the line lies to the right of the parabola (checked at $y = 0$: line gives $x = 2$, parabola gives $x = 0$). So $u(y) = y + 2$, $v(y) = y^2$:

$$A = \int_{-1}^2 [(y + 2) - y^2] dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2.$$

At $y = 2$: $2 + 4 - \frac{8}{3} = 6 - \frac{8}{3} = \frac{10}{3}$. At $y = -1$: $\frac{1}{2} - 2 + \frac{1}{3} = -\frac{7}{6}$. So

$$A = \frac{10}{3} - \left(-\frac{7}{6} \right) = \frac{20}{6} + \frac{7}{6} = \frac{27}{6} = 4.5.$$



8.6 Finding the Area Between Curves That Intersect at More Than Two Points

When two curves cross more than twice on an interval, “top minus bottom” switches roles at each crossing — using a single integral across the whole interval without accounting for this would let positive and negative contributions wrongly cancel.

Theorem 8.6.1: Area with Multiple Crossings

If f and g cross at $x = a = x_0 < x_1 < \cdots < x_n = b$, the total enclosed area is the sum of the areas on each subinterval, or equivalently a single integral of an absolute value:

$$A = \sum_{i=1}^n \left| \int_{x_{i-1}}^{x_i} [f(x) - g(x)] dx \right| = \int_a^b |f(x) - g(x)| dx.$$

AP Exam Tip

By hand, always split at every crossing point and add the absolute value of each piece — do not integrate $|f - g|$ directly unless a calculator is available, since $|f - g|$ generally has no elementary antiderivative even when $f - g$ does. Determine the sign of $f - g$ on each piece with a single test point, exactly as in a sign chart (Topic

5.3).

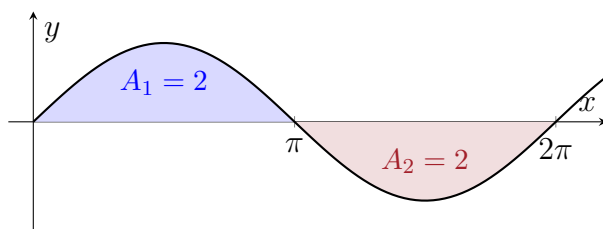
Example 8.6.1: Area with Two Crossing Points on an Interval

Find the total area between $f(x) = \sin x$ and $g(x) = 0$ on $[0, 2\pi]$.

Solution. $\sin x = 0$ at $x = 0, \pi, 2\pi$ — two subintervals. $\sin x \geq 0$ on $[0, \pi]$ and $\sin x \leq 0$ on $[\pi, 2\pi]$:

$$A = \int_0^{\pi} \sin x \, dx + \left| \int_{\pi}^{2\pi} \sin x \, dx \right| = [-\cos x]_0^{\pi} + |[-\cos x]_{\pi}^{2\pi}|.$$

First piece: $-\cos \pi - (-\cos 0) = 1 + 1 = 2$. Second piece: $-\cos 2\pi - (-\cos \pi) = -1 - 1 = -2$, so $|-2| = 2$. Total area $A = 2 + 2 = 4$.

**Remark**

This example is a useful sanity check on a familiar fact: the *signed* integral $\int_0^{2\pi} \sin x \, dx = 0$ (the two lobes cancel by symmetry), but the *enclosed area* is 4, not 0. Confusing these two questions — “what is the net signed integral” versus “what is the total enclosed area” — is one of the most common errors on this topic.

Every area computed so far has been genuinely two-dimensional — the region itself was the object of interest. The next four topics take a region in the plane and use it as the **footprint** of a three-dimensional solid, applying the identical slice-and-sum strategy one dimension up: chop the solid into thin slabs, find the (now two-dimensional) area of a representative slab, multiply by a small thickness, and integrate.

8.7 Volumes with Cross Sections: Squares and Rectangles

Imagine a solid built by standing a flat shape upright on every vertical line above a base region R in the xy -plane, with the shape’s size determined by how wide R is at that location. Slicing this solid with a plane perpendicular to the x -axis at a fixed x always exposes the same flat shape used to build it — its **cross section**. This is the governing idea behind every volume computed in this unit, disc and washer methods included.

Theorem 8.7.1: Volume by Cross Sections

If a solid lies above the interval $[a, b]$ on the x -axis, and every cross section perpendicular to the x -axis at position x has area $A(x)$, then the volume of the solid

is

$$V = \int_a^b A(x) dx.$$

Justification. Partition $[a, b]$ into n pieces of width Δx . Approximating the thin slab of the solid over the i th subinterval by a flat prism of cross-sectional area $A(x_i)$ and thickness Δx , its volume is approximately $A(x_i) \Delta x$. Summing over all slabs, $V \approx \sum_{i=1}^n A(x_i) \Delta x$ — a Riemann sum for $\int_a^b A(x) dx$, which becomes exact as $n \rightarrow \infty$. ■

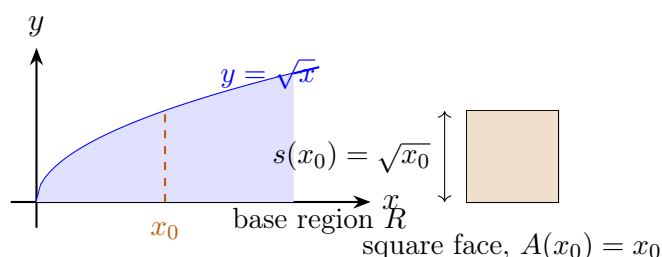
Remark

This single theorem is the parent of every volume formula in Topics 8.7–8.12. Everything that follows is really just a catalog of different shapes for $A(x)$: a square, a semicircle, or — starting in Topic 8.9 — a full circle (a disc) or a ring (a washer).

When the cross sections are squares built on a segment connecting two curves, the side length of each square is the vertical distance between the curves — precisely the integrand from Topic 8.4’s area-between-curves formula.

Theorem 8.7.2: Square and Rectangular Cross Sections

If the base region’s vertical extent at x is $s(x) = f(x) - g(x)$, then a solid with square cross sections built on that base has $A(x) = [s(x)]^2$; one with rectangular cross sections of height $k \cdot s(x)$ (for a fixed constant k) has $A(x) = k[s(x)]^2$.



Example 8.7.1: Volume with Square Cross Sections

The base of a solid is the region between $y = \sqrt{x}$ and the x -axis on $[0, 4]$. Cross sections perpendicular to the x -axis are squares. Find the volume.

Solution. The side length is $s(x) = \sqrt{x}$, so $A(x) = (\sqrt{x})^2 = x$:

$$V = \int_0^4 x dx = \left[\frac{x^2}{2} \right]_0^4 = 8.$$

8.8 Volumes with Cross Sections: Triangles and Semicircles

The same slicing strategy applies verbatim to any shape whose area formula is known — only the formula for $A(x)$ changes.

Cross-section shape	Governing dimension	$A(x)$
Square, side s	side = s	s^2
Rectangle, sides s, ks	side = s , height factor k	ks^2
Equilateral triangle, side s	side = s	$\frac{\sqrt{3}}{4}s^2$
Isosceles right triangle (leg = s)	leg = s	$\frac{1}{2}s^2$
Isosceles right triangle (hyp. = s)	hypotenuse = s	$\frac{1}{4}s^2$
Semicircle, diameter s	diameter = s	$\frac{\pi}{8}s^2$

Remark

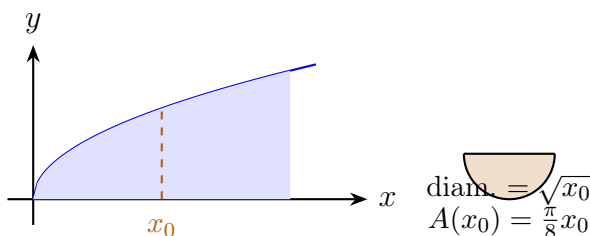
Every entry in this table comes from an ordinary geometry area formula, rewritten in terms of the single governing length s that the base region supplies at each x . For the semicircle, radius = $s/2$, so $A = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(s/2)^2 = \frac{\pi}{8}s^2$.

Example 8.8.1: Volume with Semicircular Cross Sections

The base of a solid is the region between $y = \sqrt{x}$ and the x -axis on $[0, 4]$ (as in Example 8.7.1). Cross sections perpendicular to the x -axis are semicircles with diameter along the base. Find the volume.

Solution. The diameter is $s(x) = \sqrt{x}$, so $A(x) = \frac{\pi}{8}(\sqrt{x})^2 = \frac{\pi}{8}x$:

$$V = \int_0^4 \frac{\pi}{8}x \, dx = \frac{\pi}{8} \left[\frac{x^2}{2} \right]_0^4 = \frac{\pi}{8}(8) = \pi.$$



Example 8.8.2: Volume with Equilateral Triangle Cross Sections

The base of a solid is the region between $y = 2x$ and $y = x^2$ on $[0, 2]$ (Topic 8.4). Cross sections perpendicular to the x -axis are equilateral triangles with one side along the base. Find the volume.

Solution. The base region's vertical extent is $s(x) = 2x - x^2$, so $A(x) = \frac{\sqrt{3}}{4}(2x - x^2)^2$:

$$V = \frac{\sqrt{3}}{4} \int_0^2 (2x - x^2)^2 dx = \frac{\sqrt{3}}{4} \int_0^2 (4x^2 - 4x^3 + x^4) dx.$$

$$\int_0^2 (4x^2 - 4x^3 + x^4) dx = \left[\frac{4x^3}{3} - x^4 + \frac{x^5}{5} \right]_0^2 = \frac{32}{3} - 16 + \frac{32}{5} = \frac{16}{15}.$$

$$\text{So } V = \frac{\sqrt{3}}{4} \cdot \frac{16}{15} = \frac{4\sqrt{3}}{15} \approx 0.462.$$

8.9 Volume with Disc Method: Revolving Around the x - or y -Axis

The single most important family of cross sections is also the most natural: revolve a region around an axis, and every cross section perpendicular to that axis is automatically a **circle**. This technique — the **disc method** — is Theorem 8.7.1 applied to the one shape that a rotation always produces.

Theorem 8.9.1: The Disc Method

Let f be continuous and nonnegative on $[a, b]$. Revolving the region between $y = f(x)$ and the x -axis around the **x -axis** produces a solid with volume

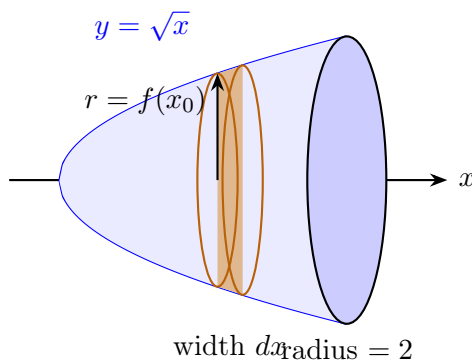
$$V = \int_a^b \pi [f(x)]^2 dx.$$

Likewise, revolving the region between $x = g(y)$ and the y -axis around the **y -axis** on $[c, d]$ produces

$$V = \int_c^d \pi [g(y)]^2 dy.$$

Remark

At each fixed x , revolving the single point $(x, f(x))$ a full 360° around the x -axis sweeps out a circle of radius $f(x)$ — so the cross section there is a disc of area $A(x) = \pi[f(x)]^2$, and Theorem 8.7.1 supplies the rest. The disc method is not a new idea; it is the square/semicircle machinery of Topics 8.7–8.8 with the cross-sectional shape fixed once and for all to be a circle.

**Remark**

The orange slab is the picture behind the Riemann sum: a thin cylinder (a **disc**) of radius $r = f(x_0)$ and thickness dx , with volume $\pi r^2 dx = \pi [f(x_0)]^2 dx$. Summing these thin cylinders across $[a, b]$ and taking the limit is precisely $\int_a^b \pi [f(x)]^2 dx$.

Example 8.9.1: Volume by the Disc Method

The region between $y = \sqrt{x}$ and the x -axis on $[0, 4]$ (the same region as Examples 8.7.1 and 8.8.1) is revolved around the x -axis. Find the volume of the resulting solid.

Solution.

$$V = \int_0^4 \pi (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = 8\pi.$$

Remark

Compare the three solids built from the exact same base region: square cross sections gave volume 8 (Example 8.7.1), semicircular cross sections gave $\pi \approx 3.14$ (Example 8.8.1), and circular (disc-method) cross sections give $8\pi \approx 25.13$. Same footprint, three entirely different solids — the cross-sectional *shape* is just as important as the base region itself.

8.10 Volume with Disc Method: Revolving Around Other Axes

Nothing about the disc method requires the axis of revolution to be a coordinate axis — only that the radius of each disc be measured as the distance from the curve to *whichever* horizontal or vertical line is the actual axis of revolution.

Theorem 8.10.1: Disc Method Around a General Horizontal or Vertical Line

Revolving the region between $y = f(x)$ and the horizontal line $y = k$ (with $f(x) \geq k$ or $f(x) \leq k$ throughout $[a, b]$) around the line $y = k$ produces

$$V = \int_a^b \pi [f(x) - k]^2 dx.$$

Revolving the region between $x = g(y)$ and the vertical line $x = h$ around the line $x = h$ produces

$$V = \int_c^d \pi [g(y) - h]^2 dy.$$

AP Exam Tip

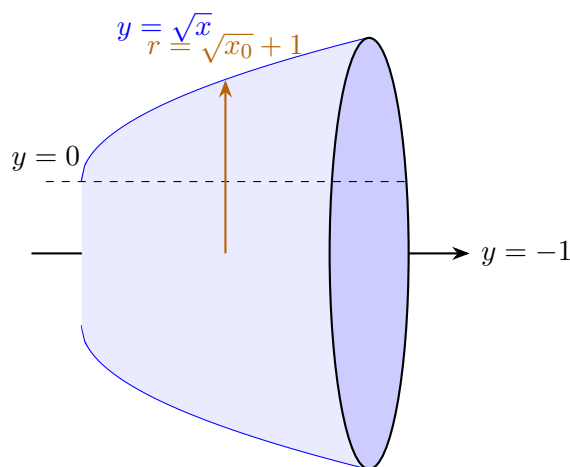
The radius is always a *difference*: (curve) $-$ (axis), or (axis) $-$ (curve) — whichever is nonnegative. Sketching the region and the axis together before writing the integral avoids sign errors far more reliably than memorizing a rule.

Example 8.10.1: Disc Method Around a Horizontal Line Other Than the x -Axis

The region between $y = \sqrt{x}$ and the x -axis on $[0, 4]$ is revolved around the line $y = -1$. Find the volume.

Solution. Every point of the region sits a vertical distance $\sqrt{x} - (-1) = \sqrt{x} + 1$ above the axis of revolution, so this is the radius:

$$\begin{aligned} V &= \pi \int_0^4 (\sqrt{x} + 1)^2 dx = \pi \int_0^4 (x + 2\sqrt{x} + 1) dx = \pi \left[\frac{x^2}{2} + \frac{4}{3}x^{3/2} + x \right]_0^4 \\ &= \pi \left(8 + \frac{32}{3} + 4 \right) = \pi \left(12 + \frac{32}{3} \right) = \frac{68\pi}{3} \approx 71.21. \end{aligned}$$



Example 8.10.2: Disc Method Around a Vertical Line

The region between $y = \sqrt{x}$ (equivalently $x = y^2$) and the y -axis on $[0, 2]$ (in y) is revolved around the line $x = -1$. Find the volume.

Solution. Rewriting the curve as $x = y^2$, each point lies a horizontal distance $y^2 - (-1) = y^2 + 1$ from the axis:

$$\begin{aligned} V &= \pi \int_0^2 (y^2 + 1)^2 dy = \pi \int_0^2 (y^4 + 2y^2 + 1) dy = \pi \left[\frac{y^5}{5} + \frac{2y^3}{3} + y \right]_0^2 \\ &= \pi \left(\frac{32}{5} + \frac{16}{3} + 2 \right) = \pi \cdot \frac{96 + 80 + 30}{15} = \frac{206\pi}{15} \approx 43.14. \end{aligned}$$

8.11 Volume with Washer Method: Revolving Around the x- or y-Axis

The disc method assumed the revolved region touched the axis of revolution directly. When a region is instead **bounded away** from the axis — trapped between an outer curve and an inner curve — revolving it sweeps out a solid with a hole down the middle. Each cross section is no longer a full disc but a **washer**: a disc with a smaller disc removed.

Theorem 8.11.1: The Washer Method

Let $f(x) \geq g(x) \geq 0$ on $[a, b]$. Revolving the region between $y = f(x)$ (outer boundary) and $y = g(x)$ (inner boundary) around the x -axis produces a solid with volume

$$V = \int_a^b \pi \left([f(x)]^2 - [g(x)]^2 \right) dx.$$

Justification. At each x , the cross section is an annulus (a ring) of outer radius $R = f(x)$ and inner radius $r = g(x)$, with area equal to the outer disc's area minus the inner disc's: $A(x) = \pi R^2 - \pi r^2 = \pi([f(x)]^2 - [g(x)]^2)$. Theorem 8.7.1 supplies the rest. ■

AP Exam Tip

The washer formula is *not* $\pi[f(x) - g(x)]^2$ — squaring the difference of the radii is a common and costly error. Each radius must be squared *separately*, since the two cross-sectional discs (outer and, notionally, the removed inner one) have areas that add and subtract independently: $\pi R^2 - \pi r^2 \neq \pi(R - r)^2$.

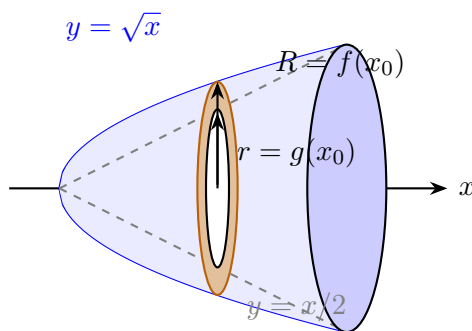
Example 8.11.1: Volume by the Washer Method

The region between $y = \sqrt{x}$ (outer) and $y = \frac{x}{2}$ (inner) on $[0, 4]$ is revolved around the x -axis. Find the volume.

Solution. The curves meet where $\sqrt{x} = \frac{x}{2}$, i.e. at $x = 0$ and $x = 4$; on $(0, 4)$,

$\sqrt{x} > \frac{x}{2}$ (checked at $x = 1$: $1 > 0.5$), so $\sqrt{x}$ is indeed the outer curve.

$$\begin{aligned} V &= \pi \int_0^4 \left[(\sqrt{x})^2 - \left(\frac{x}{2}\right)^2 \right] dx = \pi \int_0^4 \left(x - \frac{x^2}{4} \right) dx = \pi \left[\frac{x^2}{2} - \frac{x^3}{12} \right]_0^4 \\ &= \pi \left(8 - \frac{64}{12} \right) = \pi \left(8 - \frac{16}{3} \right) = \frac{8\pi}{3} \approx 8.38. \end{aligned}$$



Remark

This solid is the same trumpet-shaped solid from Example 8.9.1 with a second, smaller trumpet-shaped solid (the revolution of $y = x/2$) *removed* from its interior — visibly hollow in the middle, closing back up to a solid disc exactly at $x = 4$, where the two boundary curves meet and the “hole” pinches shut. Indeed, subtracting the two disc-method volumes gives the same answer directly:

$$8\pi - \pi \int_0^4 \left(\frac{x}{2}\right)^2 dx = 8\pi - \frac{16\pi}{3} = \frac{8\pi}{3}, \text{ matching Example 8.11.1.}$$

8.12 Volume with Washer Method: Revolving Around Other Axes

As with the disc method, the washer method needs only the radii re-measured from whatever line is the actual axis of revolution.

Theorem 8.12.1: Washer Method Around a General Axis

Revolving the region between $y = f(x)$ (outer) and $y = g(x)$ (inner) around the horizontal line $y = k$ produces

$$V = \int_a^b \pi \left([f(x) - k]^2 - [g(x) - k]^2 \right) dx,$$

with the analogous formula (integrating in y , with radii measured horizontally) for revolution around a vertical line $x = h$.

Example 8.12.1: Washer Method Around a Horizontal Line

The region between $y = \sqrt{x}$ and $y = \frac{x}{2}$ on $[0, 4]$ (Example 8.11.1) is revolved around the line $y = -1$. Find the volume.

Solution. Outer radius $R = \sqrt{x} - (-1) = \sqrt{x} + 1$; inner radius $r = \frac{x}{2} - (-1) = \frac{x}{2} + 1$.

$$R^2 - r^2 = (x + 2\sqrt{x} + 1) - \left(\frac{x^2}{4} + x + 1\right) = 2\sqrt{x} - \frac{x^2}{4}.$$

$$V = \pi \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4}\right) dx = \pi \left[\frac{4}{3}x^{3/2} - \frac{x^3}{12}\right]_0^4 = \pi \left(\frac{32}{3} - \frac{16}{3}\right) = \frac{16\pi}{3} \approx 16.76.$$

Remark

Notice how much the algebra simplifies by expanding $R^2 - r^2$ *before* integrating rather than integrating R^2 and r^2 separately and subtracting at the end — the x and 1 terms from each squared binomial cancel directly, exactly as they did (with opposite sign) in Example 8.10.1's expansion of $(\sqrt{x} + 1)^2$.

8.13 The Arc Length of a Smooth, Planar Curve and Distance Traveled

BC Only

Topic 8.13 is assessed on the BC Exam only.

Every quantity accumulated so far in this unit — average value, displacement, area, volume — reduced to summing small pieces of some two- or three-dimensional quantity. **Arc length** applies the identical strategy to the length of the curve itself: chop the curve into many tiny pieces, approximate each piece by a straight segment whose length is easy to compute, add, and take a limit.

Definition 8.13.1: Smooth Curve

A curve $y = f(x)$ is **smooth** on $[a, b]$ if f' exists and is continuous on $[a, b]$ — equivalently, the curve has a continuously turning tangent line, with no corners or vertical tangent lines.

Theorem 8.13.1: Arc Length Formula

If f is smooth on $[a, b]$, the length of the curve $y = f(x)$ from $x = a$ to $x = b$ is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

Derivation. Partition $[a, b]$ into n subintervals using points $a = x_0 < x_1 < \cdots < x_n = b$, each of width $\Delta x = \frac{b-a}{n}$, and approximate the curve on $[x_{i-1}, x_i]$ by the straight segment

connecting $(x_{i-1}, f(x_{i-1}))$ to $(x_i, f(x_i))$. By the distance formula, this segment has length

$$\ell_i = \sqrt{(\Delta x)^2 + (\Delta y_i)^2}, \quad \Delta y_i := f(x_i) - f(x_{i-1}).$$

Because f is differentiable on $[x_{i-1}, x_i]$, the Mean Value Theorem (Topic 5.2) guarantees a point $c_i \in (x_{i-1}, x_i)$ with

$$f'(c_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} = \frac{\Delta y_i}{\Delta x} \implies \Delta y_i = f'(c_i) \Delta x.$$

Substituting this into ℓ_i and factoring $(\Delta x)^2$ out of the square root (valid since $\Delta x > 0$):

$$\ell_i = \sqrt{(\Delta x)^2 + [f'(c_i)\Delta x]^2} = \sqrt{(\Delta x)^2(1 + [f'(c_i)]^2)} = \sqrt{1 + [f'(c_i)]^2} \Delta x.$$

Summing the n segment lengths approximates the total arc length:

$$L \approx \sum_{i=1}^n \ell_i = \sum_{i=1}^n \sqrt{1 + [f'(c_i)]^2} \Delta x.$$

This is exactly a Riemann sum for the function $h(x) = \sqrt{1 + [f'(x)]^2}$ on $[a, b]$, evaluated at the MVT-supplied points c_i rather than at a chosen endpoint — a valid sample point in each subinterval regardless. Because f' is continuous on $[a, b]$ (the smoothness hypothesis), h is continuous as well, so as $n \rightarrow \infty$ (equivalently $\Delta x \rightarrow 0$) this Riemann sum converges to the definite integral of h , and the polygonal approximation converges to the curve's true length:

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{1 + [f'(c_i)]^2} \Delta x = \int_a^b \sqrt{1 + [f'(x)]^2} dx. \quad \blacksquare$$

Remark

The same argument, run with the roles of x and y exchanged (slicing the curve $x = g(y)$ into horizontal steps Δy instead), gives the companion formula for a curve expressed as a function of y :

$$L = \int_c^d \sqrt{1 + [g'(y)]^2} dy.$$

Remark

Both formulas are easiest to remember through the differential $ds = \sqrt{dx^2 + dy^2}$ — the length of an infinitesimal straight piece of the curve, by the Pythagorean theorem, exactly mirroring $\ell_i = \sqrt{(\Delta x)^2 + (\Delta y_i)^2}$ above. Factoring out dx^2 or dy^2 recovers either formula:

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{or} \quad ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy, \quad L = \int ds.$$

This mnemonic previews Unit 9's parametric arc length formula $ds =$

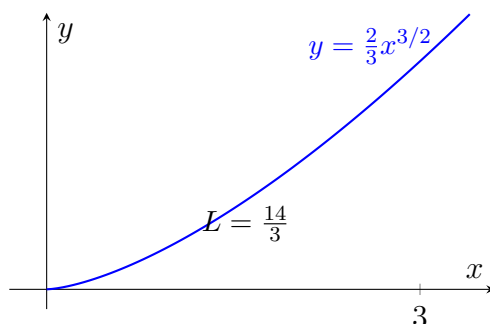
$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$ — the same Pythagorean idea, with both coordinates written in terms of a third variable t instead of one written in terms of the other.

Example 8.13.1: Computing an Arc Length

Find the length of the curve $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

Solution. $f'(x) = x^{1/2}$, so $[f'(x)]^2 = x$ and $1 + [f'(x)]^2 = 1 + x$:

$$L = \int_0^3 \sqrt{1+x} \, dx = \left[\frac{2}{3}(1+x)^{3/2} \right]_0^3 = \frac{2}{3}(4^{3/2} - 1^{3/2}) = \frac{2}{3}(8 - 1) = \frac{14}{3} \approx 4.67.$$



Example 8.13.2: Arc Length in Context: Distance Traveled Along a Path

A hiking trail follows the curve $y = \frac{2}{3}x^{3/2}$ (distances in miles) from the trailhead at $x = 0$ to a lookout at $x = 3$. Using Example 8.13.1, find the total distance hiked.

Solution. The distance hiked is exactly the arc length of the trail's graph: $L = \frac{14}{3} \approx 4.67$ miles. This is the same computation as Example 8.13.1 — the context changes only the interpretation of the answer, not the integral itself.

Remark

Most arc length integrals do not simplify as nicely as Example 8.13.1's — the square root $\sqrt{1 + [f'(x)]^2}$ rarely has an elementary antiderivative (even $y = \sin x$ produces $\int \sqrt{1 + \cos^2 x} \, dx$, which cannot be expressed in closed form). In practice, most arc length problems on the AP Exam are evaluated numerically with a calculator; the conceptual content being tested is setting up the correct integral, not necessarily evaluating it by hand.

Unit 8 completed the reinterpretation of the definite integral begun in Unit 6: the same limit-of-a-Riemann-sum construction, applied to different quantities, computes an average, a net change, an area, a volume, or a length. For students taking the BC Exam, Unit 9 extends this toolkit one more time — differentiating and integrating curves traced out parametrically or by a vector-valued function, and applying these same area and arc-length ideas to curves described in polar coordinates.

UNIT 9

Parametric Equations, Polar Coordinates, and Vector-Valued Functions

Class Periods	~10–11 (BC only)
Big Ideas	Change (CHA) Function Analysis (FUN)

BC Only

Every topic in Unit 9 is assessed on the BC Exam only.

Every curve studied so far has been the graph of $y = f(x)$: one output for each input, traced left to right as x increases. This rules out an enormous range of natural curves — a circle, a loop, a spiral, the path of a thrown ball, or the orbit of a planet — none of which are graphs of a single function of x , since a vertical line can cross them more than once. Unit 9 removes this restriction in three complementary ways. **Parametric equations** let both x and y depend separately on a third variable t (typically time), so the curve is *traced out* rather than merely plotted. **Vector-valued functions** package the same pair $(x(t), y(t))$ into a single position vector, giving velocity and acceleration natural interpretations as vectors in the plane. **Polar coordinates** describe a point by a distance r from the origin and an angle θ rather than by horizontal and vertical displacement, which is often the more natural language for curves with radial or rotational structure. In every case, the calculus built in Units 1–8 is not replaced — it is reapplied, one small step of abstraction at a time.

9.1 Defining and Differentiating Parametric Equations

Definition 9.1.1: Parametric Curve

A **parametric curve** is a pair of functions $x = x(t)$, $y = y(t)$, called **parametric equations**, where the auxiliary variable t is the **parameter**. As t ranges over an interval, the point $(x(t), y(t))$ traces out a curve in the plane.

Example 9.1.1: From Parametric to Rectangular

Identify the curve traced by $x(t) = \cos t$, $y(t) = \sin t$ for $t \in [0, 2\pi]$.

Solution. Since $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$ for every t , every point on the curve lies on the unit circle. As t increases from 0 to 2π , the point starts at $(1, 0)$ and moves *counterclockwise* around the circle exactly once. This is more information than the bare equation $x^2 + y^2 = 1$ provides: the parametrization specifies not just the shape, but the direction and starting point of the trace.

Remark

This is the essential advantage of a parametric description: it encodes *motion along* a curve, not merely the curve's shape. Two different parametrizations, such as $x = \cos t$, $y = \sin t$ and $x = \cos(2t)$, $y = \sin(2t)$, can trace the identical set of points at different speeds or in a different number of trips around — information completely invisible to the equation $x^2 + y^2 = 1$ alone.

Differentiating a parametric curve requires a new formula, but not a new idea — it is a direct application of the chain rule and the derivative of an inverse function (Topic 3.5).

Theorem 9.1.1: Derivative of a Parametrically Defined Curve

If $x(t)$ and $y(t)$ are differentiable and $\frac{dx}{dt} \neq 0$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

Proof. Since x is differentiable with $\frac{dx}{dt} \neq 0$ at the point in question, x' does not change sign in some neighborhood of that point (by the Intermediate Value Theorem applied to x' , if x' is continuous there — the case tested on the AP Exam), so $x(t)$ is strictly monotonic and therefore invertible on that neighborhood: $t = x^{-1}(x)$ is a well-defined, differentiable function of x . This makes y a genuine composite function of x :

$$y = y(t) = y(x^{-1}(x)).$$

By the chain rule,

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}.$$

By the Inverse Function Theorem (Topic 3.5), since $t = x^{-1}(x)$,

$$\frac{dt}{dx} = \frac{1}{dx/dt}.$$

Substituting,

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{dx/dt} = \frac{dy/dt}{dx/dt}. \quad \blacksquare$$

Remark

This proof is worth pausing on, because it is not a new technique in disguise — it is Topic 3.5's inverse function derivative rule and the ordinary chain rule, applied to a situation neither was originally stated for. Nothing about parametric differentiation required a new axiom; it fell out of theorems already proven.

Example 9.1.2: Computing a Parametric Derivative

Find $\frac{dy}{dx}$ for $x(t) = t^2 - 1$, $y(t) = 2t + 3$, and find the equation of the tangent line at $t = 2$.

Solution. $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 2$, so

$$\frac{dy}{dx} = \frac{2}{2t} = \frac{1}{t} \quad (t \neq 0).$$

At $t = 2$: slope $= \frac{1}{2}$, and the point is $(x(2), y(2)) = (3, 7)$. The tangent line is

$$y - 7 = \frac{1}{2}(x - 3).$$

Example 9.1.3: A Geometric Check: the Unit Circle

Find $\frac{dy}{dx}$ for $x = \cos t$, $y = \sin t$, and verify the result geometrically at $t = \frac{\pi}{4}$.

Solution.

$$\frac{dy}{dx} = \frac{\cos t}{-\sin t} = -\cot t.$$

At $t = \frac{\pi}{4}$: the point is $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and the slope is $-\cot \frac{\pi}{4} = -1$. Geometrically, the radius to this point has slope $\frac{\sqrt{2}/2}{\sqrt{2}/2} = 1$, and a tangent line to a circle is always perpendicular to the radius at the point of tangency — consistent with slopes of 1 and -1 multiplying to -1 .

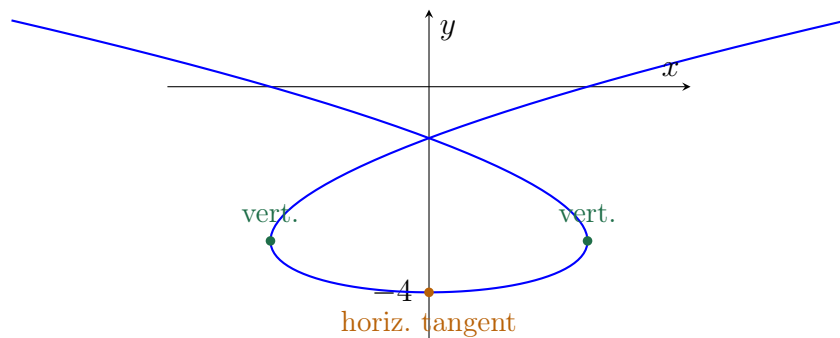
Example 9.1.4: Locating Horizontal and Vertical Tangents

Find all points where the curve $x(t) = t^3 - 3t$, $y(t) = t^2 - 4$ has a horizontal or vertical tangent line.

Solution. $\frac{dx}{dt} = 3t^2 - 3 = 3(t^2 - 1)$ and $\frac{dy}{dt} = 2t$.

Horizontal tangents occur where $\frac{dy}{dt} = 0$ while $\frac{dx}{dt} \neq 0$: $t = 0$ gives $\frac{dx}{dt} = -3 \neq 0$, so there is a horizontal tangent at $(x(0), y(0)) = (0, -4)$.

Vertical tangents occur where $\frac{dx}{dt} = 0$ while $\frac{dy}{dt} \neq 0$: $t = \pm 1$. At $t = 1$: $\frac{dy}{dt} = 2 \neq 0$, point $(1 - 3, 1 - 4) = (-2, -3)$. At $t = -1$: $\frac{dy}{dt} = -2 \neq 0$, point $(-1 + 3, 1 - 4) = (2, -3)$.



AP Exam Tip

A curve can have *both* $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$ at the same t — this typically signals a sharp corner (a **cusp**), not a horizontal or vertical tangent, and $\frac{dy}{dx}$ is indeterminate ($\frac{0}{0}$) there. Always check that only one of the two derivatives vanishes before classifying a tangent line.

9.2 Second Derivatives of Parametric Equations

Because $\frac{dy}{dx}$ from Topic 9.1 is itself a function of the parameter t , finding the second derivative $\frac{d^2y}{dx^2}$ requires differentiating *that* function with respect to x — which means applying Theorem 9.1.1 a second time.

Theorem 9.2.1: Second Derivative of a Parametric Curve

If $x(t)$ and $y(t)$ are twice differentiable and $\frac{dx}{dt} \neq 0$, then

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left[\frac{dy}{dx} \right]}{dx/dt}.$$

Proof. Let $z(t) := \frac{dy}{dx}$, itself a function of t by Theorem 9.1.1. Since $\frac{d^2y}{dx^2}$ is, by definition, the derivative of z with respect to x , and $x(t)$, $z(t)$ form another parametric pair with $\frac{dx}{dt} \neq 0$, Theorem 9.1.1 applies again — this time to the pair $(x(t), z(t))$ instead of $(x(t), y(t))$:

$$\frac{d^2y}{dx^2} = \frac{dz}{dx} = \frac{dz/dt}{dx/dt} = \frac{\frac{d}{dt} \left[\frac{dy}{dx} \right]}{dx/dt}. \quad \blacksquare$$

AP Exam Tip

$\frac{d^2y}{dx^2} \neq \frac{d^2y/dt^2}{d^2x/dt^2}$ — this is one of the most common errors on this topic. The second

derivative formula always requires differentiating $\frac{dy}{dx}$ *itself* with respect to t first, then dividing by $\frac{dx}{dt}$ once more — never a direct ratio of raw second derivatives.

Example 9.2.1: Computing a Second Derivative and Concavity

For $x(t) = t^3 - 3t$, $y(t) = t^2 - 4$ (Example 9.1.4), find $\frac{d^2y}{dx^2}$ and determine the concavity of the curve at $t = 2$.

Solution. From before, $\frac{dy}{dx} = \frac{2t}{3t^2 - 3}$. Differentiate this with respect to t using the quotient rule:

$$\frac{d}{dt} \left[\frac{2t}{3t^2 - 3} \right] = \frac{2(3t^2 - 3) - 2t(6t)}{(3t^2 - 3)^2} = \frac{-6t^2 - 6}{(3t^2 - 3)^2} = \frac{-6(t^2 + 1)}{(3t^2 - 3)^2}.$$

Dividing by $\frac{dx}{dt} = 3t^2 - 3$ once more:

$$\frac{d^2y}{dx^2} = \frac{-6(t^2 + 1)}{(3t^2 - 3)^3}.$$

At $t = 2$: $3t^2 - 3 = 9$, so $(3t^2 - 3)^3 = 729$, and $t^2 + 1 = 5$:

$$\frac{d^2y}{dx^2} = \frac{-6(5)}{729} = -\frac{30}{729} = -\frac{10}{243} \approx -0.041.$$

Since $\frac{d^2y}{dx^2} < 0$ at $t = 2$, the curve is **concave down** there.

9.3 Finding Arc Lengths of Curves Given by Parametric Equations

Topic 8.13 measured the length of a curve $y = f(x)$ by summing tiny straight-line pieces. A parametric curve admits an identical strategy, now tracking both coordinates as functions of t .

Theorem 9.3.1: Parametric Arc Length

If $x(t)$ and $y(t)$ have continuous derivatives on $[t_1, t_2]$ and the curve does not retrace itself, its length is

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

Derivation. Partition $[t_1, t_2]$ with $t_1 = \tau_0 < \tau_1 < \dots < \tau_n = t_2$, each of width Δt , and approximate the curve on $[\tau_{i-1}, \tau_i]$ by the straight segment joining $(x(\tau_{i-1}), y(\tau_{i-1}))$ to

$(x(\tau_i), y(\tau_i))$:

$$\ell_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}.$$

Applying the Mean Value Theorem to x and to y separately on $[\tau_{i-1}, \tau_i]$ gives points $c_i, d_i \in (\tau_{i-1}, \tau_i)$ with $\Delta x_i = x'(c_i) \Delta t$ and $\Delta y_i = y'(d_i) \Delta t$, so

$$\ell_i = \sqrt{[x'(c_i)]^2 + [y'(d_i)]^2} \Delta t,$$

and summing gives $L \approx \sum_{i=1}^n \sqrt{[x'(c_i)]^2 + [y'(d_i)]^2} \Delta t$. If c_i and d_i always coincided,

this would already be a Riemann sum for $\sqrt{[x'(t)]^2 + [y'(t)]^2}$ and the theorem would follow exactly as in Topic 8.13. They need not coincide, so a short additional argument is required — given in the remark below for the interested reader. Taking this gap as negligible in the limit (which the remark justifies),

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{[x'(c_i)]^2 + [y'(d_i)]^2} \Delta t = \int_{t_1}^{t_2} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt. \quad \blacksquare$$

Remark

On closing the gap rigorously. Since x' and y' are continuous on the compact interval $[t_1, t_2]$, they are *uniformly* continuous there. Given $\varepsilon > 0$, uniform continuity of x' provides $\delta > 0$ such that $|s - t| < \delta$ forces $|x'(s) - x'(t)| < \varepsilon$ for *every* pair s, t , regardless of which subinterval they lie in. Once the mesh $\Delta t < \delta$, every c_i, d_i in the same subinterval satisfy $|c_i - d_i| < \Delta t < \delta$, so $|x'(c_i) - x'(d_i)| < \varepsilon$ uniformly across all i . The reverse triangle inequality for the Euclidean norm then bounds the gap between the “mismatched” sum used above and the true Riemann sum $\sum \sqrt{[x'(d_i)]^2 + [y'(d_i)]^2} \Delta t$ by a quantity that shrinks to 0 as $\varepsilon \rightarrow 0$, uniformly in n . Hence both sums converge to the same limit, and the derivation above is fully rigorous — it simply required uniform continuity, not just continuity, of x' .

Remark

When the curve happens to be expressible as $y = f(x)$ (i.e. $\frac{dx}{dt} \neq 0$ throughout), this formula reduces to Topic 8.13's: with $dx = x'(t) dt$ and $\frac{dy}{dx} = \frac{y'(t)}{x'(t)}$,

$$\sqrt{[x'(t)]^2 + [y'(t)]^2} dt = |x'(t)| \sqrt{1 + \left[\frac{y'(t)}{x'(t)} \right]^2} dt = \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx,$$

confirming the $ds = \sqrt{dx^2 + dy^2}$ mnemonic from Topic 8.13 works identically in both settings.

Example 9.3.1: Arc Length of a Circle

Confirm that $x = \cos t$, $y = \sin t$ on $[0, 2\pi]$ has length 2π .

Solution. $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$, so

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{\sin^2 t + \cos^2 t} = 1.$$

$$L = \int_0^{2\pi} 1 \, dt = 2\pi,$$

exactly the circumference of the unit circle.

Example 9.3.2: Arc Length Requiring Substitution

Find the length of $x(t) = t^2$, $y(t) = t^3$ for $t \in [0, 1]$.

Solution. $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2$:

$$L = \int_0^1 \sqrt{(2t)^2 + (3t^2)^2} \, dt = \int_0^1 \sqrt{4t^2 + 9t^4} \, dt = \int_0^1 t\sqrt{4 + 9t^2} \, dt$$

(factoring t^2 out of the root, valid since $t \geq 0$ on $[0, 1]$). Let $u = 4 + 9t^2$, $du = 18t \, dt$; when $t = 0$, $u = 4$, and when $t = 1$, $u = 13$:

$$L = \frac{1}{18} \int_4^{13} \sqrt{u} \, du = \frac{1}{18} \cdot \frac{2}{3} \left[u^{3/2} \right]_4^{13} = \frac{1}{27} (13\sqrt{13} - 8) \approx 1.44.$$

9.4 Defining and Differentiating Vector-Valued Functions

A parametric pair $(x(t), y(t))$ can be packaged as a single object — a vector that points from the origin to the curve at each instant t . This repackaging changes none of the calculus, but it supplies exactly the right language for describing motion in the plane.

Definition 9.4.1: Vector-Valued Function

A **vector-valued function** is $\vec{r}(t) = \langle x(t), y(t) \rangle$, assigning to each t the position vector of the point $(x(t), y(t))$. When t represents time, $\vec{r}(t)$ is called the **position vector**.

Theorem 9.4.1: Differentiating a Vector-Valued Function

$\vec{r}'(t) = \langle x'(t), y'(t) \rangle$ — the derivative of a vector-valued function is computed componentwise.

Proof. By definition,

$$\vec{r}'(t) = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h} = \lim_{h \rightarrow 0} \left\langle \frac{x(t+h) - x(t)}{h}, \frac{y(t+h) - y(t)}{h} \right\rangle.$$

A vector-valued limit exists and equals $\langle L_1, L_2 \rangle$ if and only if each component converges to the corresponding L_i (since $\|\vec{v} - \langle L_1, L_2 \rangle\| \rightarrow 0$ forces each coordinate difference to 0, and conversely). Applying this componentwise,

$$\vec{r}'(t) = \left\langle \lim_{h \rightarrow 0} \frac{x(t+h) - x(t)}{h}, \lim_{h \rightarrow 0} \frac{y(t+h) - y(t)}{h} \right\rangle = \langle x'(t), y'(t) \rangle. \quad \blacksquare$$

Definition 9.4.2: Velocity, Speed, and Acceleration

For a position vector $\vec{r}(t) = \langle x(t), y(t) \rangle$:

velocity: $\vec{v}(t) = \vec{r}'(t) = \langle x'(t), y'(t) \rangle$, acceleration: $\vec{a}(t) = \vec{r}''(t) = \langle x''(t), y''(t) \rangle$,

speed: $\|\vec{v}(t)\| = \sqrt{[x'(t)]^2 + [y'(t)]^2}$.

Remark

Speed is exactly the integrand of the parametric arc length formula (Theorem 9.3.1) — a fact that will unify displacement, distance, and arc length in Topic 9.6. Geometrically, $\vec{v}(t)$ is tangent to the path at $\vec{r}(t)$ and points in the direction of motion, exactly as $\frac{dy}{dx}$ described the tangent's slope in Topic 9.1.

Example 9.4.1: Velocity, Speed, and Acceleration at a Point

For $\vec{r}(t) = \langle t^2, t^3 - 3t \rangle$, find the velocity, speed, and acceleration at $t = 1$.

Solution. $\vec{r}'(t) = \langle 2t, 3t^2 - 3 \rangle$, so $\vec{v}(1) = \langle 2, 0 \rangle$. Speed: $\|\vec{v}(1)\| = \sqrt{2^2 + 0^2} = 2$. Acceleration: $\vec{r}''(t) = \langle 2, 6t \rangle$, so $\vec{a}(1) = \langle 2, 6 \rangle$.

9.5 Integrating Vector-Valued Functions

Antidifferentiation extends to vector-valued functions exactly as differentiation did: componentwise, with a vector constant of integration.

Theorem 9.5.1: Integrating a Vector-Valued Function

$$\int \vec{r}(t) dt = \left\langle \int x(t) dt, \int y(t) dt \right\rangle + \vec{C},$$

and for the definite integral, the Fundamental Theorem of Calculus applies componentwise:

$$\vec{r}(t) = \vec{r}(t_0) + \int_{t_0}^t \vec{v}(s) ds.$$

Example 9.5.1: Recovering Position from Velocity

A particle has velocity $\vec{v}(t) = \langle 2t, 3t^2 \rangle$ and $\vec{r}(0) = \langle 1, -2 \rangle$. Find $\vec{r}(t)$ and $\vec{r}(2)$.

Solution. Antidifferentiating componentwise:

$$\vec{r}(t) = \langle t^2 + C_1, t^3 + C_2 \rangle.$$

Using $\vec{r}(0) = \langle 1, -2 \rangle$: $C_1 = 1$, $C_2 = -2$, so $\vec{r}(t) = \langle t^2+1, t^3-2 \rangle$. Then $\vec{r}(2) = \langle 5, 6 \rangle$.

9.6 Solving Motion Problems Using Parametric and Vector-Valued Functions

Topic 8.2 studied a particle moving along a *line*: position, velocity, and acceleration were all scalars, and integrating velocity gave displacement and distance. Every one of those ideas transfers directly to a particle moving anywhere in the *plane* — position, velocity, and acceleration simply become vectors.

Theorem 9.6.1: Displacement and Total Distance in the Plane

For a particle with velocity $\vec{v}(t)$ on $[a, b]$:

$$\text{displacement} = \int_a^b \vec{v}(t) dt = \vec{r}(b) - \vec{r}(a), \quad \text{total distance traveled} = \int_a^b \|\vec{v}(t)\| dt.$$

Remark

Displacement is now a *vector* — a net change in position with both magnitude and direction — while total distance is the same scalar quantity as always. Notice that the total-distance integral is exactly the arc length integral of Theorem 9.3.1: *the total distance a particle travels equals the arc length of the path it traces*. This is the payoff of building Topics 9.3 and 9.4–9.5 side by side.

Example 9.6.1: Direction of Motion

A particle moves with position $\vec{r}(t) = \langle t^3 - 3t, t^2 \rangle$. Determine the direction of motion at $t = 0.5$.

Solution. $\vec{v}(t) = \langle 3t^2 - 3, 2t \rangle$, so $\vec{v}(0.5) = \langle 3(0.25) - 3, 1 \rangle = \langle -2.25, 1 \rangle$. Since the first component is negative and the second is positive, the particle is moving **up and to the left** at this instant.

Example 9.6.2: Reconstructing Position from Acceleration

A particle has acceleration $\vec{a}(t) = \langle 6t, 2 \rangle$, initial velocity $\vec{v}(0) = \langle -1, 3 \rangle$, and initial position $\vec{r}(0) = \langle 0, 0 \rangle$. Find $\vec{r}(t)$.

Solution. Antidifferentiate $\vec{a}(t)$ to get $\vec{v}(t)$:

$$\vec{v}(t) = \langle 3t^2 + C_1, 2t + C_2 \rangle.$$

Using $\vec{v}(0) = \langle -1, 3 \rangle$: $C_1 = -1$, $C_2 = 3$, so $\vec{v}(t) = \langle 3t^2 - 1, 2t + 3 \rangle$. Antidifferentiate again:

$$\vec{r}(t) = \langle t^3 - t + D_1, t^2 + 3t + D_2 \rangle.$$

Using $\vec{r}(0) = \langle 0, 0 \rangle$: $D_1 = D_2 = 0$, so

$$\vec{r}(t) = \langle t^3 - t, t^2 + 3t \rangle.$$

Example 9.6.3: Displacement vs. Total Distance in the Plane

A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, 2\pi]$, with $\vec{r}(0) = \langle 0, 1 \rangle$. Find the displacement and the total distance traveled.

Solution. Antidifferentiating: $\vec{r}(t) = \langle \sin t + C_1, -\cos t + C_2 \rangle$. Using $\vec{r}(0) = \langle 0, 1 \rangle$: $C_1 = 0$, $-1 + C_2 = 1 \Rightarrow C_2 = 2$, so $\vec{r}(t) = \langle \sin t, 2 - \cos t \rangle$ — a circle of radius 1 centered at $(0, 2)$.

Displacement:

$$\vec{r}(2\pi) - \vec{r}(0) = \langle 0, 2 - 1 \rangle - \langle 0, 1 \rangle = \langle 0, 0 \rangle.$$

Total distance: the speed is $\|\vec{v}(t)\| = \sqrt{\cos^2 t + \sin^2 t} = 1$ (constant), so

$$\int_0^{2\pi} \|\vec{v}(t)\| dt = \int_0^{2\pi} 1 dt = 2\pi.$$

The particle returns exactly to its starting point (zero displacement) after tracing the full circle once at constant speed 1, covering a total distance equal to the circle's circumference, 2π — consistent with Example 9.3.1.

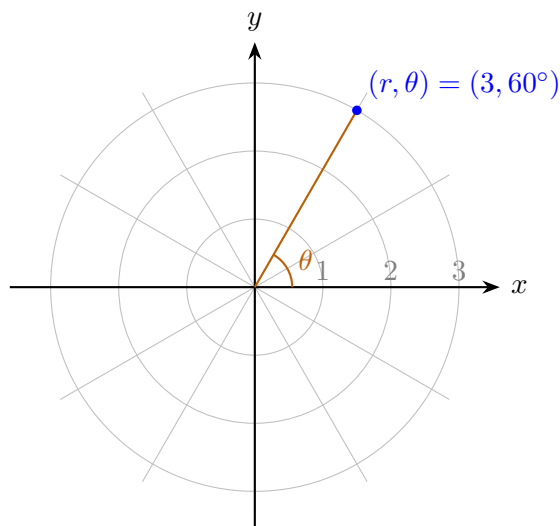
9.7 Defining Polar Coordinates and Differentiating in Polar Form

Every curve so far has been located by horizontal and vertical displacement from the origin. **Polar coordinates** instead locate a point by a distance from the origin and a direction — a natural language for anything with rotational structure: spirals, flowers, orbits, antenna patterns, and more.

Definition 9.7.1: Polar Coordinates

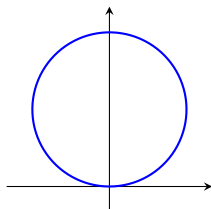
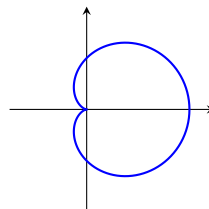
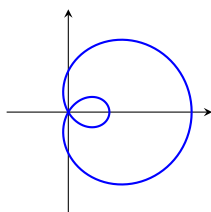
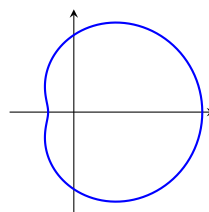
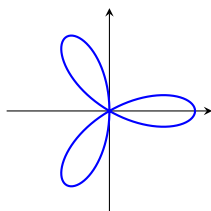
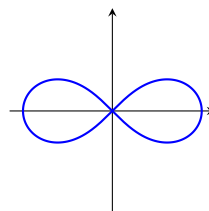
A point is described in **polar coordinates** (r, θ) by a distance r from the origin (the **pole**) and an angle θ measured counterclockwise from the positive x -axis (the **polar axis**). Polar and rectangular coordinates are related by

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x} \quad (x \neq 0).$$

**Remark**

Unlike rectangular coordinates, the same point has infinitely many polar representations: (r, θ) , $(r, \theta + 360^\circ n)$, and $(-r, \theta + 180^\circ)$ all describe the same location (the last because a *negative* radius is plotted in the *opposite* direction). This non-uniqueness is a genuine source of subtlety later, especially when locating intersections of two polar curves (Topic 9.9).

Polar curves come in a rich variety of standard families. The gallery below shows six of the most common shapes, all generated by an equation of the form $r = f(\theta)$.

circle: $r = 3 \sin \theta$ cardioid: $r = 1 + \cos \theta$ limaçon w/ loop: $r = 1 + 2 \cos \theta$ dimpled limaçon: $r = 3 + 2 \cos \theta$ rose: $r = \cos 3\theta$ lemniscate: $r^2 = 4 \cos 2\theta$ 

Remark

The limaçon family $r = a + b \cos \theta$ alone produces four qualitatively different shapes depending on the ratio b/a : a convex limaçon ($|b/a| < \frac{1}{2}$), a dimpled limaçon ($\frac{1}{2} < |b/a| < 1$), a cardioid (exactly $|b/a| = 1$), and a limaçon with an inner loop ($|b/a| > 1$). A rose $r = \cos(n\theta)$ has n petals when n is odd and $2n$ petals when n is even — a consequence of how many times $\cos(n\theta)$ completes a full sign cycle as θ ranges over $[0, 2\pi)$.

Differentiating a polar curve requires no new machinery: treating θ as the parameter converts $r = f(\theta)$ into the parametric pair $x(\theta) = f(\theta) \cos \theta$, $y(\theta) = f(\theta) \sin \theta$, and Topic 9.1 applies directly.

Theorem 9.7.1: Slope of a Polar Curve

For $r = f(\theta)$,

$$\frac{dx}{d\theta} = \frac{dr}{d\theta} \cos \theta - r \sin \theta, \quad \frac{dy}{d\theta} = \frac{dr}{d\theta} \sin \theta + r \cos \theta,$$

and consequently

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}.$$

Derivation. Since $x = r(\theta) \cos \theta$ is a product of two functions of θ , the product rule gives $\frac{dx}{d\theta} = \frac{dr}{d\theta} \cos \theta + r \cdot (-\sin \theta)$, and similarly for $y = r(\theta) \sin \theta$. Substituting both into Theorem 9.1.1's formula $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$ (with θ playing the role of the parameter t) gives the result. ■

Example 9.7.1: Tangent Line to a Cardioid

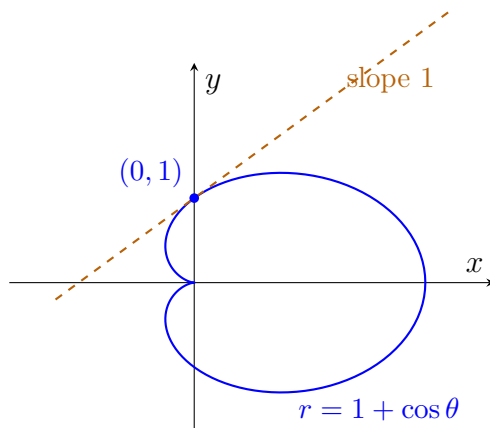
Find the slope of the tangent line to $r = 1 + \cos \theta$ at $\theta = \frac{\pi}{2}$.

Solution. $\frac{dr}{d\theta} = -\sin \theta$. At $\theta = \frac{\pi}{2}$: $r = 1 + \cos \frac{\pi}{2} = 1$ and $\frac{dr}{d\theta} = -1$.

$$\frac{dx}{d\theta} = (-1) \cos \frac{\pi}{2} - (1) \sin \frac{\pi}{2} = 0 - 1 = -1, \quad \frac{dy}{d\theta} = (-1) \sin \frac{\pi}{2} + (1) \cos \frac{\pi}{2} = -1 + 0 = -1.$$

$$\frac{dy}{dx} = \frac{-1}{-1} = 1.$$

The point of tangency is $(r \cos \theta, r \sin \theta) = (0, 1)$.



9.8 Find the Area of a Polar Region or the Area Bounded by a Single Polar Curve

Rectangular area (Topic 8.1 onward) was built from thin *rectangles* — the natural shape for a region sliced by vertical or horizontal lines. A polar region is far more naturally sliced by thin **circular sectors** radiating from the pole, and the same Riemann-sum strategy applies once the area of a single sector is known.

Theorem 9.8.1: Area of a Circular Sector

A sector of a circle of radius r subtending an angle $\Delta\theta$ (in radians) has area $\frac{1}{2}r^2 \Delta\theta$.

Justification. A full circle of radius r has area πr^2 and subtends an angle of 2π . A sector subtending $\Delta\theta$ is the fraction $\frac{\Delta\theta}{2\pi}$ of the full circle, so its area is $\pi r^2 \cdot \frac{\Delta\theta}{2\pi} = \frac{1}{2}r^2 \Delta\theta$. ■

Theorem 9.8.2: Area Enclosed by a Polar Curve

If $r = f(\theta)$ is continuous and $f(\theta) \geq 0$ on $[\alpha, \beta]$ with $\beta - \alpha \leq 2\pi$, the area of the region swept out as θ runs from α to β is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)]^2 d\theta = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta.$$

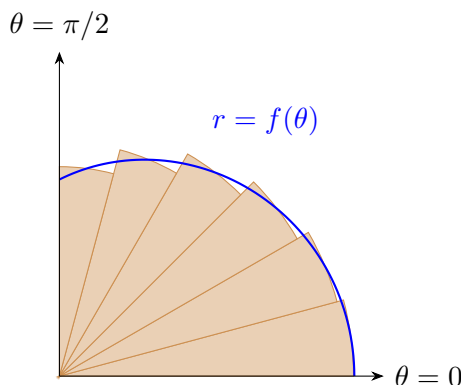
Derivation. Partition $[\alpha, \beta]$ into n subintervals of equal width $\Delta\theta$, using sample angles θ_i^* in the i th subinterval. Approximate the thin curved wedge over that subinterval by a genuine circular sector of radius $f(\theta_i^*)$ and angle $\Delta\theta$; by Theorem 9.8.1 this sector has area $\frac{1}{2}[f(\theta_i^*)]^2 \Delta\theta$. Summing over all n wedges approximates the total area:

$$A \approx \sum_{i=1}^n \frac{1}{2} [f(\theta_i^*)]^2 \Delta\theta.$$

Because f is continuous, so is f^2 , and this sum is precisely a Riemann sum for $\frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)]^2 d\theta$. As $n \rightarrow \infty$ (equivalently $\Delta\theta \rightarrow 0$), the sector approximation's error vanishes and the Rie-

Riemann sum converges to the exact area:

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{2} [f(\theta_i^*)]^2 \Delta\theta = \frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)]^2 d\theta. \quad \blacksquare$$



Remark

Compare this derivation to the disc method's (Theorem 8.9.1): there, thin *cylinders* approximated a solid and πr^2 (a circle's area) was the key building block; here, thin *sectors* approximate a region and $\frac{1}{2}r^2\Delta\theta$ (a sector's area) is the building block. Both are the identical Riemann-sum argument, adapted to whatever shape best matches the geometry at hand.

Example 9.8.1: Area Enclosed by a Cardioid

Find the area enclosed by $r = 1 + \cos \theta$.

Solution. The full cardioid is traced once as θ runs from 0 to 2π :

$$A = \frac{1}{2} \int_0^{2\pi} (1 + \cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta.$$

Using $\int_0^{2\pi} 1 d\theta = 2\pi$, $\int_0^{2\pi} 2 \cos \theta d\theta = 0$, and $\int_0^{2\pi} \cos^2 \theta d\theta = \pi$ (Topic 6.9's power-reduction identity):

$$A = \frac{1}{2}(2\pi + 0 + \pi) = \frac{3\pi}{2} \approx 4.71.$$

Example 9.8.2: Area of One Petal of a Rose

Find the area of one petal of $r = \cos(2\theta)$.

Solution. A single petal is traced as θ ranges over the values where $\cos(2\theta) \geq 0$ closest to 0: solving $\cos(2\theta) = 0$ nearest the origin gives $\theta = \pm \frac{\pi}{4}$, so one petal corresponds to $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

$$A = \frac{1}{2} \int_{-\pi/4}^{\pi/4} \cos^2(2\theta) d\theta.$$

Using $\cos^2 u = \frac{1+\cos 2u}{2}$ with $u = 2\theta$:

$$A = \frac{1}{2} \int_{-\pi/4}^{\pi/4} \frac{1 + \cos 4\theta}{2} d\theta = \frac{1}{4} \left[\theta + \frac{\sin 4\theta}{4} \right]_{-\pi/4}^{\pi/4} = \frac{1}{4} \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right) = \frac{\pi}{8}.$$

AP Exam Tip

Finding the correct bounds is usually the hardest part of a polar area problem — *always* sketch the curve, or at minimum solve $f(\theta) = 0$ to identify where the curve returns to the pole, before setting up the integral. Integrating over the wrong interval is the single most common error on this topic.

9.9 Finding the Area of the Region Bounded by Two Polar Curves

Just as Topic 8.11's washer method subtracted an inner disc's area from an outer one, the area *between* two polar curves is found by subtracting the inner sector's area from the outer sector's area at each angle.

Theorem 9.9.1: Area Between Two Polar Curves

If $R(\theta) \geq r(\theta) \geq 0$ on $[\alpha, \beta]$, the area of the region between the outer curve $\rho = R(\theta)$ and the inner curve $\rho = r(\theta)$ is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} \left([R(\theta)]^2 - [r(\theta)]^2 \right) d\theta.$$

Example 9.9.1: Area Inside One Curve and Outside Another

Find the area inside $r = 1 + \cos \theta$ and outside $r = 1$.

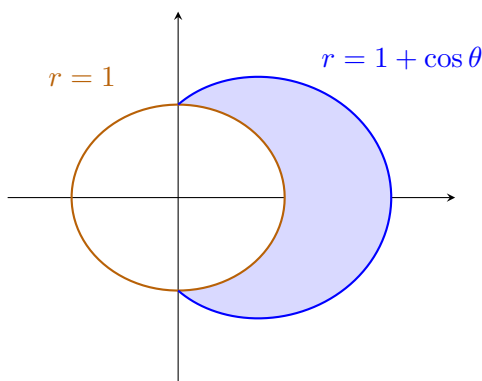
Solution. Find intersections: $1 + \cos \theta = 1 \Rightarrow \cos \theta = 0 \Rightarrow \theta = \pm \frac{\pi}{2}$. For $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, the cardioid is outer (checked at $\theta = 0$: $2 > 1$):

$$A = \frac{1}{2} \int_{-\pi/2}^{\pi/2} \left[(1 + \cos \theta)^2 - 1 \right] d\theta = \frac{1}{2} \int_{-\pi/2}^{\pi/2} (2 \cos \theta + \cos^2 \theta) d\theta.$$

By symmetry, this equals $\int_0^{\pi/2} (2 \cos \theta + \cos^2 \theta) d\theta$:

$$\int_0^{\pi/2} 2 \cos \theta d\theta = 2, \quad \int_0^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{4}.$$

$$A = 2 + \frac{\pi}{4} \approx 2.79.$$



Two curves given in polar form can also intersect at the pole for *different* values of θ — a subtlety with no rectangular analogue, caused directly by the non-uniqueness of polar coordinates (Topic 9.7's opening remark).

AP Exam Tip

Solving $f(\theta) = g(\theta)$ finds every intersection where both curves reach the *same* point at the *same* θ — but if each curve passes through the pole at a *different* θ , that shared point (the origin) will never appear as a solution to $f(\theta) = g(\theta)$. Always check separately whether $f(\theta) = 0$ and $g(\theta) = 0$ have solutions, even after solving for the ordinary intersections.

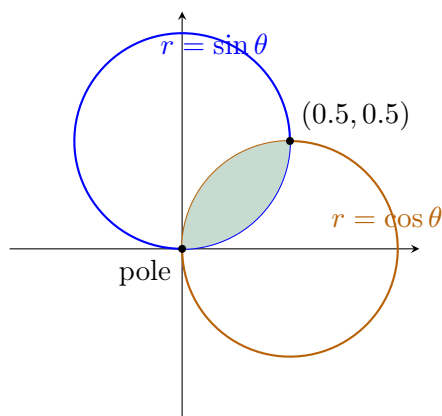
Example 9.9.2: Finding All Intersections, Including the Pole

Find every point where $r = \cos \theta$ and $r = \sin \theta$ intersect, and use the result to find the area common to both curves (both are circles of radius $\frac{1}{2}$).

Solution. Setting $\cos \theta = \sin \theta$ gives $\theta = \frac{\pi}{4}$, where $r = \frac{\sqrt{2}}{2}$, i.e. the point $(\frac{1}{2}, \frac{1}{2})$. But $r = \cos \theta = 0$ at $\theta = \frac{\pi}{2}$, while $r = \sin \theta = 0$ at $\theta = 0$ — each curve passes through the pole, just at different θ -values for its own equation. So the **pole is also a common point**, even though it never appears as a solution of $\cos \theta = \sin \theta$. There are exactly two intersection points: $(\frac{1}{2}, \frac{1}{2})$ and the origin.

For $\theta \in [0, \frac{\pi}{4}]$, $\sin \theta \leq \cos \theta$, so the region common to both disks is bounded by $r = \sin \theta$ (the smaller radius); by symmetry across $\theta = \frac{\pi}{4}$, the area for $\theta \in [\frac{\pi}{4}, \frac{\pi}{2}]$ contributes an equal amount bounded by $r = \cos \theta$. So

$$A = 2 \cdot \frac{1}{2} \int_0^{\pi/4} \sin^2 \theta \, d\theta = \int_0^{\pi/4} \sin^2 \theta \, d\theta = \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{\pi/4} = \frac{\pi}{8} - \frac{1}{4} \approx 0.14.$$



Unit 9 rebuilt the entire calculus of curves — tangent lines, arc length, and area — for three settings a single function $y = f(x)$ cannot reach: curves traced by a parameter, motion described by vectors, and regions with rotational symmetry. In every case, no new axiom was required; the Fundamental Theorem, the chain rule, and the Riemann-sum construction of the integral did all of the work once more. Unit 10 turns to a final, very different question: not the length or area of a curve, but whether an *infinite sum* of numbers can itself settle on a finite value.

UNIT 10

Infinite Sequences and Series

Class Periods	~17–18 (BC only)
Big Idea	Limits (LIM)

BC Only

Every topic in Unit 10 is assessed on the BC Exam only.

Adding finitely many numbers always produces an answer. Adding *infinitely* many is a genuinely different question, and the answer is not obviously defined at all: what could $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ possibly mean, and why should it equal 2 rather than growing without bound? Unit 10 answers this by falling back on the concept that has anchored the entire course — the limit. An infinite sum will be *defined* as the limit of its finite partial sums, converting an impossible-looking question about infinity into an ordinary limit question of the kind Unit 1 already settled.

Once that definition is in place, the unit divides naturally in two. Topics 10.1–10.10 develop a toolkit of **convergence tests** for deciding whether a given infinite sum settles on a finite value. Topics 10.11–10.15 then run the idea in reverse, using infinite sums of *powers of x* to represent familiar functions like e^x , $\sin x$, and $\ln(1+x)$ exactly — closing the course with a result of remarkable reach: that a function’s derivatives at a single point can determine its values everywhere.

10.1 Defining Convergent and Divergent Infinite Series

Definition 10.1.1: Sequence and Series

A **sequence** $\{a_n\}$ is an ordered list of numbers $a_1, a_2, a_3, \dots$. The associated **infinite series** is the formal sum

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots,$$

and its n th **partial sum** is the ordinary finite sum

$$s_n = \sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n.$$

Definition 10.1.2: Convergence of a Series

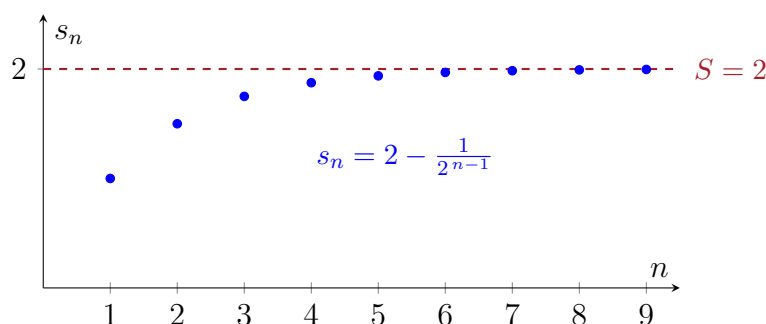
The series $\sum_{n=1}^{\infty} a_n$ **converges** to S if the *sequence of partial sums* $\{s_n\}$ converges to S :

$$\lim_{n \rightarrow \infty} s_n = S, \quad \text{in which case we write } \sum_{n=1}^{\infty} a_n = S.$$

If $\lim_{n \rightarrow \infty} s_n$ does not exist (or is infinite), the series **diverges**.

Remark

This definition is the whole conceptual content of the unit. An infinite sum is *never* evaluated by “adding infinitely many terms” — that operation is undefined. Instead, one adds finitely many terms (which is ordinary arithmetic), producing a *sequence* of answers $s_1, s_2, s_3, \dots$, and then asks the Unit 1 question: does that sequence approach a limit? Every test in this unit is ultimately a tool for answering that single question without computing the limit directly.

**Remark**

The picture above shows the partial sums of $1 + \frac{1}{2} + \frac{1}{4} + \dots$. No partial sum ever *equals* 2 — each falls short by exactly $\frac{1}{2^{n-1}}$ — yet the sequence converges to 2, which is precisely what the statement “ $\sum_{n=1}^{\infty} (\frac{1}{2})^{n-1} = 2$ ” asserts. Compare this to a vertical asymptote in Unit 1: getting arbitrarily close without arriving is exactly what a limit describes.

Example 10.1.1: A Telescoping Series

Determine whether $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges, and if so find its sum.

Solution. Partial fractions (Topic 6.12) give $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, so the partial sum collapses:

$$s_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1},$$

since every interior term cancels with its neighbor. Then

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) = 1,$$

so the series converges to 1.

Remark

Telescoping series and geometric series (Topic 10.2) are essentially the only two families for which the partial sum s_n has a usable closed form, making them the only two for which the sum can be computed directly from the definition. For every other series in this unit, the tests decide *whether* a sum exists without revealing what it is — a distinction worth keeping firmly in mind.

10.2 Working with Geometric Series

A **geometric series** is one in which each term is a fixed multiple of the previous one. It is the single most important series in the course, both because its sum can be computed exactly and because almost every later test works by comparing an unknown series to a geometric one.

Theorem 10.2.1: Geometric Series

For $a \neq 0$, the series $\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \cdots$ converges if and only if $|r| < 1$, in which case

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.$$

If $|r| \geq 1$ the series diverges.

Proof. Consider the partial sum $s_n = a + ar + ar^2 + \cdots + ar^{n-1}$. Multiply by r and subtract:

$$s_n - rs_n = (a + ar + \cdots + ar^{n-1}) - (ar + ar^2 + \cdots + ar^n) = a - ar^n,$$

since every term except the first of s_n and the last of rs_n cancels. Factoring the left side, $s_n(1-r) = a(1-r^n)$, so for $r \neq 1$,

$$s_n = \frac{a(1-r^n)}{1-r}.$$

Now take the limit. If $|r| < 1$, then $r^n \rightarrow 0$ (Topic 1.5's exponential limit behavior), so

$$\lim_{n \rightarrow \infty} s_n = \frac{a(1-0)}{1-r} = \frac{a}{1-r},$$

a finite value — the series converges. If $|r| > 1$, then $|r^n| \rightarrow \infty$ and s_n has no finite limit. If $r = 1$, the formula above does not apply, but directly $s_n = na \rightarrow \pm\infty$. If $r = -1$, then

s_n alternates between a and 0, so $\lim s_n$ does not exist. In every case with $|r| \geq 1$ the series diverges. ■

AP Exam Tip

Read the starting index carefully. The formula $\frac{a}{1-r}$ requires a to be the *first term actually written*, whatever the index. For $\sum_{n=1}^{\infty} ar^n$ the first term is ar , not a , so the sum is $\frac{ar}{1-r}$. When in doubt, write out the first two or three terms explicitly and identify a and r from them.

Example 10.2.1: Summing Geometric Series

Evaluate each series, or show it diverges.

(a) $\sum_{n=0}^{\infty} 3 \left(\frac{1}{2}\right)^n$

(b) $\sum_{n=1}^{\infty} 5 \left(\frac{2}{3}\right)^n$

(c) $\sum_{n=0}^{\infty} \left(\frac{5}{4}\right)^n$

Solution.

(a) First term $a = 3$, ratio $r = \frac{1}{2}$. Since $|r| < 1$,

$$\sum_{n=0}^{\infty} 3 \left(\frac{1}{2}\right)^n = \frac{3}{1 - \frac{1}{2}} = 6.$$

(b) The first term (at $n = 1$) is $5 \cdot \frac{2}{3} = \frac{10}{3}$, and $r = \frac{2}{3}$:

$$\sum_{n=1}^{\infty} 5 \left(\frac{2}{3}\right)^n = \frac{10/3}{1 - \frac{2}{3}} = \frac{10/3}{1/3} = 10.$$

(c) $r = \frac{5}{4}$ and $|r| > 1$, so the series **diverges**.

Example 10.2.2: A Repeating Decimal as a Geometric Series

Express $0.\overline{27} = 0.272727\ldots$ as a fraction.

Solution. Write the decimal as a sum:

$$0.\overline{27} = \frac{27}{100} + \frac{27}{100^2} + \frac{27}{100^3} + \cdots,$$

a geometric series with $a = \frac{27}{100}$ and $r = \frac{1}{100}$. Since $|r| < 1$,

$$0.\overline{27} = \frac{27/100}{1 - \frac{1}{100}} = \frac{27/100}{99/100} = \frac{27}{99} = \frac{3}{11}.$$

Remark

This example explains a fact usually asserted without justification in earlier courses: every repeating decimal equals a rational number. The repeating block generates a geometric series with ratio 10^{-k} (where k is the block length), and Theorem 10.2.1 guarantees that this sum is a ratio of integers.

10.3 The n th Term Test for Divergence

Before developing tests that confirm convergence, it is worth having one quick test that can rule it out. The reasoning behind it is direct: if the running totals are settling down to a limit, then the amounts being added must be shrinking to nothing.

Theorem 10.3.1: The n th Term Test for Divergence

If $\lim_{n \rightarrow \infty} a_n \neq 0$ (including the case where the limit fails to exist), then $\sum_{n=1}^{\infty} a_n$ **diverges**.

Proof. We prove the contrapositive: *if the series converges, then $a_n \rightarrow 0$* . Suppose $\sum_{n=1}^{\infty} a_n$ converges to S , so $\lim_{n \rightarrow \infty} s_n = S$. Since $\{s_{n-1}\}$ is the same sequence shifted by one index, it converges to the same limit: $\lim_{n \rightarrow \infty} s_{n-1} = S$ as well. Now observe that for $n \geq 2$,

$$a_n = (a_1 + \cdots + a_n) - (a_1 + \cdots + a_{n-1}) = s_n - s_{n-1}.$$

Taking limits on both sides and using the difference law for limits (Topic 1.5),

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} = S - S = 0.$$

Contrapositively, if $\lim_{n \rightarrow \infty} a_n \neq 0$ then the series cannot converge. ■

AP Exam Tip

This test can **only ever prove divergence** — never convergence. If $a_n \rightarrow 0$, the test is *inconclusive* and says nothing at all. The harmonic series $\sum \frac{1}{n}$ (Topic 10.5) has $a_n \rightarrow 0$ yet diverges, which is exactly why the converse of Theorem 10.3.1 is false. Writing “ $a_n \rightarrow 0$, therefore the series converges” is the single most heavily penalized error in this unit.

Example 10.3.1: Applying the n th Term Test

Determine what the n th Term Test says about each series.

(a) $\sum_{n=1}^{\infty} \frac{n}{2n+1}$

(b) $\sum_{n=1}^{\infty} (-1)^n$

(c) $\sum_{n=1}^{\infty} \frac{1}{n^2}$

Solution.

(a) $\lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2} \neq 0$, so the series **diverges**.

(b) $\lim_{n \rightarrow \infty} (-1)^n$ does not exist, so the series **diverges**.

(c) $\lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$, so the test is **inconclusive** — no conclusion may be drawn from it. (This series does in fact converge, as Topic 10.5 will establish by other means.)

Before developing tests that can *confirm* convergence, one structural fact does most of the work behind the scenes. A series of positive terms has increasing partial sums, and an increasing sequence has exactly two possible fates.

Theorem 10.3.2: Monotone Convergence Theorem

An increasing sequence that is **bounded above** converges. An increasing sequence that is not bounded above diverges to ∞ .

Remark

This is the crucial bridge from “bounded” to “convergent,” and it is what makes the Integral Test and both Comparison Tests work. For a series with $a_n \geq 0$, the partial sums satisfy $s_{n+1} = s_n + a_{n+1} \geq s_n$, so $\{s_n\}$ is automatically increasing. Therefore, for a positive-term series, *convergence is equivalent to the partial sums being bounded above* — and every test in Topics 10.4–10.6 is a strategy for establishing that bound by comparison with something already understood.

10.4 Integral Test for Convergence

Topic 6.13 foreshadowed this exact connection: rectangles built from a positive, decreasing function trap the function’s values against its improper integral, so the sum and the integral must share the same fate.

Theorem 10.4.1: The Integral Test

Let f be continuous, positive, and decreasing on $[1, \infty)$, and suppose $a_n = f(n)$. Then

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \int_1^{\infty} f(x) dx$$

either both converge or both diverge.

Proof. Because f is decreasing, on each subinterval $[k, k+1]$ its largest value is $f(k)$ and its smallest is $f(k+1)$. Comparing the area under f on that subinterval to the two rectangles of width 1 that sandwich it,

$$f(k+1) \cdot 1 \leq \int_k^{k+1} f(x) dx \leq f(k) \cdot 1, \quad \text{i.e.} \quad a_{k+1} \leq \int_k^{k+1} f(x) dx \leq a_k.$$

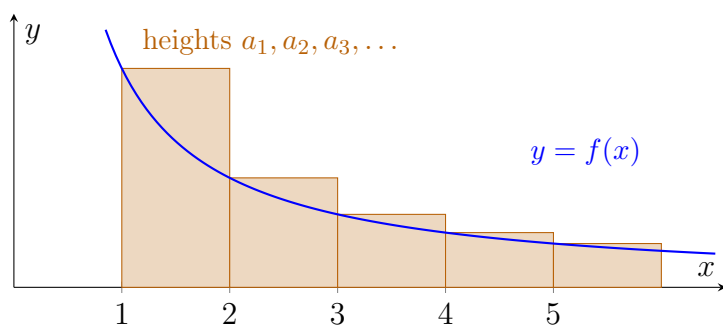
Summing these inequalities for $k = 1, 2, \dots, n-1$ and using the additivity of the integral (Topic 6.6):

$$\underbrace{a_2 + a_3 + \cdots + a_n}_{= s_n - a_1} \leq \int_1^n f(x) dx \leq \underbrace{a_1 + a_2 + \cdots + a_{n-1}}_{= s_{n-1}}.$$

Case 1: the integral converges, say to the finite value I . Since $f > 0$, $\int_1^n f \leq I$ for every n , so the left inequality gives $s_n - a_1 \leq I$, i.e. $s_n \leq a_1 + I$ for all n . The partial sums are increasing (as $a_n > 0$) and bounded above by $a_1 + I$, so by the Monotone Convergence Theorem the series converges.

Case 2: the integral diverges, i.e. $\int_1^n f \rightarrow \infty$. The right inequality gives $s_{n-1} \geq \int_1^n f$, so $s_{n-1} \rightarrow \infty$ as well and the series diverges.

Since these two cases are exhaustive and yield matching conclusions, the series and the integral always share the same behavior. ■

**Remark**

The rectangles shown overestimate the area under the curve, giving $\sum_{k=1}^{n-1} a_k \geq \int_1^n f(x) dx$ — the right-hand inequality of the proof. Shifting every rectangle one unit left produces the underestimate that supplies the left-hand inequality. Trap-

ping the integral between two versions of the same sum is the entire mechanism of the test.

AP Exam Tip

The three hypotheses — continuous, positive, and *decreasing* — must be explicitly verified on the AP Exam before the test is applied; merely naming the test earns no credit. “Decreasing” is usually confirmed by checking $f'(x) < 0$. Also note that the test only requires these conditions *eventually* (for $x \geq N$), since changing finitely many terms cannot affect convergence.

Example 10.4.1: Applying the Integral Test

Determine whether $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ converges.

Solution. Let $f(x) = \frac{1}{x \ln x}$ on $[2, \infty)$. It is continuous there (as $\ln x \neq 0$ for $x \geq 2$), positive, and decreasing (both x and $\ln x$ increase, so the denominator grows and the fraction shrinks). Applying the test with $u = \ln x$, $du = dx/x$:

$$\int_2^{\infty} \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} \left[\ln |\ln x| \right]_2^b = \lim_{b \rightarrow \infty} (\ln(\ln b) - \ln(\ln 2)) = \infty.$$

The integral diverges, so the series **diverges**.

Example 10.4.2: A Convergent Case

Determine whether $\sum_{n=1}^{\infty} n e^{-n^2}$ converges.

Solution. Let $f(x) = x e^{-x^2}$, which is continuous and positive on $[1, \infty)$; it is decreasing there since $f'(x) = e^{-x^2}(1 - 2x^2) < 0$ for $x \geq 1$. With $u = x^2$, $du = 2x dx$:

$$\int_1^{\infty} x e^{-x^2} dx = \lim_{b \rightarrow \infty} \left[-\frac{1}{2} e^{-x^2} \right]_1^b = \lim_{b \rightarrow \infty} \left(\frac{e^{-1}}{2} - \frac{e^{-b^2}}{2} \right) = \frac{1}{2e}.$$

The integral converges, so the series **converges**.

AP Exam Tip

The Integral Test tells you *whether* a series converges, never *to what*. In Example 10.4.2 the integral equals $\frac{1}{2e}$, but the series does **not** sum to $\frac{1}{2e}$ — the proof’s inequalities show only that the two quantities are within a_1 of each other. Claiming the sum equals the integral is a common and costly error.

10.5 Harmonic Series and p-Series

Applying the Integral Test to the single family $f(x) = x^{-p}$ settles an entire class of series at once — and the required integral was already computed, in full, back in Topic 6.13.

Definition 10.5.1: p-Series and the Harmonic Series

A **p-series** is a series of the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$ for a constant $p > 0$. The case $p = 1$,

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots,$$

is called the **harmonic series**.

Theorem 10.5.1: The p-Series Test

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$. In particular, the harmonic series ($p = 1$) **diverges**.

Proof. For $p > 0$ the function $f(x) = x^{-p}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test (Theorem 10.4.1) applies and the series shares the behavior of $\int_1^{\infty} x^{-p} dx$. That integral was evaluated completely in Theorem 6.13.1, where writing $b^{1-p} = e^{(1-p)\ln b}$ and analyzing the sign of the exponent $1-p$ established that it converges precisely when $p > 1$ (to the value $\frac{1}{p-1}$) and diverges when $p \leq 1$. By the Integral Test the series behaves identically. ■

Remark

The harmonic series is the most important counterexample in the unit: its terms $\frac{1}{n}$ tend to 0, yet the sum grows without bound. It does so extremely slowly — the partial sums grow like $\ln n$, so reaching a total of 20 requires roughly 2.7×10^8 terms — but grow without bound they do. This is precisely why the n th Term Test can never confirm convergence (Topic 10.3).

Example 10.5.1: Classifying p-Series

Determine the convergence of each series.

(a) $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$

(b) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

$$(c) \sum_{n=1}^{\infty} \frac{1}{n^2}$$

Solution.

(a) $p = \frac{3}{2} > 1$: **converges**.

(b) $\frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$, so $p = \frac{1}{2} \leq 1$: **diverges**.

(c) $p = 2 > 1$: **converges** — settling the case left open in Example 10.3.1(c).

10.6 Comparison Tests for Convergence

Geometric series and p -series are now completely understood. The comparison tests leverage that knowledge: an unfamiliar positive-term series can be judged by measuring it against a familiar one.

Theorem 10.6.1: The Direct Comparison Test

Suppose $0 \leq a_n \leq b_n$ for all n (or all n beyond some index).

(i) If $\sum b_n$ converges, then $\sum a_n$ converges.

(ii) If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Proof. Let $s_n = \sum_{k=1}^n a_k$ and $t_n = \sum_{k=1}^n b_k$ be the two sequences of partial sums. Since all terms are nonnegative, both sequences are increasing, and since $a_k \leq b_k$ termwise, $s_n \leq t_n$ for every n .

(i) If $\sum b_n$ converges to T , then because $\{t_n\}$ is increasing it never exceeds its limit: $t_n \leq T$ for all n . Hence $s_n \leq t_n \leq T$, so $\{s_n\}$ is an increasing sequence bounded above by T . By the Monotone Convergence Theorem (Theorem 10.3.2), $\{s_n\}$ converges — that is, $\sum a_n$ converges.

(ii) This is the contrapositive of (i): if $\sum b_n$ converged, part (i) would force $\sum a_n$ to converge, contradicting the assumption that it diverges. ■

Remark

The logic runs in only one direction each way, and the direction matters. Being smaller than a *divergent* series says nothing (a series can be smaller than $\sum \frac{1}{n}$ and still converge — for instance $\sum \frac{1}{n^2}$), and being larger than a *convergent* series says nothing either. Only “smaller than convergent” and “larger than divergent” yield conclusions.

Example 10.6.1: Direct Comparison

Determine the convergence of each series.

(a) $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$

(b) $\sum_{n=1}^{\infty} \frac{1}{2^n + n}$

(c) $\sum_{n=2}^{\infty} \frac{1}{\ln n}$

Solution.

(a) Since $n^2 + 1 > n^2 > 0$, we have $0 < \frac{1}{n^2 + 1} < \frac{1}{n^2}$. The larger series $\sum \frac{1}{n^2}$ is a convergent p -series ($p = 2$), so the smaller series **converges**.

(b) Since $2^n + n > 2^n$, $0 < \frac{1}{2^n + n} < \frac{1}{2^n}$, and $\sum \frac{1}{2^n}$ is a convergent geometric series ($r = \frac{1}{2}$). So the series **converges**.

(c) For $n \geq 2$, $\ln n < n$, so $\frac{1}{\ln n} > \frac{1}{n} > 0$. The smaller series $\sum \frac{1}{n}$ is the divergent harmonic series, so by part (ii) the larger series **diverges**.

Direct comparison requires an inequality that holds exactly, which can be awkward when the comparison is only true “in the long run.” The limit form removes that obstacle.

Theorem 10.6.2: The Limit Comparison Test

Suppose $a_n > 0$ and $b_n > 0$ for all n , and that

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L \quad \text{with} \quad 0 < L < \infty.$$

Then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

Proof. Apply the definition of the limit with the particular choice $\varepsilon = \frac{L}{2} > 0$: there exists N such that for all $n > N$,

$$\left| \frac{a_n}{b_n} - L \right| < \frac{L}{2} \quad \Longleftrightarrow \quad \frac{L}{2} < \frac{a_n}{b_n} < \frac{3L}{2}.$$

Multiplying through by $b_n > 0$ gives, for all $n > N$,

$$\frac{L}{2} b_n < a_n < \frac{3L}{2} b_n.$$

Now both directions follow from Direct Comparison. If $\sum b_n$ converges, so does $\sum \frac{3L}{2} b_n$ (a constant multiple), and the right inequality forces $\sum a_n$ to converge. If $\sum b_n$ diverges,

so does $\sum \frac{L}{2}b_n$, and the left inequality forces $\sum a_n$ to diverge. Since finitely many terms (those with $n \leq N$) cannot affect convergence, the conclusion holds for the full series. ■

AP Exam Tip

To choose the comparison series b_n , keep only the *highest-power* term in the numerator and denominator of a_n and simplify — this captures the term's long-run size. The result is almost always a p -series or a geometric series, whose behavior is already known.

Example 10.6.2: Limit Comparison

Determine the convergence of $\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{5 + n^5}}$.

Solution. For large n the numerator behaves like $2n^2$ and the denominator like $\sqrt{n^5} = n^{5/2}$, so

$$a_n \approx \frac{2n^2}{n^{5/2}} = \frac{2}{n^{1/2}}.$$

Take $b_n = \frac{1}{n^{1/2}}$, a divergent p -series ($p = \frac{1}{2} \leq 1$). Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{(2n^2 + 3n)n^{1/2}}{\sqrt{5 + n^5}} = \lim_{n \rightarrow \infty} \frac{2n^{5/2} + 3n^{3/2}}{\sqrt{5 + n^5}} = 2,$$

dividing numerator and denominator by $n^{5/2}$. Since $0 < 2 < \infty$ and $\sum b_n$ diverges, the given series **diverges**.

10.7 Alternating Series Test for Convergence

Every test so far has required positive terms, because each relied on partial sums being increasing. A series whose terms alternate in sign behaves entirely differently: its partial sums overshoot and undershoot the limit in turn, closing in on it from both sides.

Definition 10.7.1: Alternating Series

An **alternating series** has the form $\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + \cdots$ (or its negative), where $b_n > 0$ for all n .

Theorem 10.7.1: The Alternating Series Test

The alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ with $b_n > 0$ **converges** provided both:

- (i) $b_{n+1} \leq b_n$ for all n (the terms are decreasing in magnitude), and
- (ii) $\lim_{n \rightarrow \infty} b_n = 0$.

Proof. Consider the partial sums with an *even* number of terms. Grouping in consecutive pairs,

$$s_{2n} = (b_1 - b_2) + (b_3 - b_4) + \cdots + (b_{2n-1} - b_{2n}),$$

and every parenthesized group is ≥ 0 by hypothesis (i). Hence $\{s_{2n}\}$ is an *increasing* sequence. Regrouping the very same expression differently,

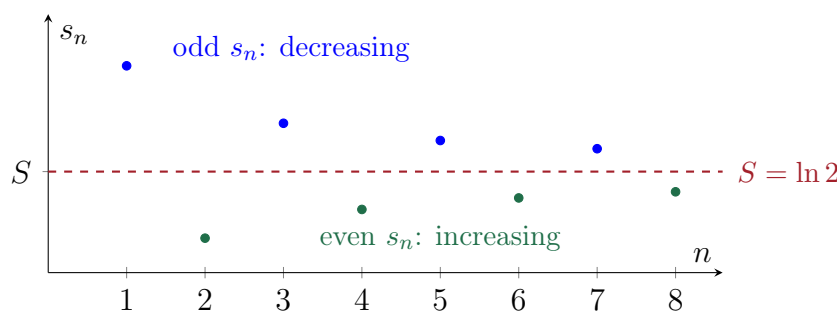
$$s_{2n} = b_1 - (b_2 - b_3) - (b_4 - b_5) - \cdots - (b_{2n-2} - b_{2n-1}) - b_{2n},$$

where again each group is ≥ 0 and $b_{2n} > 0$, so $s_{2n} \leq b_1$ for every n . Thus $\{s_{2n}\}$ is increasing and bounded above by b_1 ; by the Monotone Convergence Theorem it converges, say $\lim_{n \rightarrow \infty} s_{2n} = S$.

Now handle the odd partial sums. Since $s_{2n+1} = s_{2n} + b_{2n+1}$, hypothesis (ii) gives

$$\lim_{n \rightarrow \infty} s_{2n+1} = \lim_{n \rightarrow \infty} s_{2n} + \lim_{n \rightarrow \infty} b_{2n+1} = S + 0 = S.$$

The even-indexed and odd-indexed partial sums therefore converge to the *same* limit S . Because every partial sum is either even- or odd-indexed, the full sequence $\{s_n\}$ converges to S , so the series converges. ■



Remark

The picture shows the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \cdots$. The two subsequences squeeze the limit between them from above and below — precisely the structure the proof exploits — and the gap between consecutive partial sums is exactly the next term b_{n+1} , a fact that becomes the error bound of Topic 10.10.

Example 10.7.1: Applying the Alternating Series Test

Show that the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converges.

Solution. Here $b_n = \frac{1}{n} > 0$. Checking both conditions: (i) $\frac{1}{n+1} < \frac{1}{n}$, so the terms decrease; (ii) $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. Both hold, so the series **converges** by the Alternating Series Test.

Remark

This is a striking result when set beside Topic 10.5: the harmonic series $\sum \frac{1}{n}$

diverges, but attaching alternating signs to the very same terms produces a convergent series (with sum $\ln 2$, as Topic 10.15 will show). Convergence here depends entirely on cancellation between positive and negative terms — a fragile situation examined closely in Topic 10.9.

10.8 Ratio Test for Convergence

The ratio test compares a series to a geometric one — not by a fixed inequality, but by examining the limiting ratio between consecutive terms. It is the most broadly useful test in the unit, and the indispensable tool for power series in Topic 10.13.

Theorem 10.8.1: The Ratio Test

For a series $\sum a_n$ with $a_n \neq 0$, let

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

- (i) If $L < 1$, the series converges absolutely (hence converges).
- (ii) If $L > 1$ (or $L = \infty$), the series diverges.
- (iii) If $L = 1$, the test is **inconclusive**.

Proof of (i). Suppose $L < 1$. Choose any number r strictly between them, $L < r < 1$, and set $\varepsilon = r - L > 0$. By the definition of the limit there is an N such that for all $n \geq N$,

$$\left| \frac{a_{n+1}}{a_n} \right| < L + \varepsilon = r, \quad \text{so} \quad |a_{n+1}| < r |a_n| \quad \text{for all } n \geq N.$$

Applying this inequality repeatedly starting from $n = N$:

$$|a_{N+1}| < r |a_N|, \quad |a_{N+2}| < r |a_{N+1}| < r^2 |a_N|, \quad \dots, \quad |a_{N+k}| < r^k |a_N|.$$

Therefore the tail $\sum_{k=0}^{\infty} |a_{N+k}|$ is bounded termwise by $\sum_{k=0}^{\infty} |a_N| r^k$, a geometric series with ratio $0 < r < 1$, which converges by Theorem 10.2.1. By the Direct Comparison Test, $\sum_k |a_{N+k}|$ converges, and adding back the finitely many terms before index N leaves

convergence unaffected. So $\sum |a_n|$ converges — the series converges absolutely, and hence converges by Theorem 10.9.1 below.

Proof of (ii). If $L > 1$, then for all sufficiently large n we have $|a_{n+1}| > |a_n|$, so the terms are increasing in magnitude and cannot approach 0. By the n th Term Test (Theorem 10.3.1), the series diverges. ■

Remark

Part (iii) is not a gap in the proof but a genuine limitation. Both $\sum \frac{1}{n}$ (divergent)

and $\sum \frac{1}{n^2}$ (convergent) produce $L = 1$, so no conclusion could possibly follow from that value alone. Whenever $L = 1$, abandon the ratio test and try a comparison or the integral test instead.

AP Exam Tip

Reach for the Ratio Test first whenever the terms contain **factorials** or **n th powers** ($n!$, 2^n , n^n), since these simplify dramatically in the ratio $\frac{a_{n+1}}{a_n}$. For terms that are ratios of polynomials, the ratio tends to 1 and the test is useless — use p -series comparison there instead.

Example 10.8.1: Ratio Test with a Factorial

Determine the convergence of $\sum_{n=1}^{\infty} \frac{2^n}{n!}$.

Solution.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2 \cdot 2^n}{(n+1)n!} \cdot \frac{n!}{2^n} = \frac{2}{n+1}.$$

Then $L = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0 < 1$, so the series **converges absolutely**.

Example 10.8.2: Ratio Test with an Exponential

Determine the convergence of $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$.

Solution.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} = \frac{1}{2} \left(\frac{n+1}{n} \right)^2 = \frac{1}{2} \left(1 + \frac{1}{n} \right)^2.$$

So $L = \frac{1}{2}(1+0)^2 = \frac{1}{2} < 1$, and the series **converges absolutely**.

10.9 Determining Absolute or Conditional Convergence

A convergent series with mixed signs may owe its convergence to genuine smallness of its terms, or merely to cancellation between them. The distinction is important enough to have its own vocabulary.

Definition 10.9.1: Absolute and Conditional Convergence

A series $\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges. It **converges conditionally** if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Theorem 10.9.1: Absolute Convergence Implies Convergence

If $\sum |a_n|$ converges, then $\sum a_n$ converges.

Proof. For every n , the inequality $-|a_n| \leq a_n \leq |a_n|$ holds; adding $|a_n|$ throughout gives

$$0 \leq a_n + |a_n| \leq 2|a_n|.$$

The right-hand series $\sum 2|a_n|$ converges (a constant multiple of a convergent series), so by the Direct Comparison Test — applicable because the middle expression is nonnegative — the series $\sum (a_n + |a_n|)$ converges. Finally, write a_n as a difference of two convergent series:

$$\sum a_n = \sum [(a_n + |a_n|) - |a_n|] = \sum (a_n + |a_n|) - \sum |a_n|,$$

and the difference of two convergent series converges. ■

Remark

The trick of adding $|a_n|$ to force nonnegativity is what makes the Comparison Test — a positive-term tool — applicable to a series with mixed signs. Note that the converse fails: the alternating harmonic series converges (Example 10.7.1) while $\sum \frac{1}{n}$ diverges, so convergence does not imply absolute convergence.

AP Exam Tip

To classify a convergent series, test $\sum |a_n|$ *first*. If it converges, the original converges absolutely and no further work is needed. If $\sum |a_n|$ diverges, you must still check the original series separately (usually with the Alternating Series Test) before declaring conditional convergence.

Example 10.9.1: Classifying Convergence

Classify each series as absolutely convergent, conditionally convergent, or divergent.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$

(b) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$

(c) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} n}{n+1}$

Solution.

(a) $\sum \left| \frac{(-1)^{n-1}}{n^2} \right| = \sum \frac{1}{n^2}$ converges ($p = 2 > 1$), so the series converges **absolutely**.

- (b) $\sum \left| \frac{(-1)^{n-1}}{n} \right| = \sum \frac{1}{n}$ diverges (harmonic), but the original series converges by the Alternating Series Test (Example 10.7.1). So it converges **conditionally**.
- (c) $\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0$, so $\lim_{n \rightarrow \infty} a_n$ does not exist and the series **diverges** by the n th Term Test. (The Alternating Series Test does not apply, since condition (ii) fails.)

10.10 Alternating Series Error Bound

For most convergent series, knowing the exact sum is impossible — but approximating it by a partial sum is easy, and for alternating series the resulting error can be bounded with remarkable simplicity.

Theorem 10.10.1: Alternating Series Error Bound

Suppose $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ satisfies the hypotheses of the Alternating Series Test and converges to S . Then for every n ,

$$|S - s_n| \leq b_{n+1},$$

i.e. the error in approximating S by the n th partial sum is at most the magnitude of the *first omitted term*.

Proof. The proof of Theorem 10.7.1 established that the even partial sums increase to S while the odd partial sums decrease to S ; consequently S always lies *between* any two consecutive partial sums:

$$S \text{ lies between } s_n \text{ and } s_{n+1} \text{ for every } n.$$

Therefore the distance from S to s_n is no greater than the distance from s_{n+1} to s_n :

$$|S - s_n| \leq |s_{n+1} - s_n|.$$

But consecutive partial sums differ by exactly one term: $s_{n+1} - s_n = (-1)^n b_{n+1}$, whose magnitude is b_{n+1} . Hence $|S - s_n| \leq b_{n+1}$. ■

Example 10.10.1: Bounding the Error

Approximate $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ using four terms, and bound the error.

Solution. The fourth partial sum is

$$s_4 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12} \approx 0.5833.$$

The first omitted term is $b_5 = \frac{1}{5}$, so

$$|S - s_4| \leq \frac{1}{5} = 0.2.$$

(The true sum is $\ln 2 \approx 0.6931$, and indeed $|0.6931 - 0.5833| \approx 0.110 \leq 0.2$, consistent with the bound.)

Example 10.10.2: Choosing n to Meet an Accuracy Requirement

How many terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$ are needed to approximate the sum with error less than 0.01?

Solution. The error after n terms is at most $b_{n+1} = \frac{1}{(n+1)^2}$. Requiring

$$\frac{1}{(n+1)^2} < 0.01 \iff (n+1)^2 > 100 \iff n+1 > 10 \iff n > 9.$$

So $n = 10$ terms suffice.

AP Exam Tip

This bound applies *only* to alternating series meeting both conditions of Theorem 10.7.1. For a general series, the corresponding tool is the Lagrange Error Bound (Topic 10.12), which is considerably less forgiving to compute.

10.11 Finding Taylor Polynomial Approximations of Functions

Topic 4.1 approximated a function near a point by its tangent line — a degree-1 polynomial matching the function's value and slope at that point. Nothing prevents matching more: a degree- n polynomial has $n+1$ coefficients to spend, so it can match the function's value and its first n derivatives simultaneously, hugging the curve far more closely and over a wider interval.

Definition 10.11.1: Taylor and Maclaurin Polynomials

The **n th-degree Taylor polynomial** of f centered at $x = a$ is

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

When $a = 0$ it is called a **Maclaurin polynomial**.

Remark

The coefficients are forced, not chosen. Requiring $P_n^{(k)}(a) = f^{(k)}(a)$ for each k determines them uniquely: differentiating the term $c_k(x-a)^k$ exactly k times and evaluating at $x = a$ leaves $k!c_k$ (every lower-degree term vanishes under enough differentiation, and every higher-degree term still carries a factor of $(x-a)$). Setting $k!c_k = f^{(k)}(a)$ gives $c_k = \frac{f^{(k)}(a)}{k!}$ — the factorial in the denominator is exactly what the repeated differentiation produces.

Example 10.11.1: Maclaurin Polynomial for e^x

Find $P_3(x)$ for $f(x) = e^x$ centered at $a = 0$.

Solution. Every derivative of e^x is e^x , so $f^{(k)}(0) = e^0 = 1$ for all k . Hence

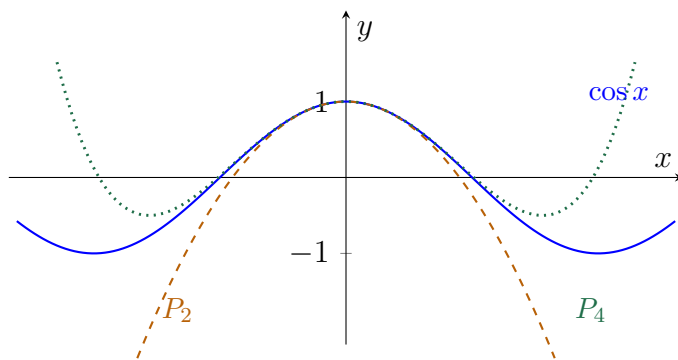
$$P_3(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$

Example 10.11.2: Maclaurin Polynomial for $\cos x$

Find $P_4(x)$ for $f(x) = \cos x$ centered at $a = 0$.

Solution. The derivatives cycle: $f = \cos x$, $f' = -\sin x$, $f'' = -\cos x$, $f''' = \sin x$, $f^{(4)} = \cos x$. Evaluating at 0 gives 1, 0, -1, 0, 1. So

$$P_4(x) = 1 + 0 \cdot x - \frac{x^2}{2!} + 0 \cdot x^3 + \frac{x^4}{4!} = 1 - \frac{x^2}{2} + \frac{x^4}{24}.$$

**Remark**

Notice how each successive polynomial tracks the curve accurately over a wider interval before peeling away. This is the qualitative behavior that Topic 10.12 makes quantitative, and Topic 10.14 pushes to its limit: letting the degree go to infinity produces a series that represents $\cos x$ *exactly*, for every real x .

10.12 Lagrange Error Bound

A Taylor polynomial is only useful as an approximation if the size of its error can be controlled. The remainder is the difference between the function and the polynomial, and it admits a bound strikingly similar in form to the first omitted term.

Definition 10.12.1: Taylor Remainder

The **remainder** after the n th-degree Taylor polynomial is $R_n(x) = f(x) - P_n(x)$, so that $f(x) = P_n(x) + R_n(x)$.

Theorem 10.12.1: Lagrange Error Bound

If f has $n + 1$ derivatives on an interval containing a and x , and $|f^{(n+1)}(t)| \leq M$ for all t between a and x , then

$$|R_n(x)| \leq \frac{M |x - a|^{n+1}}{(n + 1)!}.$$

Remark

The bound is exactly the form of the *next* term of the Taylor polynomial, with the unknown derivative $f^{(n+1)}$ replaced by its maximum possible magnitude M on the relevant interval. This mirrors the Alternating Series Error Bound (Theorem 10.10.1), where the error was bounded by the first omitted term — here the same principle holds, but only after accounting for how large the next derivative could possibly be. The result follows from the Mean Value Theorem applied repeatedly (or from a generalized version of it); the case $n = 0$ is precisely the Mean Value Theorem of Topic 5.2, since $|f(x) - f(a)| = |f'(c)||x - a| \leq M|x - a|$.

AP Exam Tip

Finding M is where most of the work lies: M must be an upper bound for $|f^{(n+1)}|$ across the *entire* interval between a and x , not merely at one endpoint. Any valid upper bound may be used — it need not be the smallest — so choosing a convenient overestimate is both legitimate and usually easier.

Example 10.12.1: Bounding the Error in Approximating $e^{0.1}$

Use $P_2(x)$ for e^x centered at 0 to approximate $e^{0.1}$, and bound the error.

Solution. From Example 10.11.1, $P_2(x) = 1 + x + \frac{x^2}{2}$, so

$$e^{0.1} \approx P_2(0.1) = 1 + 0.1 + 0.005 = 1.105.$$

For the bound, $f^{(3)}(t) = e^t$, which is increasing, so on $[0, 0.1]$ its maximum is $e^{0.1} < 1.2$; take $M = 1.2$. Then

$$|R_2(0.1)| \leq \frac{1.2 |0.1|^3}{3!} = \frac{1.2(0.001)}{6} = 0.0002.$$

(The true value is $e^{0.1} \approx 1.10517$, and the actual error ≈ 0.00017 does satisfy the bound.)

10.13 Radius and Interval of Convergence of Power Series

Letting a Taylor polynomial's degree grow without bound produces an infinite series in powers of $(x - a)$. Such a series may converge for some values of x and diverge for others, so the first question is always: for which x does it converge at all?

Definition 10.13.1: Power Series

A **power series centered at** a is a series of the form

$$\sum_{n=0}^{\infty} c_n(x - a)^n = c_0 + c_1(x - a) + c_2(x - a)^2 + \cdots.$$

Theorem 10.13.1: Convergence of a Power Series

For any power series $\sum c_n(x - a)^n$, exactly one of the following holds:

- (i) it converges only at $x = a$ (radius of convergence $R = 0$);
- (ii) it converges for all real x ($R = \infty$); or
- (iii) there is a number $R > 0$, the **radius of convergence**, such that the series converges absolutely for $|x - a| < R$ and diverges for $|x - a| > R$.

The **interval of convergence** is the set of all x for which the series converges — the interval $(a - R, a + R)$ together with whichever endpoints converge.

Remark

The Ratio Test is what produces R . Applying it to $\sum c_n(x - a)^n$ gives a limit of the form $L = |x - a| \cdot \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$, and requiring $L < 1$ yields an inequality of exactly the form $|x - a| < R$. Because the Ratio Test is inconclusive when $L = 1$ — that is, precisely at $x = a \pm R$ — the **endpoints must always be tested separately**, by substituting each into the original series and applying a different test.

Example 10.13.1: Finding a Radius and Interval of Convergence

Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(x - 2)^n}{n 3^n}$.

Solution. Apply the Ratio Test:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x - 2|^{n+1}}{(n + 1)3^{n+1}} \cdot \frac{n 3^n}{|x - 2|^n} = \frac{|x - 2|}{3} \cdot \frac{n}{n + 1} \xrightarrow{n \rightarrow \infty} \frac{|x - 2|}{3}.$$

Convergence requires $\frac{|x-2|}{3} < 1$, i.e. $|x-2| < 3$, so $\boxed{R=3}$ and the open interval is $(-1, 5)$.

Endpoints. At $x = 5$: the series becomes $\sum \frac{3^n}{n3^n} = \sum \frac{1}{n}$, the divergent harmonic series. At $x = -1$: it becomes $\sum \frac{(-3)^n}{n3^n} = \sum \frac{(-1)^n}{n}$, which converges by the Alternating Series Test. So $x = -1$ is included and $x = 5$ is not:

$$\text{interval of convergence} = [-1, 5).$$

Example 10.13.2: Infinite Radius of Convergence

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

Solution.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|^{n+1}}{(n+1)!} \cdot \frac{n!}{|x|^n} = \frac{|x|}{n+1} \xrightarrow{n \rightarrow \infty} 0$$

for *every* real x . Since $0 < 1$ always, the series converges for all x : $R = \infty$, and the interval of convergence is $(-\infty, \infty)$.

10.14 Finding Taylor or Maclaurin Series for a Function

Taking the Taylor polynomial's degree to infinity produces the **Taylor series** — and when the remainder vanishes in the limit, that series represents the function exactly.

Definition 10.14.1: Taylor and Maclaurin Series

The **Taylor series** of f centered at a is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Centered at $a = 0$ it is the **Maclaurin series**.

Theorem 10.14.1: When a Taylor Series Represents Its Function

If $\lim_{n \rightarrow \infty} R_n(x) = 0$ for all x in an interval I , then on that interval

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Justification. By definition $f(x) = P_n(x) + R_n(x)$ for every n . Taking $n \rightarrow \infty$, the left side is constant while $P_n(x)$ approaches the Taylor series' sum and $R_n(x) \rightarrow 0$ by hypothesis. Hence $f(x)$ equals the series' sum. ■

Example 10.14.1: The Maclaurin Series for e^x

Derive the Maclaurin series for e^x and show it converges to e^x for all real x .

Solution. As in Example 10.11.1, $f^{(n)}(0) = 1$ for every n , so the Maclaurin series is

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots,$$

which converges for all x by Example 10.13.2. To confirm it converges to e^x itself, fix any x and let I be the interval between 0 and x . Since $f^{(n+1)}(t) = e^t$ is increasing, $M = e^{|x|}$ bounds it on I , and the Lagrange bound gives

$$|R_n(x)| \leq \frac{e^{|x|} |x|^{n+1}}{(n+1)!}.$$

For fixed x , the factorial in the denominator outgrows the power in the numerator, so this bound $\rightarrow 0$ as $n \rightarrow \infty$. By Theorem 10.14.1, $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for every real x .

Example 10.14.2: The Maclaurin Series for $\sin x$

Find the Maclaurin series for $\sin x$.

Solution. The derivatives of $\sin x$ cycle through $\sin x$, $\cos x$, $-\sin x$, $-\cos x$, so at $x = 0$ they take the values 0, 1, 0, -1 repeating. Only odd powers survive, with alternating signs:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

valid for all real x (the same Lagrange argument applies, since every derivative of $\sin x$ is bounded by $M = 1$).

10.15 Representing Functions as Power Series

Deriving each new series from scratch by computing infinitely many derivatives is rarely necessary. A handful of known series can be transformed — by substitution, differentiation, integration, or multiplication — into a great many others.

Theorem 10.15.1: Operations on Power Series

If $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ on $|x-a| < R$, then on that same interval the series may be differentiated and integrated termwise:

$$f'(x) = \sum_{n=1}^{\infty} n c_n(x-a)^{n-1}, \quad \int f(x) dx = C + \sum_{n=0}^{\infty} \frac{c_n(x-a)^{n+1}}{n+1},$$

and both resulting series have the same radius of convergence R (though endpoint behavior may change). Substituting any expression for x is likewise valid wherever the substituted series converges.

Remark

The starting point for nearly every manipulation is the geometric series of Topic 10.2, rewritten as a function:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1,$$

which is simply Theorem 10.2.1 with $a = 1$ and ratio $r = x$.

Example 10.15.1: Building a Series by Substitution

Find a power series for $\frac{1}{1+x^2}$.

Solution. Substitute $-x^2$ for x in the geometric series:

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \cdots,$$

valid when $|-x^2| < 1$, i.e. $|x| < 1$, so $R = 1$.

Example 10.15.2: Building a Series by Integration

Use the previous example to find the Maclaurin series for $\arctan x$.

Solution. Since $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$ (Topic 3.4), integrate the series termwise:

$$\arctan x = \int \frac{dx}{1+x^2} = C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}.$$

Setting $x = 0$ gives $\arctan 0 = 0 = C$, so $C = 0$ and

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad R = 1.$$

Remark

Substituting $x = 1$ above (legitimate here, since the series converges at that endpoint by the Alternating Series Test) yields the celebrated identity

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots,$$

an exact expression for π as an infinite sum of rational numbers — a fitting illustration of how much reach the machinery of this unit provides.

10.15.1 Summary: Common Power Series and Their Radii of Convergence

The series below are worth committing to memory. Every one can be derived from the definition, but recognizing them on sight — and transforming them by the operations of Theorem 10.15.1 — is far faster than recomputing derivatives.

Function	Maclaurin series	Radius R
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$	1
$\frac{1}{1+x}$	$\sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + \cdots$	1
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	∞
$\sin x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	∞
$\cos x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	∞
$\ln(1+x)$	$\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$	1
$\arctan x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$	1
$(1+x)^k$	$\sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!} x^2 + \cdots$	1

Remark

The three series with $R = \infty$ are exactly the ones whose derivatives are uniformly bounded or tamed by factorials (e^x , $\sin x$, $\cos x$), which is why the Lagrange remainder vanishes everywhere for them. The four with $R = 1$ all trace back to the geometric series, whose convergence genuinely fails at $|x| = 1$ — and the last row, the **binomial series**, reduces to an ordinary finite binomial expansion whenever k is a nonnegative integer, in which case it terminates and converges for all x .

AP Exam Tip

Endpoint behavior varies even among series sharing a radius: $\sin x$ and $\cos x$ converge everywhere, $\arctan x$ and $\ln(1+x)$ converge at $x = 1$ but not $x = -1$, and $\frac{1}{1-x}$ converges at neither endpoint. The radius never determines the endpoints — they must always be checked individually.

Unit 10 closes the course by turning its central tool on itself. The limit, introduced in Unit 1 to make sense of instantaneous rates, has now been used to make sense of infinite addition — and the payoff is that a function’s behavior at a single point, encoded in its derivatives there, can determine its values across an entire interval. The derivative and the integral, built in Units 2–9 as tools for analyzing functions, turn out in Unit 10 to be tools for *constructing* them.